

Contour instruments for partially linear models with cumulant-separated treatment noise

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Abstract

This paper studies estimation of the partially linear coefficient θ_0 , the treatment coefficient, when the treatment innovation is independent of covariates, sub-Gaussian, and separated from Gaussianity by a fixed nonzero cumulant. The identifying resource is a zero of the treatment-innovation moment-generating function: the associated polynomial-exponential weight annihilates real shifts induced by treatment-code error, and a contour average of observable residual transforms identifies θ_0 under the stated boundary zero-freeness, nuisance zero-freeness, and positive-count conditions.

The statistical theorem is a fixed-code result. For fixed primitive constants, i.i.d. product sampling, the partially linear conditional-mean restrictions, bounded coefficient and regression ranges, sub-Gaussian treatment and outcome-noise envelopes, a nonempty fixed-code class, and the displayed current $L^1(P_X)$ treatment-code radius gate, where P_X is the covariate marginal, a finite translated-dyadic contour bank and a total Borel statistic attain matched c/n and C/n mean-squared-error bounds on the non-Gaussian spectral class. For $\gamma \in (1/2, 1)$, where γ is the probability level, the same statistic controls the generalized lower $(1 - \gamma)$ -quantile of absolute error by order $(\gamma n)^{-1/2}$.

The same fixed-separation rate holds on the aligned Jin–Mackey–Syrkanis ACE comparison class. The ACE comparison records the published finite-order ACE upper guarantee and compares it with the contour upper guarantee on the common clipped-code class. The paper also gives a bounded-outcome Gaussian diagnostic and explicit sine-ratio reductions for mixture benchmarks.

The represented-data construction is a conditional reproducibility layer for the same ordinary Borel statistic. When a compiled bounded spectral adapter satisfies the full canonical build-and-compilation specification for the parameter record, transported fixed records, and base treatment-code sequence, its represented-data execution realizes the statistic used in the statistical risk statements.

1 Introduction

Partially linear models are a standard way to separate a low-dimensional structural coefficient from flexible covariate regressions. In the model studied here, the observed data are triples $O = (X, T, Y)$, the treatment regression is $g_0(X) = E[T \mid X]$, the outcome regression is $q_0(X) = E[Y \mid X]$, and the target is the partially linear coefficient θ_0 in the conditional mean

$$E[Y \mid X, T] = q_0(X) + \theta_0\{T - g_0(X)\}.$$

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Classical semiparametric theory and modern double/debiased machine learning estimate this coefficient by orthogonalizing treatment and outcome variation around nuisance regressions [Robinson, 1988, Newey, 1990, 1994, Newey and McFadden, 1994, Bickel et al., 1993, van der Vaart, 1998, van der Vaart and Wellner, 1996, Belloni et al., 2014, Chernozhukov et al., 2018]. Higher-order orthogonality and ACE methods refine this idea by exploiting additional structure in nuisance errors and treatment residuals [Robins et al., 2008, Liu et al., 2017, Newey and Robins, 2018, Liu et al., 2016, Mackey et al., 2018, Jin et al., 2025]. This paper develops a complementary fixed-separation construction based on non-Gaussian treatment innovations.

The statistical decision problem fixes the supplied clipped treatment and outcome codes at the current sample size. The risk bounds compare laws compatible with that same code pair; if the codes are obtained from an external training stage, the theorem applies after conditioning on the training output. The finite-arithmetic material supplies a conditional reproducibility clause for represented-data executions that satisfy the displayed build contract.

Four statements organize the paper. First, the contour identification theorem recovers θ_0 from an observable population contour ratio under the displayed zero-free and positive-count conditions. Second, under the theorem’s fixed-code nonemptiness, sampling, and radius conditions, the fixed-code minimax theorem gives matched c/n and C/n mean-squared-error bounds on the broad non-Gaussian spectral class and the aligned Jin–Mackey–Syrkanis ACE class. Under the same fixed-code hypotheses, the same contour statistic also has a generalized lower $(1 - \gamma)$ -quantile upper bound of order $(\gamma n)^{-1/2}$. Third, the ACE alignment proposition compares the published finite-order ACE upper guarantee with the contour upper guarantee on the common clipped-code class. Fourth, the bounded-outcome Gaussian diagnostic characterizes the comparison class by its implied target and fixed-code risks. Here P_X denotes the covariate marginal, r the ACE order, δ the cumulant-separation level, γ the probability level, and σ^2 the Gaussian innovation variance.

The identifying resource is a zero of the treatment-innovation moment-generating function. Let $\eta = T - g_0(X)$ denote the treatment innovation and let $M(z) = E[e^{z\eta}]$. When M has a zero z_0 of multiplicity ℓ , the polynomial-exponential weight $J_{z_0, \ell}(w) = w^{\ell-1}e^{z_0 w}$ annihilates every shifted innovation $\eta + d$. This shift invariance is useful for code residuals: if $Z_n = T - \bar{g}_n(X) = \eta + \{g_0(X) - \bar{g}_n(X)\}$, where $\bar{g}_n$ is the clipped supplied treatment code, then the same zero generates conditional moments for Z_n . Theorem 1 gives the resulting known-zero ratio, and Lemma 1 and Theorem 2 turn the ratio into an observable contour identity.

The contour identity is the paper’s population mechanism. The residual transform $F_n(z) = E[e^{zZ_n}]$ factors as $M(z)H_n(z)$, where $H_n(z) = E[e^{z\{g_0(X) - \bar{g}_n(X)\}}]$ is the nuisance transform. The outcome-weighted transform $G_n(z) = E[Ye^{zZ_n}]$ decomposes into a derivative term carrying θ_0 and an analytic contamination term. On any circle where F_n is zero-free on the boundary, H_n is zero-free on the closed disk, and the enclosed zero count is positive, the normalized contour integral of G_n/F_n equals θ_0 . This places the argument in the transform-based identification tradition [Feuerverger and McDunnough, 1981, Singleton, 2001, Chacko and Viceira, 2003, Carrasco et al., 2007, Carrasco, 2017, Kagan et al., 1973, Mattner, 1992, D’Haultfoeulle, 2011, Darolles et al., 2011, Hu and Shiu, 2022].

Estimator in words. The statistic splits the sample deterministically. The pilot fold scans a fixed finite bank of circles, keeps circles whose empirical residual transform has a positive boundary modulus and a positive decoded winding count, and chooses an admissible circle by the displayed lower-modulus rule with an index tie-break. The evaluation fold computes the

normalized contour ratio of the outcome-weighted transform to the residual transform on that selected circle. Clipping to the fixed coefficient range makes the statistic total on every ordinary sample.

The main statistical result is fixed-separation and fixed-code. Under the non-Gaussian spectral class in [Definition 3](#), the treatment innovation is independent of covariates, sub-Gaussian, and has a fixed nonzero $(r + 1)$ st cumulant bounded below by δ . [Lemma 3](#) converts this cumulant separation and the treatment-tail scale into an explicit disk containing a transform zero. [Lemma 8](#) constructs a finite translated-dyadic bank of circles with a positive dyadic boundary-modulus certificate. The $L^1(P_X)$ treatment-code condition in [Assumption 16](#) keeps H_n separated from zero on the search disk, as shown in [Lemma 10](#). Empirical transform concentration in [Lemma 9](#) then supplies the stochastic control needed for the selected contour ratio.

The resulting estimator is the total Borel statistic $\hat{\theta}_{\text{spec},n}$ in [Definition 18](#). [Theorem 3](#) states that, for fixed primitive constants and sufficiently small current $L^1(P_X)$ treatment-code radius, this statistic attains

$$\sup_{P \in \mathcal{P}_{\text{NG},n}(p;\text{base})} E_P\{(\hat{\theta}_{\text{spec},n} - \theta_0(P))^2\} \leq C/n,$$

while the fixed-code non-Gaussian minimax risk is bounded below by c/n . Thus the fixed-code mean-squared-error rate is $\Theta(n^{-1})$. For $\gamma \in (1/2, 1)$, the same theorem controls the generalized lower $(1 - \gamma)$ -quantile of absolute error by order $(\gamma n)^{-1/2}$, and it gives the parallel fixed-code result on the Jin–Mackey–Syrngkanis ACE comparison class. [Theorem 4](#) records the corresponding sequence-level statement for experiments sharing the same primitive constants.

The comparison with ACE is stated as a comparison of upper guarantees on a common cumulant-separated domain. [Lemma 5](#) aligns the published ACE class with the broad spectral class and its $L^s(P_X)$ -restricted subclass. [Proposition 1](#) then records the published order- r ACE generalized-quantile guarantee from [Jin et al. \[2025\]](#) and the spectral guarantee on the same clipped-code ACE class. Along the displayed nuisance-dominance sequences, the ratio of the contour upper guarantee to the imported finite-order ACE upper guarantee tends to zero. This is an upper-bound comparison on the common clipped-code class. The minimax lower bound used for the ACE class is class-level over all estimators on that fixed-code class, as in [Theorem 3](#).

The Gaussian comparison characterizes the bounded-outcome Gaussian intersection in [Definition 4](#). That class combines a nondegenerate Gaussian treatment innovation with bounded observed outcomes and the exact partially linear conditional mean. [Proposition 2](#) shows that these restrictions force $\theta_0(P) = 0$, and consequently the corresponding fixed-code Gaussian mean-squared and generalized-quantile minimax risks are zero whenever the class intersection is nonempty. This result is a diagnostic for that simultaneous set of source assumptions.

Two explicit one-dimensional benchmarks illustrate the contour mechanism. [Proposition 3](#) treats the symmetric mixture $\frac{1}{2}N(-1, 1) + \frac{1}{2}N(1, 1)$, whose transform has a known imaginary zero and whose sine-ratio estimator attains $\bar{C}/n$ risk under the stated range, tail, independence, and $L^1(P_X)$ treatment-code conditions. [Lemma 7](#) and [Theorem 5](#) give a Gaussian–Rademacher path and a local ACE oracle envelope, making the dependence on a shrinking fourth-cumulant magnitude explicit. These benchmarks connect the analysis to non-Gaussian identification ideas in independent component analysis and structural econometrics [[Comon, 1994](#), [Hyvärinen and Oja, 1997](#), [Hyvärinen et al., 2001](#), [Lanne et al., 2017](#), [Lee and Mesters, 2024](#), [Hoesch et al., 2024](#), [Reizinger et al., 2025](#)].

The computational statement is a conditional output-correspondence theorem. [Definitions 17](#) and [18](#) use the same primitive records for the ordinary Borel statistic and for represented-

data execution. For any compiled bounded spectral adapter satisfying the full canonical build-and-compilation specification for the parameter record, transported fixed records, and base treatment-code sequence, [Theorem 3](#) states that the represented-data execution realizes the same statistic. The appendix records proof and deferred-detail information for the paper’s formal statements.

The paper proceeds as follows. The setup section defines the partially linear experiment, supplied-code restrictions, law classes, quantile criterion, and minimax risks. The zero-instrument section proves the known-zero and contour-identification identities. The main fixed-separation section gives zero localization, the contour statistic, and the fixed-code minimax theorem. The comparison section aligns the spectral, ACE, and Gaussian benchmark classes. The mixture section gives explicit sine reductions and local benchmarks. The limitations and future-work section states the local-to-Gaussian adaptation agenda. The appendices collect analytic, empirical-process, lower-bound, certified-construction, verification, and deferred-proof details.

2 Related work

The partially linear model originates in semiparametric regression, where a parametric coefficient is estimated alongside a nonparametric function of covariates [[Engle et al., 1986](#), [Speckman, 1988](#)]; [Robinson \[1988\]](#) gave the root- n consistent residualized estimator that remains the reference construction, [Härdle and Mammen \[1993\]](#) developed the associated specification testing, and [Härdle et al. \[2000\]](#) give the monograph treatment. Partially linear models are a central workhorse for semiparametric estimation of low-dimensional structural coefficients in the presence of high-dimensional or otherwise flexible nuisance functions. The classical formulation and efficiency theory connect the coefficient θ_0 , the partially linear coefficient, to residualized treatment variation after removing the treatment regression and outcome regression [[Hansen, 1982](#), [Newey, 1990, 1994](#), [Newey and McFadden, 1994](#), [Bickel et al., 1993](#), [van der Vaart, 1998](#), [van der Vaart and Wellner, 1996](#)]. Modern double/debiased machine learning builds on the same orthogonalization principle by allowing flexible first-stage estimates while preserving first-order stability of the target moment [[Belloni et al., 2014](#), [Chernozhukov et al., 2018](#)]. Within that tradition, the present paper focuses on a partially linear coefficient estimated from supplied treatment and outcome codes, and studies the additional identifying information carried by non-Gaussian treatment innovations.

A closely related line of work studies higher-order orthogonality and influence-function expansions for nuisance-robust semiparametric estimation. Higher-order constructions refine the first-order orthogonal-score idea by exploiting additional smoothness or distributional structure in nuisance errors [[Robins et al., 1994](#), [van der Laan and Rubin, 2006](#), [Robins et al., 2008](#), [Liu et al., 2017](#), [Newey and Robins, 2018](#), [Liu et al., 2016](#), [Chernozhukov et al., 2022](#)]. For the partially linear model, [Mackey et al. \[2018\]](#) identify the role of non-Gaussian treatment residuals in higher-order orthogonal learning. [Jin et al. \[2025\]](#) give the closest finite-order comparison point: their published order- r ACE guarantee, with ACE order r , controls the $1 - \gamma$ error quantile, where γ is the probability level, by a constant multiple of

$$\varepsilon_{1,n}^r \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1} + (\gamma n)^{-1/2},$$

where $\varepsilon_{1,n}$ and $\varepsilon_{2,n}$ are the treatment- and outcome-code radii, C_θ is the coefficient-range bound, and n is the sample size. The explicit prefactor $r! 16^r \delta^{-1}$, with δ denoting the cumulant-

separation level, makes the comparison sensitive to the finite order and to the amount of non-Gaussian separation.

The direct $L^1(P_X)$ gate in the contour theorem plays a different operational role from the JMS pair of $L^r(P_X)$ treatment and outcome radii. The contour gate controls the treatment-code perturbation of the residual transform and preserves the zero geometry; the ACE eligibility quantities in [Definitions 8](#) and [10](#) also use the outcome-code radius and the order- r finite-difference scale. On bounded covariate-regression ranges, an $L^r(P_X)$ treatment radius at level $\varepsilon_{1,n}$ gives the corresponding $L^1(P_X)$ treatment control, so the common ACE class lies inside the contour class through [Lemma 5](#). The contour result therefore asks the treatment code to meet the displayed small-radius gate, while the published ACE guarantee asks the treatment and outcome radii to meet the JMS eligibility inequalities.

Feature	Contour construction	JMS ACE comparison
Input accuracy	Current $L^1(P_X)$ treatment-code radius and fixed primitive constants	Current $L^r(P_X)$ treatment and outcome radii with JMS eligibility
Analytic resource	A localized zero of the treatment-innovation transform and a selected contour ratio	A finite order- r ACE expansion under cumulant separation
Risk criterion compared	Fixed-code MSE minimax bounds and a generalized-quantile upper bound for the contour statistic	Published generalized-quantile upper guarantee and class-level fixed-code MSE lower bound
Fixed-separation dependence	Constants depend on the fixed cumulant-separation level and primitive envelopes	The displayed imported bound carries the finite-order prefactor and δ^{-1} dependence

This translation gives concrete nuisance-rate regimes for the comparison in [Proposition 1](#). If $\varepsilon_{1,n} = n^{-a}$ and $\varepsilon_{2,n} = n^{-b}$, the ACE nuisance part dominates sampling when $\min\{ra + b, (r + 1)a\} < 1/2$, subject to the JMS eligibility and contour small-radius gates; in that regime the ratio of the contour upper guarantee to the imported finite-order ACE upper guarantee tends to zero on the common clipped-code class. If $\min\{ra + b, (r + 1)a\} \geq 1/2$, the ACE upper guarantee and the contour upper guarantee are both governed by the sampling term up to constants and eligibility factors. These comparisons concern the available upper bounds on the aligned class.

The contour approach developed here represents the same source of identifying variation through zeros of the treatment-innovation moment-generating function and converts those zeros into instruments and contour ratios. Transform methods have a long history in econometrics and statistics, including characteristic-function estimation, empirical transform methods, and moment-generating-function arguments [[Feuerverger and McDunnough, 1981](#), [Singleton, 2001](#), [Chacko and Viceira, 2003](#), [Carrasco et al., 2007](#), [Carrasco, 2017](#)]. The classical foundation for reading distributional structure off a transform is Marcinkiewicz’s theorem, that a characteristic function of the form $\exp(P)$ with P a polynomial forces $\deg P \leq 2$ and hence a Gaussian law [[Marcinkiewicz, 1939](#)]. In this paper’s centered sub-Gaussian envelope, the fixed cumulant-separation condition yields the explicit zero-localization conclusion in [Lemma 3](#): the treatment-innovation MGF has a zero inside the displayed disk. The zero-based step also connects to classical results on distributions, transforms, and completeness [[Kagan et al., 1973](#), [Mattner,](#)

1992, D’Haultfoeuille, 2011, Darolles et al., 2011, Hu and Shiu, 2022]. In the present setting, those tools are applied to residualized treatment variation formed with a supplied treatment code, so the analytic question becomes stability of transform zeros under $L^1(P_X)$ treatment-code perturbations.

Non-Gaussianity is also a recurring identifying resource in independent component analysis, structural shock recovery, and related econometric models [Comon, 1994, Hyvärinen and Oja, 1997, Hyvärinen et al., 2001, Lanne et al., 2017, Lee and Mesters, 2024, Hoesch et al., 2024, Reizinger et al., 2025]. Relatedly, Andrews [2017] exhibits families that are L^2 -complete or boundedly complete, the property that makes a vanishing transform functional informative under the relevant class. The construction here uses non-Gaussianity through localized transform-zero geometry: it localizes a transform zero and integrates the observable ratio on a selected circle around the enclosed zero structure. The paper applies that resource in a partially linear estimation problem: cumulant separation supplies a transform zero, and the contour construction converts the resulting analytic geometry into a coefficient estimator. This places the contribution between higher-order semiparametric learning and non-Gaussian identification, with the comparison to ACE made on the common cumulant-separated class where both sets of assumptions are imposed.

The fixed-separation statistical statement is expressed in minimax terms. Under fixed cumulant separation and stable supplied treatment codes, the contour estimator attains the parametric n^{-1} mean-squared-error order over the non-Gaussian spectral class, while the comparison with Jin et al. [2025] records the published ACE generalized-quantile upper guarantee on the aligned ACE class. The sequence comparison in Proposition 1 is an upper-guarantee comparison on the common clipped-code class: when the displayed ACE nuisance terms dominate $n^{-1/2}$, the ratio of the fixed-separation contour upper guarantee to the imported finite-order ACE upper guarantee tends to zero.

Finally, the finite-construction component draws on computable analysis and interval arithmetic. The relevant background treats real-number computation, constructive numerical representations, and certified interval enclosures [Weihrauch, 2000, Ko, 1991, Moore, 1966]. Those tools provide the arithmetic substrate for the contour bank and represented-data statistic, while the statistical exposition keeps the identification and risk arguments in standard econometric form.

3 Setup and assumptions

We use generic constants c and C only in later bounds, where their dependence is stated locally. All norms are taken under the law displayed in the surrounding statement, and $L^p(P_X)$ denotes integration with respect to the covariate marginal. The sample index is fixed within each finite- n experiment, while class constants and code-radius sequences are carried by the parameter record. Asymptotic comparisons in later sections vary the sample size through these records, with the order pair, tail scales, cumulant separation, and range bounds held according to the relevant theorem hypotheses.

The basic experiment is a partially linear model with deterministic supplied codes for the treatment and outcome regressions. The first definition records the numerical constants, the split convention, and the exact sub-Gaussian norm convention used throughout.

3.1 Fixed-code decision problem

The supplied treatment and outcome codes are inputs to the current finite- n decision problem. At sample size n , the estimator uses the clipped pair $(\bar{g}_n, \bar{q}_n)$, and the risks below compare laws whose current clipped code pair agrees with that same input. The data used in the risk bound are then the i.i.d. product sample from the law under evaluation, with the code pair treated as fixed for that experiment.

This convention is the one used by the minimax criteria in [Definition 9](#) and by the fixed-code classes in [Theorem 3](#). When the supplied codes come from an external training stage, the theorem applies through the deterministic code pair obtained after conditioning on that training output, or through any deterministic code sequence fixed before the product-law sample is drawn. The theorem's code-radius assumptions are assumptions about that current clipped pair under the covariate marginal P_X .

Definition 1. `[def:model-parameters]` (Parameter record and standing conventions) *A parameter record fixes a sample size $n \geq 1$, an order pair $r, k \in \mathbb{N}$ with*

$$k = r + 1, \quad r \geq 2,$$

a comparison exponent $s \in [0, \infty]$ with $s \geq r$, a probability level $\gamma \in (1/2, 1)$, positive constants

$$C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma > 0,$$

and two code-radius sequences $\varepsilon_1, \varepsilon_2 : \mathbb{N} \rightarrow \mathbb{R}$ that are nonnegative and nonincreasing on positive indices; their values at n are written $\varepsilon_{1,n}$ and $\varepsilon_{2,n}$.

Two conventions are used throughout: $N(m, v)$ denotes the Gaussian law on $\mathbb{R}$ with mean m and variance v , and $\mathbb{P}_n f = n^{-1} \sum_{i=1}^n f(O_i)$ denotes the empirical average of f over the sample. The sample is split into the two deterministic folds

$$I_0 = \{i : i < \lfloor n/2 \rfloor\}, \quad I_1 = \{i : i \geq \lfloor n/2 \rfloor\},$$

written I_a for $a \in \{0, 1\}$.

For a real random variable W and $t > 0$, the Luxemburg sub-Gaussian convention is that $\|W\|_{\psi_2} \leq t$ means $E[\exp(W^2/t^2)] \leq 2$, and its conditional form $\|W\|_{\psi_2|X} \leq t$ means $E[\exp(W^2/t^2) | X] \leq 2$ almost surely.

The observation-level model follows the usual partially linear decomposition, with g_0 and q_0 denoting the treatment and outcome regressions and with supplied code sequences clipped to the fixed ranges in [Definition 1](#).

Definition 2. `[def:plm-model]` (Observed-data partially linear model) *An observation is the triple $O = (X, T, Y)$, where X takes values in a measurable covariate space $\mathcal{X}$ and $T, Y \in \mathbb{R}$. A model over a parameter record consists of a probability law P for O under which T and Y are integrable, a real coefficient θ_0 , measurable regression functions g_0, q_0 on the covariate space with*

$$g_0(X) = E_P[T | X], \quad q_0(X) = E_P[Y | X]$$

P -almost surely, and deterministic measurable supplied code sequences $\hat{g} = (\hat{g}_n)_{n \in \mathbb{N}}$ and $\hat{q} = (\hat{q}_n)_{n \in \mathbb{N}}$ for g_0 and q_0 . The treatment and outcome innovations are

$$\eta = T - g_0(X), \quad \xi = Y - q_0(X) - \theta_0 \eta,$$

and the clipped supplied codes at index n are

$$\bar{g}_n(x) = \min\{\max\{\hat{g}_n(x), -C_g\}, C_g\}, \quad \bar{q}_n(x) = \min\{\max\{\hat{q}_n(x), -C_q\}, C_q\}.$$

The covariate marginal of P is written P_X . A model is written m , and when the law has to be displayed the coefficient is written $\theta_0(P)$.

The sampling and structural restrictions are collected next. They separate the finite-sample product experiment, the partially linear conditional mean, the bounded ranges, and the tail and cumulant conditions that define the statistical classes.

Assumption 1. `[ass:iid-sampling]` (I.i.d. sampling) *Under [Definitions 1](#) and [2](#), the observations satisfy*

$$O_1, \dots, O_n \stackrel{\text{iid}}{\sim} P.$$

The i.i.d. sampling condition is the standard finite-sample product-law setup for orthogonal and semiparametric procedures [[Chernozhukov et al., 2018](#)]. It fixes the probability space on which all estimators and generalized quantiles below are evaluated.

Assumption 2. `[ass:independent-treatment-noise]` (Independent treatment noise) *Under [Definitions 1](#) and [2](#), the treatment innovation is independent of the covariates:*

$$\eta \perp X.$$

Independent treatment noise is the standard residual independence condition used in the cumulant-based partially linear comparison [[Jin et al., 2025](#)]. In this paper it also makes the treatment-innovation transform a law-level object separate from the covariate distribution.

Assumption 3. `[ass:outcome-mean-independence]` (Outcome mean independence) *Under [Definitions 1](#) and [2](#), the outcome innovation has conditional mean zero given the covariates and treatment:*

$$E_P[\xi \mid X, T] = 0$$

almost surely.

Outcome mean independence is the standard partially linear conditional mean restriction [[Robinson, 1988](#)]. It pins down the coefficient through the conditional mean of Y given X and T .

Assumption 4. `[ass:theta-range]` (Coefficient range) *Under [Definitions 1](#) and [2](#), the partially linear coefficient satisfies*

$$|\theta_0| \leq C_\theta.$$

The coefficient range is the standard bounded target range condition in the JMS comparison framework [[Jin et al., 2025](#)]. It supplies a common compact parameter range for minimax risks and clipped estimators.

Assumption 5. `[ass:g-range]` (Treatment regression range) *Under [Definitions 1](#) and [2](#), the treatment regression satisfies*

$$\|g_0\|_\infty \leq C_g.$$

The bounded treatment regression condition is standard in the same comparison class [Jin et al., 2025]. It also aligns the supplied treatment code with the clipping radius used in Definition 2.

Assumption 6. `[ass:q-range]` (Outcome regression range) *Under Definitions 1 and 2, the outcome regression satisfies*

$$\|q_0\|_\infty \leq C_q.$$

The bounded outcome regression condition is the standard common sub-Gaussian PLM range condition [Jin et al., 2025]. Together with the treatment and coefficient bounds, it fixes the envelope in which the supplied outcome code is compared.

Assumption 7. `[ass:bounded-gaussian-outcome]` (Bounded outcome) *In the model of Definition 2 with constants as in Definition 1, the outcome is uniformly bounded:*

$$|Y| \leq C_q$$

almost surely.

The bounded observed-outcome restriction is the standard bounded- Y condition in the simultaneous Gaussian comparison experiment [Jin et al., 2025]. It is used only for that Gaussian intersection class.

Assumption 8. `[ass:eta-subgaussian]` (Sub-Gaussian treatment noise) *In the model of Definition 2 with constants as in Definition 1, the Luxemburg sub-Gaussian norm is*

$$\|W\|_{\psi_2} = \inf\{u > 0 : E_P[\exp(W^2/u^2)] \leq 2\}.$$

Under this convention, the treatment innovation satisfies

$$\|\eta\|_{\psi_2} \leq \psi_\eta.$$

The treatment-tail condition is the standard uniform sub-Gaussian treatment-noise requirement under the exact Luxemburg convention [Jin et al., 2025]. The convention matches the concentration and transform-envelope bounds used later; background sub-Gaussian facts follow the usual conventions in Vershynin [2018] and Wainwright [2019].

Assumption 9. `[ass:xi-subgaussian]` (Sub-Gaussian outcome noise) *In the model of Definition 2 with constants as in Definition 1, the outcome innovation satisfies*

$$\|\xi\|_{\psi_2|X} \leq \psi_\xi$$

almost surely.

The outcome-tail condition is the standard uniform conditional sub-Gaussian outcome-noise restriction [Jin et al., 2025]. It provides the conditional exponential-square control needed for empirical outcome-weighted transforms.

Assumption 10. `[ass:cumulant-separation]` (Cumulant separation) *In the model of Definition 2 with parameters as in Definition 1, the k th cumulant of the treatment innovation satisfies*

$$|\kappa_k(\eta)| \geq \delta.$$

Cumulant separation is the standard Jin–Mackey–Syrkanis nonzero cumulant condition with a uniform separation level [Jin et al., 2025]. In the contour analysis, the same lower bound supplies the non-Gaussian signal that later yields transform zeros.

The supplied-code restrictions specify how the deterministic first-stage codes enter the statistical classes. The $L^s(P_X)$ conditions give a common language for comparison with ACE, the $L^r(P_X)$ versions match the published order- r condition, the Gaussian condition defines the comparison experiment, and the $L^1(P_X)$ condition is the stability requirement used by the contour class.

Assumption 11. [ass:treatment-code-radius] (Treatment code radius) *In the model of Definition 2 with parameters as in Definition 1, for every n ,*

$$\|\bar{g}_n - g_0\|_{L^s(P_X)} \leq \varepsilon_{1,n}.$$

The $L^s(P_X)$ treatment-code radius is a standard structure-agnostic supplied-code uncertainty set [Jin et al., 2025]. It measures the current-index treatment-code error under the covariate marginal.

Assumption 12. [ass:outcome-code-radius] (Outcome code radius) *In the model of Definition 2 with parameters as in Definition 1, for every n ,*

$$\|\bar{q}_n - q_0\|_{L^s(P_X)} \leq \varepsilon_{2,n}.$$

The matching $L^s(P_X)$ outcome-code radius is the standard supplied-code uncertainty set for the outcome regression [Jin et al., 2025]. It places the outcome code and treatment code on the same comparison scale.

Assumption 13. [ass:jms-treatment-code-radius] (Treatment code radius) *In the notation of Definitions 1 and 2, for every $n \geq 1$,*

$$\|\bar{g}_n - g_0\|_{L^r(P_X)} \leq \varepsilon_{1,n}.$$

The $L^r(P_X)$ treatment-code radius is the standard Theorem 5.4 treatment-code uncertainty set in the JMS ACE guarantee [Jin et al., 2025]. It is stated separately because the published finite-order bound is indexed by the ACE order r .

Assumption 14. [ass:jms-outcome-code-radius] (Outcome code radius) *In the notation of Definitions 1 and 2, for every $n \geq 1$,*

$$\|\bar{q}_n - q_0\|_{L^r(P_X)} \leq \varepsilon_{2,n}.$$

The $L^r(P_X)$ outcome-code radius is the corresponding standard Theorem 5.4 outcome-code uncertainty set [Jin et al., 2025]. It is paired with Assumption 13 in the ACE comparison class.

Assumption 15. [ass:gaussian-treatment-noise] (Gaussian treatment noise) *In the notation of Definitions 1 and 2, the treatment innovation satisfies*

$$\eta \sim N(0, \sigma^2).$$

The Gaussian treatment-noise condition is the standard known-variance Gaussian experiment used in Theorem 3.2 of Jin et al. [2025]. It defines the Gaussian benchmark class against which the non-Gaussian contour class is compared.

Assumption 16. `[ass:l1-treatment-code-radius]` (L^1 treatment radius) *In the notation of Definitions 1 and 2, for every $n \geq 1$,*

$$\|\bar{g}_n - g_0\|_{L^1(P_X)} \leq \varepsilon_{1,n}.$$

The $L^1(P_X)$ treatment radius is specific to this analysis. It is the stability condition that controls the code-residual perturbation of the treatment transform in the contour argument. Throughout, a *code residual* means $Z_n = T - \bar{g}_n(X)$ formed with the deterministic supplied clipped code $\bar{g}_n$. All risk statements condition on deterministic supplied clipped treatment and outcome codes and evaluate fixed-code performance at the current sample index.

We now package these restrictions into the law classes used by the main theorem and by the ACE and Gaussian comparisons. The broad spectral class keeps the cumulant-separated treatment noise and the $L^1(P_X)$ treatment-code control that are needed for the contour construction.

Definition 3. `[def:non-gaussian-class]` (Non-Gaussian law class $\mathcal{P}_{\text{NG},n}$) *In the notation of Assumptions 2 to 6 and 8 to 10 and Definitions 1 and 2, let P be an observed-data law on $O = (X, T, Y)$. Define*

$$g_0(x) = E_P[T \mid X = x], \quad q_0(x) = E_P[Y \mid X = x],$$

let $\theta_0(P)$ be the constant coefficient in the partially linear conditional mean

$$E_P[Y \mid X, T] = q_0(X) + \theta_0(P)\{T - g_0(X)\},$$

and set

$$\eta = T - g_0(X), \quad \xi = Y - q_0(X) - \theta_0(P)\eta.$$

For each $n \geq 1$, $\mathcal{P}_{\text{NG},n}$ is the class of such laws P satisfying:

- *(Treatment independence.)* $\eta \perp X$.
- *(Outcome mean.)* $E_P[\xi \mid X, T] = 0$.
- *(Coefficient range.)* $|\theta_0(P)| \leq C_\theta$.
- *(Treatment regression range.)* $\|g_0\|_\infty \leq C_g$.
- *(Outcome regression range.)* $\|q_0\|_\infty \leq C_q$.
- *(Treatment tail.)* $\|\eta\|_{\psi_2} \leq \psi_\eta$.
- *(Outcome tail.)* $\|\xi\|_{\psi_2|X} \leq \psi_\xi$.
- *(Cumulant separation.)* $|\kappa_{r+1}(\eta)| \geq \delta$.
- *(Treatment code control.)* $\|\bar{g}_n - g_0\|_{L^1(P_X)} \leq \varepsilon_{1,n}$.

The bounded-outcome Gaussian class uses the same partially linear residual functionals but imposes Gaussian treatment noise and the supplied-code restrictions at exponent s . It is the source-stated Gaussian comparison class for the diagnostic and minimax criteria below.

Definition 4. `[def:gaussian-class]`

$$\mathcal{P}_{G,n}^{\text{JMS}} := \left\{ P : \begin{array}{l} 1 \leq n, \ E_P[\xi \mid X, T] = 0 \text{ } P\text{-a.s.}, \ \xi \in L^1(P), \ |\theta_0| \leq C_\theta, \ |Y| \leq C_q \text{ } P\text{-a.s.}, \\ \eta \perp X, \ \eta \sim N(0, \sigma^2), \ |g_0(x)| \leq C_g \text{ } P_X\text{-a.e.}, \\ \|\bar{g}_n - g_0\|_{L^s(P_X)} \leq \varepsilon_{1,n}, \ \|\bar{q}_n - q_0\|_{L^s(P_X)} \leq \varepsilon_{2,n} \end{array} \right\}.$$

Here g_0, q_0, η, ξ are the residual functionals determined by P in [Definition 3](#).

The ACE comparison class records the published cumulant-separated setup at order r . Its member conditions mirror the preceding assumptions while using the $L^r(P_X)$ code radii that enter the JMS finite-order guarantee.

Definition 5. `[def:jms-ace-class]` (JMS ACE comparison class $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$) *Fix an admissible parameter record with positive sample-size coordinate and*

$$k = r + 1, \quad 2 \leq r, \quad r \leq s, \quad \gamma \in (1/2, 1),$$

with

$$0 < C_\theta, \quad 0 < C_g, \quad 0 < C_q, \quad 0 < \psi_\eta, \quad 0 < \psi_\xi, \quad 0 < \delta, \quad 0 < \sigma,$$

and with radius sequences satisfying

$$0 \leq \varepsilon_{j,b} \leq \varepsilon_{j,a} \quad (j \in \{1, 2\}, \ 1 \leq a \leq b).$$

At any positive class index n , the JMS ACE comparison class $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ consists of all model records m whose observed-data component P is a probability law on $O = X \times \mathbb{R} \times \mathbb{R}$, carrying a real θ_0 , measurable functions $g_0, q_0 : X \rightarrow \mathbb{R}$, and deterministic measurable supplied code sequences whose clipped versions are denoted $\bar{g}_\ell$ and $\bar{q}_\ell$. The coordinates T and Y are P -integrable, and the conditional-mean identities

$$g_0(X) = E_P[T \mid X], \quad q_0(X) = E_P[Y \mid X]$$

hold P -almost surely. Writing $\eta = T - g_0(X)$ and $\xi = Y - q_0(X) - \theta_0\eta$, the member conditions in [Assumptions 2 to 6](#) and [8 to 10](#) hold in the following form:

$$\xi \in L^1(P), \quad E_P[\xi \mid X, T] = 0, \quad \eta \perp X,$$

with the conditional-mean identity for ξ holding P -almost surely;

$$|\theta_0| \leq C_\theta, \quad |g_0(x)| \leq C_g, \quad |q_0(x)| \leq C_q,$$

with the displayed bounds on g_0 and q_0 holding for P_X -almost every x ;

$$\exp(\eta^2/\psi_\eta^2) \in L^1(P), \quad \int \exp(\eta^2/\psi_\eta^2) dP \leq 2,$$

$$\exp(\xi^2/\psi_\xi^2) \in L^1(P), \quad E_P[\exp(\xi^2/\psi_\xi^2) \mid X] \leq 2,$$

with the conditional tail bound holding P -almost surely; and

$$|\kappa_{r+1}(\eta)| \geq \delta, \quad \|\bar{g}_n - g_0\|_{L^r(P_X)} \leq \varepsilon_{1,n}, \quad \|\bar{q}_n - q_0\|_{L^r(P_X)} \leq \varepsilon_{2,n}.$$

Risk comparisons use a generalized lower quantile for absolute estimation error. Because $\gamma \in (1/2, 1)$, the controlled level $1 - \gamma$ lies below one half, so $Q_{P,1-\gamma}$ is a lower quantile of the absolute error and every quantile bound below is to be read on that scale. The definition is stated for an arbitrary real sample statistic so it can be applied both to estimator errors and to the published ACE quantile bound.

Definition 6. `[def:generalized-quantile]` (Generalized quantile $Q_{P,1-\gamma}(W)$) *In the sampling setting of [Assumption 1](#) and [Definitions 1](#) and [2](#), let P be an observed-data law, let $n \geq 1$, and let $W = W(O_1, \dots, O_n)$ be a real-valued statistic of the sample. Under $P^{\otimes n}$, write*

$$F_W(t) = \Pr(W \leq t)$$

for the distribution function of W . For $\gamma \in (1/2, 1)$, the generalized lower $(1 - \gamma)$ -quantile of W is

$$Q_{P,1-\gamma}(W) = \inf\{t \in \mathbb{R} : F_W(t) \geq 1 - \gamma\}.$$

Equivalently, $Q_{P,1-\gamma}(W)$ is the left-continuous generalized inverse of F_W evaluated at level $1 - \gamma$.

The contour factorization in the next section is expressed through the covariate-level error made by the supplied treatment code. The following notation isolates that error and the corresponding outcome-contamination transform.

Definition 7. `[def:outcome-contamination]` (Treatment-code error D_n , outcome contamination b_n , and its transform B_n) *Fix a law P , an index n , and the clipped supplied treatment code $\bar{g}_n$. The treatment-code error and the outcome contamination are the covariate functions*

$$D_n(X) = g_0(X) - \bar{g}_n(X), \quad b_n(X) = q_0(X) - \theta_0\{g_0(X) - \bar{g}_n(X)\} = q_0(X) - \theta_0 D_n(X),$$

and the contamination transform is their weighted exponential transform

$$B_n(z) = E_P[b_n(X)e^{zD_n(X)}], \quad z \in \mathbb{C}.$$

For comparison with [Jin et al. \[2025\]](#), we also record the eligibility quantities that enter the published order- r ACE condition. They combine code radii, sample size, tail scales, coefficient range, and cumulant separation into the finite-order admissibility check.

Definition 8. `[def:jms-eligibility-quantities]` (JMS eligibility quantities $a_{1,n}, b_{1,n}, a_2, b_{2,n}$) *Under the notation and conditions in [Assumptions 4, 5, 8 to 10, 13](#) and [14](#) and [Definitions 1](#) and [2](#), the four quantities entering the Jin–Mackey–Syrkanis order- r eligibility condition at index n are*

$$a_{1,n} = 2 \log \left(\frac{6(C_g + \psi_\eta)}{\varepsilon_{1,n}} \right), \quad b_{1,n} = \log \left(\frac{\gamma n}{9} \right), \quad a_2 = 4(C_g + \psi_\eta),$$

$$b_{2,n} = \frac{200 \min\{1, C_\theta\} \delta}{\max\{\varepsilon_{1,n}, \varepsilon_{2,n}, (\gamma n)^{-1/2}(\psi_\xi + C_\theta \psi_\eta)\}}.$$

The minimax criteria are class-indexed and fix the supplied codes at the current sample size. This keeps the decision problem focused on laws compatible with the same clipped code pair.

Definition 9. [def:minimax-risks] (Minimax risks $\mathfrak{R}_n^{\text{NG}}, \mathfrak{R}_n^{\text{G}}, \mathfrak{M}_{n,1-\gamma}^{\text{G}}$) For fixed supplied code sequences, let $\bar{g}_n$ and $\bar{q}_n$ denote their current-index clippings to $[-C_g, C_g]$ and $[-C_q, C_q]$. All infima below range over measurable estimators based on $O_1, \dots, O_n$, and each supremum is restricted to laws in the displayed class whose clipped treatment and outcome codes at index n are $\bar{g}_n$ and $\bar{q}_n$. Given a parameter record p and supplied code sequences $\hat{g}, \hat{q}$, a class $\mathcal{C}$ of laws carries the notation $\mathcal{C}(p; \bar{g}_n, \bar{q}_n)$ for the subset cut out by this fixed-code restriction, and we write $\mathfrak{R}_n^{\text{NG}}(p; \hat{g}, \hat{q})$, $\mathfrak{R}_n^{\text{G}}(p; \hat{g}, \hat{q})$, and $\mathfrak{M}_{n,1-\gamma}^{\text{G}}(p; \hat{g}, \hat{q})$ for the three risks defined next whenever their dependence on the parameter record and the supplied codes has to be displayed. Define the non-Gaussian minimax mean-squared risk by

$$\mathfrak{R}_n^{\text{NG}} = \inf_{\tilde{\theta}(O_1, \dots, O_n; \bar{g}_n, \bar{q}_n)} \sup_{P \in \mathcal{P}_{\text{NG},n}} E_P \left[(\tilde{\theta} - \theta_0(P))^2 \right].$$

Define the Gaussian JMS minimax mean-squared risk by

$$\mathfrak{R}_n^{\text{G}} = \inf_{\tilde{\theta}(O_1, \dots, O_n; \bar{g}_n, \bar{q}_n)} \sup_{P \in \mathcal{P}_{\text{G},n}^{\text{JMS}}} E_P \left[(\tilde{\theta} - \theta_0(P))^2 \right].$$

For $1/2 < \gamma < 1$, define the Gaussian JMS minimax generalized lower-quantile risk by

$$\mathfrak{M}_{n,1-\gamma}^{\text{G}} = \inf_{\tilde{\theta}} \sup_{P \in \mathcal{P}_{\text{G},n}^{\text{JMS}}} Q_{P,1-\gamma}(|\tilde{\theta} - \theta_0(P)|).$$

The order- r ACE guarantee is invoked in later comparisons through its eligibility predicate. The predicate below is the finite-sample condition formed from the four quantities in [Definition 8](#).

Definition 10. [def:jms-eligibility] (JMS eligibility condition $\mathbf{E}_n^{\text{JMS}}$) Let $a_{1,n}, b_{1,n}, a_{2,n}, b_{2,n}$ be the quantities in [Definition 8](#). For every index n for which the following expressions are defined, $\mathbf{E}_n^{\text{JMS}}$ is the condition

$$r \leq \min \left\{ \frac{b_{1,n}}{a_{1,n}} - \frac{1}{a_{1,n}} \log \left(\frac{b_{1,n}}{a_{1,n}} \right), \frac{b_{2,n}}{a_{2,n} \log(a_{2,n} b_{2,n})} \right\}.$$

Finally, the common ACE comparison class is obtained by adding the $L^s(P_X)$ treatment- and outcome-code restrictions to the broad spectral class. This subclass is the domain on which the contour and ACE upper guarantees are aligned in the comparison section.

Definition 11. [def:ace-comparison-subclass] (ACE comparison subclass $\mathcal{P}_{\text{NG},n}^{\text{ACE}}$) Relative to [Definition 3](#), define

$$\mathcal{P}_{\text{NG},n}^{\text{ACE}} = \{P \in \mathcal{P}_{\text{NG},n} : \|\bar{g}_n - g_0\|_{L^s(P_X)} \leq \varepsilon_{1,n}, \quad \|\bar{q}_n - q_0\|_{L^s(P_X)} \leq \varepsilon_{2,n}\}.$$

Thus $\mathcal{P}_{\text{NG},n}^{\text{ACE}}$ is the subclass of the broad spectral class that satisfies both supplied-code restrictions.

4 Zero instruments and contour identification

The analytic mechanism starts from a zero of the treatment-innovation transform and turns it into an instrument for the residualized treatment. In the partially linear model, the supplied treatment code enters through the residual $Z_n = T - \bar{g}_n(X)$; the point of the construction is that a transform zero of the innovation η continues to generate usable population moments after

the residual has been shifted by the treatment-code error. The use of transform zeros follows a classical identification tradition for moment-generating and characteristic functions [Kagan et al., 1973, Mattner, 1992, D’Haultfoeuille, 2011, Darolles et al., 2011, Hu and Shiu, 2022], while the contour formulation below is aligned with transform-based estimation in econometrics [Feuerverger and McDunnough, 1981, Singleton, 2001, Carrasco et al., 2007, Carrasco, 2017].

The first object is the treatment-innovation transform $M(z)$ and its zero-based instrument. If z_0 is a zero of multiplicity ℓ , differentiating the exponential tilt at that zero produces the polynomial-exponential weight used throughout this section.

Definition 12. [def:zero-instrument] (Treatment-innovation transform M) *Let $M : \mathbb{C} \rightarrow \mathbb{C}$, let $z_0 \in \mathbb{C}$, and let $\ell \in \mathbb{N}$ satisfy $1 \leq \ell$, $M(z_0) = 0$, and analytic order ℓ for M at z_0 . Define*

$$J_{z_0, \ell}(w) = w^{\ell-1} e^{z_0 w}, \quad w \in \mathbb{C}.$$

The instrument $J_{z_0, \ell}(w)$ is indexed by the complex zero z_0 and the positive multiplicity ℓ . Its role is local in the transform variable but global in the residual shift: the same weight will be evaluated at shifted innovations and at the code residual.

Theorem 1. [thm:known-zero-instrument] (Known zero instrument) *Fix $n \geq 1$ and a law $P \in \mathcal{P}_{\text{NG}, n}$ as in Definition 3. Let*

$$M(z) = E_P[e^{z\eta}], \quad Z_n = T - \bar{g}_n(X), \quad H_n(z) = E_P[e^{z\{g_0(X) - \bar{g}_n(X)\}}].$$

Suppose that:

- **(Order.)** $\ell \geq 1$.
- **(Zero.)** $z_0 \in \mathbb{C}$ satisfies $M(z_0) = 0$.
- **(Multiplicity.)** The analytic order of M at z_0 is ℓ .

For the zero-based instrument $J_{z_0, \ell}(w) = w^{\ell-1} e^{z_0 w}$ from Definition 12, the following identities hold:

$$\begin{aligned} \forall d \in \mathbb{R}, \quad E_P[J_{z_0, \ell}(\eta + d)] &= 0, \\ E_P[J_{z_0, \ell}(Z_n) \mid X] &= 0 \quad P\text{-a.s.}, \end{aligned}$$

and

$$E_P[Z_n J_{z_0, \ell}(Z_n)] = M^{(\ell)}(z_0) H_n(z_0).$$

Consequently, whenever $E_P[Z_n J_{z_0, \ell}(Z_n)] \neq 0$,

$$\theta_0 = \frac{E_P[Y J_{z_0, \ell}(Z_n)]}{E_P[Z_n J_{z_0, \ell}(Z_n)]}.$$

Theorem 1 gives the moment-ratio form of the argument when a valid zero is known. The identity $E_P[J_{z_0, \ell}(\eta + d)] = 0$ applies uniformly over real shifts d , so conditioning on X converts the residualized treatment into a valid zero moment. The denominator identity identifies the relevant scale as the product of the nonzero derivative at the transform zero and the nuisance transform $H_n(z)$; when that scale is nonzero, the partially linear coefficient is the displayed population ratio.

For data-driven contour selection, the paper uses observable transforms of the code residual. The treatment-residual transform $F_n(z)$ carries the zero geometry, the outcome-weighted transform $G_n(z)$ carries the numerator information, and a contour average of G_n/F_n extracts the coefficient over all enclosed zeros at once.

Definition 13. [def:contour-functional] (Contour functional $\Theta_{C_j}(F_n, G_n)$) For $Z_n = T - \bar{g}_n(X)$, define

$$F_n(z) = E_P[e^{zZ_n}], \quad G_n(z) = E_P[Ye^{zZ_n}], \quad H_n(z) = E_P[e^{z\{g_0(X) - \bar{g}_n(X)\}}].$$

Let C_j be a positively oriented circle centered at the origin with positive radius, and suppose F_n is analytic on a neighborhood of the closed disk bounded by C_j . Let $N_{C_j}(F_n)$ denote the number of zeros of F_n strictly inside C_j , counted with multiplicity. When F_n is zero-free on C_j and $N_{C_j}(F_n) \geq 1$, define

$$\Theta_{C_j}(F_n, G_n) = \{N_{C_j}(F_n)2\pi i\}^{-1} \oint_{C_j} \frac{G_n(z)}{F_n(z)} dz.$$

Here C_j is the positively oriented circle used for the contour integral, $N_{C_j}(F_n)$ is the enclosed zero count, and $\Theta_{C_j}(F_n, G_n)$ is the normalized contour functional. The normalization by the zero count makes the functional an average over the enclosed zero structure.

The next identity links this observable contour world to the latent treatment innovation. It factors the residual transform through the innovation transform and isolates the covariate-only contamination term that enters the outcome-weighted transform.

Lemma 1. [lem:observable-factorization] (Observable transform factorization) Let $n \geq 1$, and let $P \in \mathcal{P}_{\text{NG},n}$ satisfy [Definition 3](#). Put

$$Z_n = T - \bar{g}_n(X), \quad D_n(X) = g_0(X) - \bar{g}_n(X),$$

and let $b_n(X) = q_0(X) - \theta_0 D_n(X)$, as in [Definition 7](#). Define, for $z \in \mathbb{C}$,

$$F_n(z) = E_P[e^{zZ_n}], \quad M(z) = E_P[e^{z\eta}], \quad H_n(z) = E_P[e^{zD_n(X)}],$$

and

$$G_n(z) = E_P[Ye^{zZ_n}], \quad B_n(z) = E_P[b_n(X)e^{zD_n(X)}].$$

Then, for every $z \in \mathbb{C}$,

$$F_n(z) = M(z)H_n(z)$$

and

$$G_n(z) = \theta_0 F'_n(z) + M(z)B_n(z).$$

The proof is deferred to [Section E](#).

[Lemma 1](#) is the bridge from residual contamination to contour identification. The factorization $F_n = MH_n$ states that the observable residual transform inherits zeros from the treatment-innovation transform whenever the nuisance transform stays away from zero. The second display decomposes G_n/F_n into the derivative term carrying θ_0 and a nuisance ratio; integrated around a valid contour, the latter contribution is analytic inside the contour and the former contributes through the enclosed zero count.

The population identification statement packages these ingredients into the contour ratio used later by the statistical construction. The contour must be valid for the observable residual transform and for the nuisance transform on the same disk.

Theorem 2. [thm:exact-contour-identification] (Exact contour identification) *Fix $n \geq 1$, a law $P \in \mathcal{P}_{\text{NG},n}$ as in Definition 3, and a certified contour-bank input $\mathbf{p}_\star$ as in Definition 17. For any bank index $j \in \{0, \dots, J_{\text{base}}\}$, let C_j be the positively oriented circle with radius ρ_j , and let*

$$F_n(z) = E_P[e^{zZ_n}], \quad G_n(z) = E_P[Ye^{zZ_n}], \quad H_n(z) = E_P[e^{z\{g_0(X) - \bar{g}_n(X)\}}].$$

Assume:

- **(Residual zero-freeness.)** $F_n(z) \neq 0$ for every $z \in C_j$.
- **(Nuisance zero-freeness.)** $H_n(z) \neq 0$ for every z on or inside C_j .
- **(Positive contour count.)** The zero count $N_{C_j}(F_n)$, counted with multiplicity inside C_j , satisfies $N_{C_j}(F_n) \geq 1$.

Then the contour functional of Definition 13 identifies the partially linear coefficient and is given by the normalized contour integral:

$$(\theta_0(P) : \mathbb{C}) = \Theta_{C_j}(F_n, G_n) = \{N_{C_j}(F_n) 2\pi i\}^{-1} \oint_{C_j} \frac{G_n(z)}{F_n(z)} dz.$$

Theorem 2 establishes the population contour identity on any bank circle satisfying the stated zero-free and positive-count conditions. Residual zero-freeness makes G_n/F_n well defined on the integration path; nuisance zero-freeness keeps the zeros enclosed by the residual transform aligned with the innovation-zero geometry supplied by Lemma 1; and the positive count gives a nontrivial normalization. Under those conditions, the contour average of the observable ratio is exactly the partially linear coefficient.

The section therefore reduces identification with code residuals to a geometric task: find a circle on which the observable residual transform is separated from zero, whose interior contains at least one residual-transform zero, and on whose closed disk the nuisance transform is separated from zero. The next section supplies fixed-separation conditions and a finite contour bank under which such circles are available with enough empirical stability to yield the stated fixed-code risk bounds.

5 Main fixed-separation result

The contour identity in Theorem 2 turns identification into a finite geometric problem: find a circle on which the observable residual transform is separated from zero, whose interior contains at least one transform zero, and whose associated empirical transforms are stable enough for estimation. This section gives the fixed- δ statistical version of that construction. The key inputs are an explicit transform envelope, a numerical zero-localization radius, and an outcome-weighted transform envelope on the contour domain. Together they support a finite contour bank and the total Borel contour statistic used in the main minimax statement.

The first analytic bound converts the exact Luxemburg convention from Definition 1 into a moment-generating-function envelope. This keeps the constants in the zero-localization step tied to the scale convention used in the law classes.

Lemma 2. [lem:luxemburg-mgf-envelope] (Luxemburg MGF envelope) *Let P be a probability law and let $\psi > 0$. Let W be a real-valued random variable under P . Suppose that:*

- **(Regularity.)** W is measurable and $W \in L^1(P)$.
- **(Centering.)** $E_P W = 0$.
- **(Luxemburg envelope.)** Under the Luxemburg sub-Gaussian convention of [Definition 1](#), $\exp(W^2/\psi^2) \in L^1(P)$ and

$$E_P \exp(W^2/\psi^2) \leq 2.$$

Then, for every $t \in \mathbb{R}$,

$$E_P e^{tW} \leq \exp(4\psi^2 t^2).$$

The proof is deferred to [Section E](#).

[Lemma 2](#) supplies the real-axis exponential control used to localize zeros of the treatment-innovation transform. The exact constant is part of the statement, so later radius formulas have a numerical envelope constant.

Cumulant separation then forces the transform geometry to contain a zero inside an explicit disk. Write A_k for the numerical factor in the following statement and R_0 for the zero-localization radius.

Lemma 3. [\[lem:zero-localization\]](#) Fix a primitive parameter tuple p and a law P for $O = (X, T, Y)$. Assume:

- $k = r + 1$ with $r \geq 2$;
- $\eta = T - g_0(X)$ is centered;
- $\|\eta\|_{\psi_2} \leq \psi_\eta$ under the convention in [Assumption 8](#); and
- $|\kappa_k(\eta)| \geq \delta$ as in [Assumption 10](#).

Define

$$A_k = (2^{k+4}k!)^{1/(k-2)}, \quad R_0 = A_k(\psi_\eta^2/\delta)^{1/(k-2)}, \quad M(z) = E_P[e^{z\eta}].$$

Then there exists $z_0 \in \mathbb{C}$ such that $|z_0| \leq R_0$ and $M(z_0) = 0$.

The proof is deferred to [Section E](#).

[Lemma 3](#) is the fixed-separation source of the contour bank. The cumulant lower bound δ and the treatment-tail scale ψ_η determine a disk that contains a transform zero, while the analytic technology is the same general moment-transform geometry used in zero and random-polynomial arguments [[Michelen and Sahasrabudhe, 2019](#), [Eremenko and Fryntov, 2021](#), [Dinh et al., 2021](#)]. The outer search radius R_1 used below enlarges this localization region for contour selection and empirical control.

The remaining population input bounds the outcome-weighted transform on that outer region. It gives a uniform envelope for the numerator side of the contour ratio, with constants depending only on the parameter record and the outer search radius.

Lemma 4. [\[lem:population-numerator-envelope\]](#) (Population numerator envelope) Let p be a parameter record as in [Definition 1](#), with sample size n , primitive constants $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$, and outer search radius R_1 as in [Definition 17](#). Define

$$C_G(p) = [64\{C_q^4 + 4C_\theta^4\psi_\eta^4 + 4\psi_\xi^4\} + 2\exp\{16R_1C_g + 16R_1^2\psi_\eta^2\}]^{1/2}.$$

Let m be a model with observed-data law P as in [Definition 2](#). Suppose that P belongs to the non-Gaussian class $\mathcal{P}_{\text{NG},n}$ of [Definition 3](#), where n is the sample size in p . Then, for every $z \in \mathbb{C}$ with $|z| \leq R_1$, the observable outcome-weighted transform $G_n(z) = E_P[Y e^{zZ_n}]$ from [Definition 13](#) satisfies

$$|G_n(z)| \leq C_G(p).$$

The proof is deferred to [Section E](#).

The envelope in [Lemma 4](#) is used on the same contour domain as the denominator transform. It combines the bounded regression ranges with the sub-Gaussian treatment and outcome-noise scales, giving the empirical-transform analysis a deterministic population scale in the sense of standard empirical-process risk arguments [[Bickel et al., 1993](#), [van der Vaart, 1998](#), [van der Vaart and Wellner, 1996](#)].

We can now state the fixed-separation estimator result. The statistic $\hat{\theta}_{\text{spec},n}$ is the total Borel contour statistic defined in [Definition 18](#); the contour bank is fixed by the primitive constants before the law and sample vary. The risk statements apply to the deterministic clipped code pair in force at sample size n . This includes code pairs obtained by conditioning on any external training stage, after which the product-law sample used in the theorem is evaluated with those clipped codes held fixed. The represented-execution clause is a conditional reproducibility statement for that same ordinary Borel statistic.

Theorem 3. `[thm:adaptive-rootn-minimax]` (Fixed-code root- n minimaxity under fixed cumulant separation) *Fix $r \geq 2$ and positive constants $\delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$. Then there are constants $c > 0$ and $C > 0$, depending only on these fixed primitive constants, such that the following statements hold.*

For every measurable covariate space, let p_0 be a parameter record whose primitive entries $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$ equal the fixed constants above, and fix one experiment-wide record consisting of the certified contour-bank input and the represented range input for p_0 . For every parameter record p with the same primitive entries as p_0 , and for every base model satisfying the usual measurability, conditional-mean, and integrability requirements, write n for the sample size of p . Let $\hat{\theta}_{\text{spec},n}$ be the total Borel statistic of [Definition 18](#), constructed from the transported fixed records and the base treatment-code sequence. Define the fixed-code non-Gaussian class

$$\mathcal{P}_{\text{NG},n}(p; \text{base}) = \{P : P \in \mathcal{P}_{\text{NG},n} \text{ of } \text{Definition 3}, \bar{g}_n(P) = \bar{g}_n(\text{base}), \bar{q}_n(P) = \bar{q}_n(\text{base})\}.$$

Define the fixed-code JMS ACE class

$$\mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p; \text{base}) = \{P : P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \text{ of } \text{Definition 5}, \bar{g}_n(P) = \bar{g}_n(\text{base}), \bar{q}_n(P) = \bar{q}_n(\text{base})\}.$$

- **(Measurable statistic.)** *The statistic $\hat{\theta}_{\text{spec},n}$ is measurable.*
- **(Represented execution.)** *For every compiled bounded spectral adapter satisfying the full canonical build-and-compilation specification for p , the transported fixed records, and the base treatment-code sequence, the represented-data execution realizes the same $\hat{\theta}_{\text{spec},n}$ as in [Definition 18](#).*
- **(Class relations.)** *Every law in $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ belongs to $\mathcal{P}_{\text{NG},n}$. If $s = r$, then $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ coincides with the $L^s(P_X)$ -restricted subclass $\mathcal{P}_{\text{NG},n}^{\text{ACE}}$ of [Definition 11](#).*

- **(Non-Gaussian fixed-code risk.)** If the sample law is the i.i.d. product law of [Assumption 1](#), $\mathcal{P}_{\text{NG},n}(p; \text{base})$ is nonempty, $n \geq 2$, and

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

where R_1 is the search radius from the contour-bank construction in [Definition 18](#), then

$$\frac{c}{n} \leq \mathfrak{R}_n^{\text{NG}} \leq \sup_{P \in \mathcal{P}_{\text{NG},n}(p; \text{base})} E_P \left[(\hat{\theta}_{\text{spec},n} - \theta_0(P))^2 \right] \leq \frac{C}{n},$$

and

$$\sup_{P \in \mathcal{P}_{\text{NG},n}(p; \text{base})} Q_{P,1-\gamma} \left(|\hat{\theta}_{\text{spec},n} - \theta_0(P)| \right) \leq \sqrt{\frac{C}{\gamma n}},$$

with generalized lower quantiles as in [Definition 6](#).

- **(ACE fixed-code risk.)** If $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p; \text{base})$ is nonempty, $n \geq 2$, and

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

then

$$\frac{c}{n} \leq \inf_{\tilde{\theta}} \sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p; \text{base})} E_P \left[(\tilde{\theta} - \theta_0(P))^2 \right] \leq \sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p; \text{base})} E_P \left[(\hat{\theta}_{\text{spec},n} - \theta_0(P))^2 \right] \leq \frac{C}{n},$$

where the infimum is over measurable real-valued estimators of $O_1, \dots, O_n$. Moreover,

$$\sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p; \text{base})} Q_{P,1-\gamma} \left(|\hat{\theta}_{\text{spec},n} - \theta_0(P)| \right) \leq \sqrt{\frac{C}{\gamma n}}.$$

[Theorem 3](#) is the main fixed-separation risk statement. Under fixed primitive constants and the displayed $L^1(P_X)$ treatment-code radius condition, the contour statistic is measurable and attains the matched n^{-1} minimax mean-squared-error order over the fixed-code non-Gaussian class. For $\gamma \in (1/2, 1)$, the same statistic controls the generalized lower $(1 - \gamma)$ -quantile of absolute error by order $(\gamma n)^{-1/2}$, with the quantile defined in [Definition 6](#); the ACE fixed-code clause states the parallel $\Theta(n^{-1})$ MSE result on the common JMS ACE class. The executable correspondence is conditional on the full canonical build-and-compilation specification for the parameter record, transported fixed records, and base treatment-code sequence, while the risk inequalities are ordinary sample-law conclusions for the fixed clipped codes.

The intuition follows the population geometry from the previous section. Cumulant separation supplies at least one transform zero in the radius R_0 disk, and the $L^1(P_X)$ control on $\bar{g}_n - g_0$ keeps the code-residual transform within the stability region used by the contour bank. Empirical transform concentration on the fixed bank, combined with the selected-contour perturbation bound, yields the C/n mean-squared-error upper bound under fixed primitive constants [[Bickel et al., 1993](#), [van der Vaart, 1998](#), [Chernozhukov et al., 2018](#), [Newey and Robins, 2018](#)].

The final result packages two sequence-level consequences. Along non-Gaussian experiments with the same fixed primitive constants, the finite bank can be transported across the sequence. On the bounded-outcome Gaussian comparison class, the target itself is identified by the class restrictions as zero, which gives the corresponding minimax risks directly.

Theorem 4. [thm:common-experiment-dichotomy] (Common Experiment Dichotomy) Fix $r \geq 2$ and positive constants $\delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$. For every measurable covariate space, the following hold.

- **(Non-Gaussian sequence.)** Let p_0 have these fixed values of $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$, and fix the primitive records $\mathbf{p}_\star$ and $\mathbf{c}_{\theta,\star}$ of [Definitions 17](#) and [18](#). Let (p_j) be any parameter sequence with the same fixed values of $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$ as p_0 , and with common supplied radii $\varepsilon_{1,n}$ and $\varepsilon_{2,n}$. Let $(\hat{g}_n, \hat{q}_n)$ be any measurable deterministic code pair, and write $(\bar{g}_n, \bar{q}_n)$ for the clipped pair. Suppose that $n_j = p_j \cdot n \rightarrow \infty$ and, for all sufficiently large j ,

$$\mathcal{P}_{\text{NG},n_j}(p_j; \bar{g}_{n_j}, \bar{q}_{n_j}) \neq \emptyset, \quad n_j \geq 2, \quad \varepsilon_{1,n_j} \leq \left(4R_1(p_j) \exp\{2C_g R_1(p_j)\}\right)^{-1}.$$

Then there are constants $c, C > 0$ such that, with B the translated-dyadic contour bank generated from p_0 and $\mathbf{p}_\star$, for all sufficiently large j the transported primitive bank for p_j generates the same bank B , the statistic $\hat{\theta}_{\text{spec},n_j}$ of [Definition 18](#) is measurable, every compiled bounded spectral adapter satisfying the full canonical build and compilation specification is represented by the corresponding execution on the same transported primitive records, and

$$\frac{c}{n_j} \leq \mathfrak{R}_{n_j}^{\text{NG}}(p_j; \hat{g}, \hat{q}) \leq \frac{C}{n_j}.$$

Moreover, the same statistic witnesses the upper bound:

$$\sup_{P \in \mathcal{P}_{\text{NG},n_j}(p_j; \bar{g}_{n_j}, \bar{q}_{n_j})} \mathbb{E}_P \left[(\hat{\theta}_{\text{spec},n_j} - \theta_0(P))^2 \right] \leq \frac{C}{n_j}.$$

- **(Gaussian class.)** For every parameter record p , writing n for its sample size, every $P \in \mathcal{P}_{\text{G},n}^{\text{JMS}}$ in [Definition 4](#) satisfies

$$\theta_0(P) = 0.$$

For every measurable deterministic code pair $(\hat{g}_n, \hat{q}_n)$, whenever the clipped-code intersection of $\mathcal{P}_{\text{G},n}^{\text{JMS}}$ is nonempty,

$$\mathfrak{R}_n^{\text{G}}(p; \hat{g}, \hat{q}) = 0, \quad \mathfrak{M}_{n,1-\gamma}^{\text{G}}(p; \hat{g}, \hat{q}) = 0.$$

[Theorem 4](#) records the common-experiment form of the fixed-separation result. The non-Gaussian sequence clause states that one translated-dyadic bank serves all sufficiently large experiments sharing the primitive constants, and that the resulting displayed-parameter minimax risk is of order n_j^{-1} . The Gaussian clause gives a separate diagnostic for the bounded-outcome Gaussian comparison class: its member restrictions force $\theta_0(P) = 0$, so both the mean-squared and generalized-quantile minimax risks equal zero on every nonempty fixed-code intersection.

The fixed- δ message is therefore geometric and statistical at once. Cumulant separation creates a zero inside a known disk; treatment-code stability preserves the contour geometry for code residuals; empirical-process control of the transform ratio yields parametric MSE over the fixed-code non-Gaussian class; and the same construction aligns with the ACE comparison class under the class relation stated in [Theorem 3](#). The finite-record clauses give the reproducibility interface for the same statistic, with the construction details deferred to the certified-construction appendix.

6 Comparison with ACE and Gaussian benchmarks

The fixed-separation theorem in [Theorem 3](#) gives a contour guarantee on the broad spectral class and on the fixed-code ACE class. This section records the comparison with the published order- r ACE guarantee of [Jin et al. \[2025\]](#) on the same clipped-code experiment. The comparison has two parts: first, the class relations align the cumulant-separated spectral and ACE restrictions; second, the two generalized-quantile upper bounds can be read on the common class under the same JMS eligibility condition.

The following proposition states the alignment and the resulting upper-guarantee comparison. The constant C_{spec} is fixed by the primitive constants in the contour construction, while C_γ and the estimator $\hat{\theta}_{\text{ACE},r,n}$ come from the published ACE handle. The displayed quantity $B_{\text{ACE},n,\gamma}$ is the order- r generalized-quantile upper guarantee from [Jin et al. \[2025\]](#). The sequence clause compares the displayed upper bounds on the common clipped-code class.

Proposition 1. `[prop:jms-ace-alignment]` (ACE and spectral alignment)¹ *Fix an ACE order $r \geq 2$ and positive constants $\delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$. There is a constant $C_{\text{spec}} > 0$, depending only on these fixed constants, such that the following statements hold.*

Let $\gamma \in (1/2, 1)$, let $C_\gamma > 0$ be the constant in the published order- r ACE generalized-quantile guarantee, and write $\hat{\theta}_{\text{ACE},r,n}$ for the estimator supplied by that published ACE handle. Let p_0 be a parameter record whose $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$, and γ coordinates equal the displayed constants, and fix the certified primitive bank and range records associated with p_0 . For any parameter record p sharing the same fixed experiment constants as p_0 and the same probability level γ , and for any supplied code sequences whose clipped versions are $\bar{g}_n$ and $\bar{q}_n$, the following hold, under the usual measurability and integrability conditions.

- **Class relations.** *At the sample size n of p ,*

$$\mathcal{P}_{\text{ACE},n}^{\text{JMS}} \subseteq \mathcal{P}_{\text{NG},n}, \quad \mathcal{P}_{\text{NG},n}^{\text{ACE}} \subseteq \mathcal{P}_{\text{ACE},n}^{\text{JMS}}.$$

If the comparison exponent of p satisfies $s = r$, then the common clipped-code convention gives

$$\mathcal{P}_{\text{ACE},n}^{\text{JMS}} = \mathcal{P}_{\text{NG},n}^{\text{ACE}}.$$

Here the classes and the comparison subclass are those of [Definitions 3, 5 and 11](#).

- **ACE upper guarantee.** *If a law $P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ uses the clipped supplied treatment and outcome codes $\bar{g}_n, \bar{q}_n$, if the JMS eligibility condition $\mathbf{E}_n^{\text{JMS}}$ of [Definitions 8 and 10](#) holds, and if $\varepsilon_{1,n} > 0$ and $\varepsilon_{2,n} \geq 0$, then*

$$Q_{P,1-\gamma}(|\hat{\theta}_{\text{ACE},r,n} - \theta_0(P)|) \leq B_{\text{ACE},n,\gamma},$$

where $Q_{P,1-\gamma}$ is the generalized quantile in [Definition 6](#) and

$$\begin{aligned} B_{\text{ACE},n,\gamma} = & C_\gamma r! 16^r \delta^{-1} \left[\varepsilon_{1,n}^r \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1} \right. \\ & \left. + 64(C_g + \psi_\eta)^r \{r^2(C_g + \psi_\eta) + \psi_\xi + C_\theta \psi_\eta\} (\gamma n)^{-1/2} \right]. \end{aligned}$$

- **Spectral upper guarantee.** *If $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ contains at least one law using the clipped supplied codes, $n \geq 2$, $\mathbf{E}_n^{\text{JMS}}$ holds, $\varepsilon_{1,n} > 0$, and*

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

where R_1 is the outer contour radius of [Definition 17](#) determined by the parameter record in force, then the certified contour statistic $\hat{\theta}_{\text{spec},n}$ of [Definition 18](#), built from the fixed primitive records, satisfies

$$\sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}} Q_{P,1-\gamma}(|\hat{\theta}_{\text{spec},n} - \theta_0(P)|) \leq \sqrt{\frac{C_{\text{spec}}}{\gamma n}},$$

where the supremum ranges over laws in the published ACE class whose clipped supplied treatment and outcome codes are $\bar{g}_n$ and $\bar{q}_n$.

- **Upper-guarantee separation along sequences.** For any sequence (p_n) of parameter records whose sample size at index n is n , with the same fixed experiment constants as p_0 , the same radius sequences $\varepsilon_{1,\cdot}, \varepsilon_{2,\cdot}$, and $p_n \cdot \gamma = \gamma$, suppose that eventually $n \geq 2$, $\mathbf{E}_n^{\text{JMS}}$ holds, $\varepsilon_{1,n} > 0$, $\varepsilon_{2,n} \geq 0$,

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

and the published ACE class contains a law using the clipped supplied codes. If

$$\frac{\varepsilon_{1,n}^r \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1}}{n^{-1/2}} \longrightarrow \infty,$$

then

$$\frac{\sqrt{C_{\text{spec}}/(\gamma n)}}{B_{\text{ACE},n,\gamma}} \longrightarrow 0.$$

[Proposition 1](#) places the spectral and ACE guarantees on a shared statistical domain. The class relations use the fact that the ACE $L^r(P_X)$ code control implies the $L^1(P_X)$ treatment-code control needed by the spectral class, while the $L^s(P_X)$ -restricted spectral subclass matches the published ACE code restrictions when $s = r$. Under JMS eligibility, the ACE bound retains its finite-order nuisance terms, and the contour bound is governed by the fixed-separation sampling term $(\gamma n)^{-1/2}$. Along the displayed nuisance-dominance sequences, the ratio of the contour upper guarantee to the imported finite-order ACE upper guarantee tends to zero. The conclusion is an upper-bound comparison on the common clipped-code class; the ACE-related lower-bound statement used in the paper is the class-minimax lower bound over all measurable estimators on the fixed-code class.

For later reference, the class relation used inside [Proposition 1](#) is also recorded separately. This isolates the comparison-class bookkeeping from the estimator-specific upper bounds.

Lemma 5. `[lem:jms-ace-class-relations]` (ACE class relations) *Fix a parameter tuple p with ACE order r , comparison exponent s , cumulant order $k = r + 1$, $2 \leq r$, $r \leq s$, probability level $\gamma \in (1/2, 1)$, strictly positive constants $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma$, and nonnegative nonincreasing code-radius sequences $(\varepsilon_{1,n})$ and $(\varepsilon_{2,n})$ on positive indices. For every class index n and every law P with its supplied regression codes satisfying the usual measurability and integrability conditions,*

$$P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \implies P \in \mathcal{P}_{\text{NG},n},$$

¹**Formalization scope.** This result uses the published conclusion [[Jin et al., 2025](#)] (Theorem 5.4, equations (21)–(22)). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

where the two classes are those of [Definition 5](#) and [Definition 3](#). Moreover,

$$P \in \mathcal{P}_{\text{NG},n}^{\text{ACE}} \implies P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}},$$

where $\mathcal{P}_{\text{NG},n}^{\text{ACE}}$ is the subclass in [Definition 11](#). If $s = r$, then for every such P ,

$$P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \iff P \in \mathcal{P}_{\text{NG},n}^{\text{ACE}}.$$

The proof is deferred to [Section E](#).

[Lemma 5](#) is the set-inclusion component of the comparison. It identifies which restrictions are shared by the broad cumulant-separated class and the published ACE class, and it pins down the exact equality case under the common clipped-code convention. This makes the comparison in [Proposition 1](#) an upper-guarantee comparison on the same law class and the same clipped-code experiment.

The second benchmark is the bounded-outcome Gaussian comparison class of [Definition 4](#). [Proposition 2](#) gives a bounded-outcome Gaussian diagnostic for the intersection of the exact partially linear conditional mean, bounded observed outcome, and nondegenerate Gaussian treatment innovation. In every nonempty fixed-code intersection of that class, the decision-theoretic mean-squared and generalized-quantile risks are zero because the member restrictions determine the target value itself.

Proposition 2. [[prop:bounded-outcome-gaussian-degeneracy](#)] (Gaussian class degeneracy)

Fix a parameter tuple with:

- **(Sample size.)** A positive sample size $n \geq 1$.
- **(Orders and exponent.)** Integers $r, k \in \mathbb{N}$ satisfying $k = r + 1$ and $r \geq 2$, and an exponent $s \geq r$.
- **(Probability level.)** A level $\gamma \in (1/2, 1)$.
- **(Positive constants.)** Constants $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma > 0$.
- **(Code-radius sequences.)** Sequences $\varepsilon_1, \varepsilon_2 : \mathbb{N} \rightarrow \mathbb{R}$ whose values are nonnegative on positive indices and nonincreasing on positive indices.

On any measurable covariate space, the following hold.

- **(Degenerate target.)** Every model m in the bounded-outcome Gaussian JMS comparison class $\mathcal{P}_{\text{G},n}^{\text{JMS}}$ satisfying [Definition 4](#) has

$$\theta_0(P) = 0.$$

- **(Zero minimax risks.)** For any supplied deterministic code sequences $\hat{g} = (\hat{g}_j)_{j \in \mathbb{N}}$ and $\hat{q} = (\hat{q}_j)_{j \in \mathbb{N}}$, suppose there is a model $m \in \mathcal{P}_{\text{G},n}^{\text{JMS}}$ satisfying [Definition 4](#) whose clipped supplied treatment and outcome codes agree pointwise at index n with

$$x \mapsto \min\{\max\{\hat{g}_n(x), -C_g\}, C_g\} \quad \text{and} \quad x \mapsto \min\{\max\{\hat{q}_n(x), -C_q\}, C_q\}.$$

Then

$$\mathfrak{R}_n^{\text{G}}(p; \hat{g}, \hat{q}) = 0 \quad \text{and} \quad \mathfrak{M}_{n,1-\gamma}^{\text{G}}(p; \hat{g}, \hat{q}) = 0.$$

Proposition 2 gives a diagnostic for the source-stated Gaussian comparison class. A bounded observed outcome has bounded conditional support, while the partially linear term $\theta_0\eta$ is driven by nondegenerate Gaussian treatment noise under **Definition 4**; the proposition records the resulting target value and the induced zero minimax risks on each nonempty fixed-code intersection. Thus the Gaussian benchmark characterizes that simultaneous set of restrictions, whereas the fixed-separation comparison with ACE in **Proposition 1** is the active finite-sample upper-guarantee comparison on the cumulant-separated class.

Together, **Propositions 1** and **2** and **Lemma 5** separate the two comparison roles. The ACE alignment places the published finite-order guarantee of [Jin et al. \[2025\]](#) and the contour guarantee on a common cumulant-separated domain, with the sequence clause identifying the regime in which the ratio of displayed upper guarantees tends to zero. The Gaussian proposition explains the bounded-outcome Gaussian class by its implied target and the corresponding decision-theoretic risks, matching the diagnostic role already summarized in **Theorem 4**.

7 Explicit mixture reductions and local benchmarks

The preceding comparison keeps the cumulant separation level fixed. This section records two constructive weak-non-Gaussian benchmarks that make the transform-zero mechanism explicit in one dimension. The first benchmark treats a symmetric two-component Gaussian mixture, where the relevant zero is known and the contour search collapses to a sine moment. The second follows a Gaussian–Rademacher path indexed by a mixture amplitude, so the denominator scale and the fourth cumulant can be read from the same parameter. These examples connect the contour construction to non-Gaussian identification ideas in independent-component and structural-shock settings [[Comon, 1994](#), [Hyvärinen and Oja, 1997](#), [Hyvärinen et al., 2001](#), [Lanne et al., 2017](#), [Lee and Mesters, 2024](#), [Hoesch et al., 2024](#), [Reizinger et al., 2025](#)].

The first construction is the clipped sine-ratio estimator $\hat{\theta}_{\sin,n}$. It uses the known imaginary zero of the symmetric mixture transform and estimates the corresponding ratio directly from the empirical moments of $Y \sin(\pi Z_n/2)$ and $Z_n \sin(\pi Z_n/2)$.

Definition 14. [def:sine-estimator] (Sine-ratio estimator $\hat{\theta}_{\sin,n}$) Write $\Pi_{[-C_\theta, C_\theta]}(u) = \min\{\max\{u, -C_\theta\}, C_\theta\}$ for the projection of a real number onto $[-C_\theta, C_\theta]$. For $n \geq 1$ observed triples $O_i = (X_i, T_i, Y_i)$, let $\mathbb{P}_n f = n^{-1} \sum_{i=1}^n f(O_i)$ and let $Z_n = T - \bar{g}_n(X)$, where $\bar{g}_n$ is the supplied treatment code clipped to $[-C_g, C_g]$. The clipped sine-ratio estimator is

$$\hat{\theta}_{\sin,n} = \begin{cases} \Pi_{[-C_\theta, C_\theta]} \left\{ \frac{\mathbb{P}_n[Y \sin(\pi Z_n/2)]}{\mathbb{P}_n[Z_n \sin(\pi Z_n/2)]} \right\}, & \mathbb{P}_n[Z_n \sin(\pi Z_n/2)] \geq e^{-\pi^2/8}/4, \\ 0, & \mathbb{P}_n[Z_n \sin(\pi Z_n/2)] < e^{-\pi^2/8}/4. \end{cases}$$

The threshold in **Definition 14** is tied to the population denominator scale that appears in the next proposition. In the symmetric mixture experiment, the treatment innovation has a known transform zero at $i\pi/2$, and $L^1(P_X)$ control of the supplied treatment code keeps the sine denominator bounded away from zero.

Proposition 3. [prop:symmetric-mixture-reduction] (Symmetric mixture reduction) For any real constants $C_\theta, C_g, C_q, \psi_\xi$, there is a constant $C > 0$, depending only on these four constants, such that the following holds. Let p be a parameter record whose constants $C_\theta, C_g, C_q, \psi_\xi$ are the ones just fixed, write n for its sample size, and let P be a law in the corresponding model. Assume the usual measurability and integrability conditions, and suppose that

- **(Sampling.)** $O_1, \dots, O_n$ are drawn according to the i.i.d. product law in [Assumption 1](#).
- **(Symmetric mixture.)** The treatment innovation $\eta = T - g_0(X)$ has law

$$\frac{1}{2}N(-1, 1) + \frac{1}{2}N(1, 1).$$

- **(Independence and ranges.)** The treatment innovation is independent of X as in [Assumption 2](#), the outcome innovation satisfies [Assumption 3](#), and the range and tail conditions in [Assumptions 4 to 6](#) and [9](#) hold.
- **(Treatment code.)** The supplied treatment code obeys

$$\|\bar{g}_n - g_0\|_{L^1(P_X)} \leq \varepsilon_{1,n}, \quad \varepsilon_{1,n} \leq \frac{1}{\pi}.$$

Then the fourth cumulant of η is

$$\kappa_4(\eta) = -2,$$

the treatment-innovation moment-generating function $M(z) = E_P[e^{z\eta}]$ satisfies

$$M(i\pi/2) = 0,$$

and, writing $Z_n = T - \bar{g}_n(X)$ and $D_n = g_0(X) - \bar{g}_n(X)$,

$$E_P \left[Z_n \sin \left(\frac{\pi Z_n}{2} \right) \right] = e^{-\pi^2/8} E_P \left[\cos \left(\frac{\pi D_n}{2} \right) \right],$$

with

$$\frac{e^{-\pi^2/8}}{2} \leq E_P \left[Z_n \sin \left(\frac{\pi Z_n}{2} \right) \right].$$

Moreover, for the sine estimator $\hat{\theta}_{\sin,n}$ in [Definition 14](#),

$$E_P \left[(\hat{\theta}_{\sin,n} - \theta_0(P))^2 \right] \leq \frac{C}{n}.$$

[Proposition 3](#) gives a fully explicit root- n benchmark for a symmetric non-Gaussian treatment innovation. The mixture has fourth cumulant -2 , its transform vanishes at $i\pi/2$, and the sine denominator equals a positive contamination factor times $e^{-\pi^2/8}$. The resulting empirical ratio attains C/n mean-squared error under the stated range, tail, independence, and $L^1(P_X)$ treatment-code conditions.

The next elementary inequality is the algebraic device behind the sine-risk calculations. It separates numerator fluctuation from denominator fluctuation after clipping, using only a lower bound on the population denominator mean.

Lemma 6. `[lem:clipped-ratio-risk-decomposition]` (Clipped ratio risk decomposition) *Let Ω be any observation space. For a sample $O_1, \dots, O_n \in \Omega$, write the empirical average as $\mathbb{P}_n f = n^{-1} \sum_{i=1}^n f(O_i)$, as in [Definition 1](#). Let $W, R: \Omega \rightarrow \mathbb{R}$, and let θ, C_θ, A , and μ_W be real numbers. Suppose that*

- **(Sample size.)** $n \geq 1$.
- **(Clipping range.)** The target value θ lies inside the clipping radius C_θ : $|\theta| \leq C_\theta$.

- (**Denominator scale.**) $A > 0$.
- (**Population denominator mean.**) $\mu_W \geq A/2$.

Define the clipped ratio statistic at threshold $A/4$ by

$$\hat{\theta} = \begin{cases} \min \left\{ \max \left\{ \frac{\theta \mathbb{P}_n W + \mathbb{P}_n R}{\mathbb{P}_n W}, -C_\theta \right\}, C_\theta \right\}, & A/4 \leq \mathbb{P}_n W, \\ 0, & A/4 > \mathbb{P}_n W. \end{cases}$$

Then, for every sample $O_1, \dots, O_n \in \Omega$,

$$(\hat{\theta} - \theta)^2 \leq \frac{16}{A^2} \left\{ (\mathbb{P}_n R)^2 + C_\theta^2 (\mathbb{P}_n (W - \mu_W))^2 \right\}.$$

The proof is deferred to [Section E](#).

[Lemma 6](#) is deterministic. In applications, W is the sine-weighted residualized treatment, R is the centered outcome-noise contribution to the sine numerator, and A is the population denominator scale. Once the sine moment gives $\mu_W \geq A/2$, the ratio error is controlled by two empirical averages.

The Gaussian–Rademacher path provides a local family approaching Gaussian treatment noise through an explicit fourth-cumulant parameter. Let η_a denote the centered Gaussian–Rademacher treatment innovation in the next statement. Its first positive transform zero, denominator scale, and cumulant magnitude are all explicit functions of a .

Lemma 7. [[lem:gaussian-rademacher-l1-benchmark](#)] (Gaussian–Rademacher sine benchmark) *Fix real constants $C_\theta, C_g, C_q, \psi_\xi$. There is a constant $C > 0$, depending only on these four constants, such that the following statement holds. Let p be an admissible parameter record whose constants $C_\theta, C_g, C_q, \psi_\xi$ are the ones just fixed and whose cumulant order is $k = 4$, let $a \in (0, 1]$, and let P be a law in the model m . Assume:*

- (**Gaussian–Rademacher path.**) *The treatment innovation $\eta = T - g_0(X)$ has the law of*

$$\eta_a = \sqrt{1 - a^2} G_\circ + aS,$$

where $G_\circ \sim N(0, 1)$, S is symmetric Rademacher, and $G_\circ$ and S are independent.

- (**Sampling and model restrictions.**) *The law satisfies [Assumptions 1 to 6](#) and [9](#).*
- (**Current treatment-code radius.**) *For the clipped supplied treatment code $\bar{g}_n$,*

$$\int |\bar{g}_n(x) - g_0(x)| dP_X(x) \leq \varepsilon_{1,n},$$

under the usual integrability condition for this $L^1(P_X)$ quantity.

Define

$$t_a = \frac{\pi}{2a}, \quad A_a = a \exp \left\{ -\frac{(1 - a^2)\pi^2}{8a^2} \right\}, \quad \Delta_a = 2a^4.$$

Then the treatment-innovation moment-generating function satisfies

$$M_a(z) = \exp \left(\frac{1 - a^2}{2} z^2 \right) \cosh(az),$$

the fourth cumulant satisfies $\kappa_4(\eta_a) = -2a^4$, and $t_a > 0$ is the first positive zero of the characteristic transform: $M_a(it_a) = 0$ and $M_a(iu) \neq 0$ for every $u \in (0, t_a)$. Moreover, for every $d \in \mathbb{R}$,

$$E_P[\sin\{t_a(\eta + d)\}] = 0.$$

Let $Z_n = T - \bar{g}_n(X)$ and $D_n = g_0(X) - \bar{g}_n(X)$. If $t_a \varepsilon_{1,n} \leq 1/2$, then

$$E_P[Z_n \sin(t_a Z_n)] = A_a E_P[\cos(t_a D_n)]$$

and

$$E_P[Z_n \sin(t_a Z_n)] \geq A_a/2.$$

With $\mathbb{P}_n$ denoting empirical averaging over $O_1, \dots, O_n$, define

$$\hat{\theta}_{a,n} = \begin{cases} \Pi_{[-C_\theta, C_\theta]} \frac{\mathbb{P}_n[Y \sin(t_a Z_n)]}{\mathbb{P}_n[Z_n \sin(t_a Z_n)]}, & \mathbb{P}_n[Z_n \sin(t_a Z_n)] \geq A_a/4, \\ 0, & \text{otherwise,} \end{cases}$$

where $\Pi_{[-C_\theta, C_\theta]}$ clips to $[-C_\theta, C_\theta]$. Its mean squared error obeys

$$E_P[(\hat{\theta}_{a,n} - \theta_0)^2] \leq C \min \left\{ 1, \frac{1}{n} a^{-2} \exp \left(\frac{(1 - a^2)\pi^2}{4a^2} \right) \right\}.$$

The same smallness condition also gives

$$\varepsilon_{1,n} \leq \frac{\Delta_a^{1/4}}{2^{1/4}\pi}$$

and the equivalent cumulant-parameterized bound

$$E_P[(\hat{\theta}_{a,n} - \theta_0)^2] \leq C \min \left\{ 1, \frac{1}{n} \Delta_a^{-1/2} \exp \left(\frac{\pi^2 \sqrt{2}}{4} \Delta_a^{-1/2} \right) \right\}.$$

The proof is deferred to [Section E](#).

[Lemma 7](#) shows how the explicit sine reduction behaves as the treatment-noise law moves toward Gaussianity. The first zero is $t_a = \pi/(2a)$, while the denominator scale A_a decays exponentially as a decreases. Expressing the same bound through $\Delta_a = 2a^4$ makes the dependence on fourth-cumulant magnitude explicit. The condition $t_a \varepsilon_{1,n} \leq 1/2$ is exactly the treatment-code accuracy gate used by this pathwise calculation.

The section closes with a local benchmark statement. It combines the Gaussian–Rademacher path just described with a localized version of the published ACE oracle envelope from [Jin et al. \[2025\]](#). The sequence $(\delta_n)_{n \in \mathbb{N}}$ supplies the shrinking cumulant threshold, and the ACE clause evaluates the published bound after replacing the fixed separation level by the local value $\Delta = \delta_n$.

Theorem 5. `[thm:local-to-gaussian-partial-benchmarks]` (Local Gaussian Benchmarks)
Let $(\delta_n)_{n \in \mathbb{N}}$ be a real sequence with $\delta_n > 0$ for every n , nonincreasing in n , and $\delta_n \rightarrow 0$. Fix $C_\theta, C_g, C_q, \psi_\xi \in \mathbb{R}$. The following two assertions hold.

- **(Gaussian–Rademacher path.)** There is a constant $C > 0$, depending only on $C_\theta, C_g, C_q, \psi_\xi$, such that the following holds. Let p be any parameter record whose constants $C_\theta, C_g, C_q, \psi_\xi$ are the ones just fixed and whose orders are $k = 4$ and $r = 3$. For any $a \in (0, 1]$ and any law P in a model m , assume that the treatment noise satisfies

$$\eta \sim \sqrt{1 - a^2} G_\circ + aS,$$

where $G_\circ \sim N(0, 1)$, S is symmetric Rademacher, and $G_\circ \perp S$. Assume also [Assumptions 1 to 6](#) and [9](#). At the sample size n of p , assume the supplied clipped treatment code $\bar{g}_n$ satisfies

$$x \mapsto |\bar{g}_n(x) - g_0(x)| \in L^1(P_X), \quad \int |\bar{g}_n(x) - g_0(x)| dP_X(x) \leq \varepsilon_{1,n}.$$

Then the conclusion of [Lemma 7](#) — the explicit mean-squared-error bound along the Gaussian–Rademacher path — holds with constant C for p, m, a .

- **(Local ACE oracle envelope.)** For every published ACE handle, every $\gamma \in \mathbb{R}$, and every $C_\gamma \in \mathbb{R}$, if the handle satisfies the conclusion of Theorem 5.4 of [Jin et al. \[2025\]](#) at (γ, C_γ) , then $C_\gamma > 0$ and the following local ACE statement holds. For every parameter record p whose probability level is γ , writing n for its sample size, if $\varepsilon_{1,n} > 0$ and $\varepsilon_{2,n} > 0$, then for every deterministic treatment code $\hat{g}$, deterministic outcome code $\hat{q}$, and every model, whenever

$$\bar{g}_n = \Pi_{[-C_g, C_g]} \hat{g}_n, \quad \bar{q}_n = \Pi_{[-C_q, C_q]} \hat{q}_n,$$

and, writing r and k for the orders of p and $\Delta = \delta_n$, the local ACE class conditions

$$1 \leq n, \quad \left| \operatorname{Re} \left(\frac{d^k}{dz^k} \log M(z) \right) \Big|_{z=0} \right| \geq \Delta,$$

hold together with [Assumptions 2 to 6](#) and [9](#), the treatment-noise exponential envelope

$$o \mapsto \exp\{\eta(o)^2/\psi_\eta^2\} \in L^1(P), \quad \int \exp\{\eta(o)^2/\psi_\eta^2\} dP(o) \leq 2,$$

and the nuisance-radius bounds

$$\|\bar{g}_n - g_0\|_{L^r(P_X)} \leq \varepsilon_{1,n}, \quad \|\bar{q}_n - q_0\|_{L^r(P_X)} \leq \varepsilon_{2,n},$$

and whenever the local JMS eligibility quantities

$$a_{1,n} = 2 \log(6(C_g + \psi_\eta) \varepsilon_{1,n}^{-1}), \quad b_{1,n} = \log(\gamma n / 9),$$

$$a_2 = 4(C_g + \psi_\eta), \quad b_{2,n}(\Delta) = \frac{200 \min\{1, C_\theta\} \Delta}{\max\{\varepsilon_{1,n}, \varepsilon_{2,n}, (\gamma n)^{-1/2}(\psi_\xi + C_\theta \psi_\eta)\}}$$

satisfy

$$a_{1,n} \neq 0, \quad 0 < \frac{b_{1,n}}{a_{1,n}}, \quad 0 < a_2 b_{2,n}(\Delta), \quad a_2 \log\{a_2 b_{2,n}(\Delta)\} \neq 0,$$

and

$$r \leq \min \left\{ \frac{b_{1,n}}{a_{1,n}} - a_{1,n}^{-1} \log \left(\frac{b_{1,n}}{a_{1,n}} \right), \frac{b_{2,n}(\Delta)}{a_2 \log\{a_2 b_{2,n}(\Delta)\}} \right\},$$

the published order- r ACE estimator obeys

$$Q_{P,1-\gamma}\left(\left|\hat{\theta}_{\text{ACE},r,n} - \theta_0(P)\right|\right) \leq B_{\text{ACE},n,\gamma}(\delta_n; C_\gamma),$$

where

$$B_{\text{ACE},n,\gamma}(\Delta; C_\gamma) = C_\gamma r! 16^r \Delta^{-1} \left(\varepsilon_{1,n}^r \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1} + 64(C_g + \psi_\eta)^r (r^2(C_g + \psi_\eta) + \psi_\xi + C_\theta \psi_\eta)(\gamma n)^{-1/2} \right).$$

Theorem 5 gives two procedure-specific local upper benchmarks. The Gaussian–Rademacher clause reuses [Lemma 7](#) for the $k = 4, r = 3$ path under the stated $L^1(P_X)$ treatment-code condition. The ACE clause specializes the published order- r generalized-quantile guarantee of [Jin et al. \[2025\]](#) to a local cumulant threshold $\Delta = \delta_n$, with the displayed eligibility quantities and the resulting envelope $B_{\text{ACE},n,\gamma}(\delta_n; C_\gamma)$. The last algebraic clause records how an oracle upper bound that is valid under two sets of hypotheses can be summarized by the smaller of the two displayed upper inputs.

These local statements complement the fixed-separation results by making the dependence on the non-Gaussian signal explicit in tractable submodels. For the symmetric mixture, the sine denominator is bounded by a fixed positive scale and the risk is C/n . Along the Gaussian–Rademacher path, the same sine construction has a denominator scale A_a and a cumulant magnitude Δ_a , yielding the explicit exponential dependence displayed in [Lemma 7](#). The localized ACE envelope gives the corresponding published finite-order benchmark under its stated eligibility and nuisance-radius conditions.

8 Limitations and future work

The fixed-separation results in [Theorems 3](#) and [4](#) use a cumulant-separation constant that is fixed across the experiment. The local benchmarks in [Theorem 5](#) point to a complementary triangular-array program in which the non-Gaussian signal is allowed to shrink with the sample size. This section records that agenda separately from the fixed- δ contour theorem and from the procedure-specific local upper bounds of the preceding section.

A useful starting point for that program is a finite, sample-facing object that stores the contour information available at a given library depth. The following definition names the empirical handle used to organize candidate contours, empirical winding information, contour moments, and angular variability.

Definition 15. `[def:local-gaussian-handle]` (Local Gaussian handle $\mathcal{C}_m$) *Fix a parameter tuple with positive sample size n , fixed ACE order $r \geq 2$, cumulant order $k = r + 1$, $s \geq r$, $\gamma \in (1/2, 1)$, strictly positive $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma$, and nonnegative nonincreasing sequences ε_{1n} and ε_{2n} on positive indices. Work on a measurable covariate space with observations $O = (X, T, Y)$, and fix a probability law P with real functional $\theta_0(P)$, measurable conditional means $g_0(X) = E_P[T \mid X]$ and $q_0(X) = E_P[Y \mid X]$, integrable T and Y , and deterministic measurable supplied regression-code sequences for treatment and outcome.*

For each library depth d , let $\mathcal{C}_d$ be the deterministic finite library of rational circles represented by the positive rational radii $\rho = j+1$, $0 \leq j < d$. For a realized sample indexed by $\{0, \dots, n-1\}$, form the empirical transforms $\hat{F}_{a,n}$ and $\hat{G}_{a,n}$ on the prespecified deterministic fold $a \in \{0, 1\}$, using indices below $n/2$ for $a = 0$ and indices at least $n/2$ for $a = 1$. For a circle $C \in \mathcal{C}_d$, $C =$

$\{\rho e^{it} : 0 \leq t \leq 2\pi\}$, on which the empirical boundary modulus is certified positive, the candidate record consists of quantities definitionally computed from the empirical contour integrands

$$W_{a,C}(t) = \rho e^{it} \frac{\hat{F}'_{a,n}(\rho e^{it})}{\hat{F}_{a,n}(\rho e^{it})}, \quad V_{a,C}(t) = \rho e^{it} \frac{\hat{G}_{a,n}(\rho e^{it})}{\hat{F}_{a,n}(\rho e^{it})}.$$

Specifically, the record contains a rational interval enclosure of the boundary modulus $\inf_{z \in C} |\hat{F}_{a,n}(z)|$, a complex rational interval enclosure for $(2\pi)^{-1} \int_0^{2\pi} W_{a,C}(t) dt$, a complex rational interval enclosure for $(2\pi)^{-1} \int_0^{2\pi} V_{a,C}(t) dt$, and a rational interval enclosure of the angular variance

$$\hat{v}_{a,C} = \frac{1}{2\pi} \int_0^{2\pi} \left| V_{a,C}(t) - \frac{1}{2\pi} \int_0^{2\pi} V_{a,C}(u) du \right|^2 dt.$$

Each enclosure is accompanied by its soundness certificate for the displayed empirical quantity, and the actual empirical transforms on the stated deterministic fold are the transform inputs of the schema. These records are candidate data for one realized sample, one deterministic inference fold, and one rational circle.

[Definition 15](#) provides a concrete data structure for studying adaptive contour selection in local regimes. It keeps the deterministic library, fold-specific empirical transforms, certified boundary separation, contour moments, and angular variance in one record. For shrinking cumulant separation, such records give a way to describe what a selector observes before choosing among contour-based procedures and more classical alternatives.

The open statistical question is naturally triangular. Let (δ_n) denote shrinking positive cumulant-separation thresholds. The fixed-separation theorem controls risk when the primitive separation constant is fixed, while the local benchmarks in [Theorem 5](#) describe particular upper envelopes when the non-Gaussian signal is weak. The remaining problem is to identify the sharp joint rate and the corresponding inference theory across the ordinary DML, finite-order ACE, and contour regimes [[Chernozhukov et al., 2018](#), [Newey and Robins, 2018](#), [Mackey et al., 2018](#), [Jin et al., 2025](#), [Lee and Mesters, 2024](#), [Hoesch et al., 2024](#)].

Remark 1. `[oeq:local-to-gaussian-frontier]` (Local-to-Gaussian frontier) *For triangular treatment-noise laws with $|\kappa_k(\eta_n)| \geq \delta_n$ and $\delta_n \downarrow 0$ under the same supplied code sequences, a natural next question is whether there exists a data-driven selector among ordinary DML, finite-order ACE, and global-contour procedures that attains the sharp minimax mean-squared-error rate as a function of $(n, \varepsilon_{1,n}, \varepsilon_{2,n}, \delta_n)$, yields uniformly valid inference across these regimes, and admits a matching local minimax lower bound. Resolving this question requires specifying a sharp-rate functional, a uniform-inference or coverage criterion, an existential selector, a confidence procedure, and a matching lower-bound witness.*

[Remark 1](#) states the future-work target as an open research agenda rather than as an identification or risk theorem. Its rate arguments are the sample size, the two supplied-code radii, and the local cumulant-separation threshold. The agenda also includes a uniform-inference criterion and a local minimax lower-bound construction, so any completed theory would need to align estimation, coverage, selection, and lower-bound witnesses on the same triangular experiment.

This separation clarifies the scope of the paper's established results. The fixed-separation contour theorem gives the $\Theta(n^{-1})$ MSE characterization under fixed cumulant separation and the stated treatment-code stability condition. The local mixture and ACE benchmarks give constructive upper comparisons under their own displayed hypotheses. The shrinking-separation

program in [Remark 1](#) is the natural next step for a unified adaptive theory across Gaussian-adjacent and fixed non-Gaussian regimes.

Appendices

A Proofs for identification and analytic localization

This appendix collects the analytic statements that support the identification argument in [Theorems 1 and 2](#) and [Lemma 1](#) and the fixed-bank localization used in [Theorem 3](#). The common theme is that zeros of moment-generating functions can be controlled by classical complex-analytic arguments and then transferred to residualized-treatment transforms. The zero and completeness perspective is standard in identification arguments based on transforms [[Kagan et al., 1973](#), [Mattner, 1992](#)], and the finite-bank construction uses the same analytic geometry that underlies modern zero-localization bounds for moment-generating functions [[Michelen and Sahasrabudhe, 2019](#), [Eremenko and Fryntov, 2021](#), [Dinh et al., 2021](#)].

For [Theorem 1](#), the analytic order of a zero of M determines which polynomial-exponential weight annihilates shifted treatment innovations. Independence of the treatment innovation from X , as imposed in [Assumption 2](#), then turns the shifted identity into a conditional moment for the code residual Z_n . The denominator calculation is the corresponding differentiated transform identity, and the partially linear conditional-mean restriction in [Assumption 3](#) supplies the numerator relation.

The factorization in [Lemma 1](#) isolates the effect of treatment-code contamination. Since $Z_n = \eta + D_n(X)$, independence separates the residual transform into the treatment-innovation transform and the nuisance transform. The outcome-weighted transform has the derivative term carrying θ_0 plus the covariate-only contamination transform from [Definition 7](#). This decomposition is the algebraic input for [Theorem 2](#): on a contour where the residual transform is separated from zero and the nuisance transform has no zero on the closed disk, the logarithmic-derivative contribution counts the enclosed zeros and the analytic contamination contribution integrates to zero.

The remaining analytic step is uniform localization. [Lemma 3](#) gives a disk, determined by the fixed cumulant order, sub-Gaussian scale, and separation constant, that contains at least one zero of the treatment-innovation transform. The finite-bank statement below turns that disk-level information into a deterministic list of circles, with explicit dyadic bookkeeping and a positive boundary-modulus certificate. Its role is to provide the law-blind contour geometry used later by the empirical statistic.

Lemma 8. [`lem:finite-contour-bank`] (Finite contour bank) *Fix the primitive parameter record p , the positive certified-real input record $\mathbf{p}_\star$ of [Definition 17](#), an index n with $n \geq 1$, and a law P in $\mathcal{P}_{\text{NG},n}$ as specified by [Definition 3](#). Let B be the contour-bank data computed from $(p, \mathbf{p}_\star)$, with constants $U_\psi, U_R, N_{\text{cert}}, m_\star, d_\star, h_\star, e_\star, d_\star^{\text{den}}, u_\star, a_\star$, total size J , and radii ρ_j .*

Then the following properties hold:

- (*Positive certificate.*) $a_\star > 0$.

- **(Bookkeeping identities.)**

$$\begin{aligned} N_{\text{cert}} &= 4U_\psi^2(U_R + 2)^3, & m_\star &= N_{\text{cert}} + 3, & J &= 2^{m_\star-1} + 1, \\ d_\star &= 2^{-m_\star}, & h_\star &= d_\star/3, & e_\star &= 8U_R U_\psi^2(U_R + 1)^2, \\ d_\star^{\text{den}} &= 3(2U_R + 1), & u_\star &= 2e_\star + N_{\text{cert}}(m_\star + d_\star^{\text{den}}), & a_\star &= 2^{-u_\star}. \end{aligned}$$

- **(Bank geometry.)** The radii are strictly increasing and satisfy

$$R_0 < \rho_j < R_1$$

for every bank index j .

- **(Uniform treatment-noise certificate.)** For the treatment-noise moment-generating function $M(z) = E_P[e^{z\eta}]$, there is a bank index j such that, with $C_j = \{z \in \mathbb{C} : |z| = \rho_j\}$,

$$N_{C_j}(M) \geq 1$$

and

$$a_\star \leq |M(z)| \quad \text{for every } z \in C_j.$$

Proof of Lemma 8. Let B be the bank produced from $(p, \mathbf{p}_\star)$. The construction in Definition 17 gives

$$\begin{aligned} N_{\text{cert}} &= 4U_\psi^2(U_R + 2)^3, & m_\star &= N_{\text{cert}} + 3, & J &= 2^{m_\star-1} + 1, \\ d_\star &= 2^{-m_\star}, & h_\star &= d_\star/3, \\ e_\star &= 8U_R U_\psi^2(U_R + 1)^2, & d_\star^{\text{den}} &= 3(2U_R + 1), \end{aligned}$$

and

$$u_\star = 2e_\star + N_{\text{cert}}(m_\star + d_\star^{\text{den}}), \quad a_\star = 2^{-u_\star}.$$

Thus $a_\star > 0$ and the displayed bookkeeping identities hold.

The radii have the explicit form

$$\rho_j = R_0 + \frac{1}{4} + j 2^{-m_\star}, \quad 0 \leq j \leq 2^{m_\star-1}.$$

Hence $j \mapsto \rho_j$ is strictly increasing. Since

$$\frac{1}{4} + j 2^{-m_\star} \leq \frac{1}{4} + 2^{m_\star-1} 2^{-m_\star} = \frac{3}{4} < 1,$$

and $R_1 = R_0 + 1$, each bank radius satisfies

$$R_0 < \rho_j < R_1.$$

It remains to prove the treatment-noise certificate. The error-one certified refinements and the ceiling construction give

$$\psi_\eta \leq U_\psi, \quad R_1 \leq U_R.$$

Let

$$M(z) = E_P e^{z\eta}, \quad R_{\text{fac}} = R_1 + 1, \quad R_{\text{out}} = R_1 + 2.$$

The conditional-mean identity for g_0 centers η . The treatment-tail condition in $\mathcal{P}_{\text{NG},n}$, together with [Lemma 2](#), gives for every $t \in \mathbb{R}$

$$E_P e^{t\eta} \leq \exp(4\psi_\eta^2 t^2).$$

The same real exponential integrability gives the entire complex moment-generating function, normalization at the origin follows from the probability law, and the complex modulus estimate yields

$$M(0) = 1, \quad M \text{ is entire}, \quad |M(z)| \leq \exp(4\psi_\eta^2 |z|^2) \quad (z \in \mathbb{C}).$$

Let

$$N_0 = N_{\{|z| < R_{\text{fac}}\}}(M)$$

be the number of zeros of M in the open disk of radius R_{fac} , counted with multiplicity. Jensen's zero-counting bound applied on R_{out} gives

$$N_0 \log\left(\frac{R_{\text{out}}}{R_{\text{fac}}}\right) \leq 4\psi_\eta^2 R_{\text{out}}^2.$$

Since $R_{\text{out}} = R_{\text{fac}} + 1$,

$$\frac{1}{R_{\text{out}}} \leq \log\left(\frac{R_{\text{out}}}{R_{\text{fac}}}\right).$$

Therefore

$$N_0 \leq 4\psi_\eta^2 R_{\text{out}}^3 \leq 4U_\psi^2 (U_R + 2)^3 = N_{\text{cert}}.$$

List the N_0 zeros in $\{|z| < R_{\text{fac}}\}$, with multiplicity, as

$$a_1, \dots, a_{N_0}.$$

The finite Blaschke factorization on the disk of radius R_{fac} gives a holomorphic function g , zero-free in $\{|z| < R_{\text{fac}}\}$, such that

$$M(z) = Q(z)g(z), \quad |z| \leq R_{\text{fac}},$$

where

$$Q(z) = \prod_{\nu=1}^{N_0} \frac{R_{\text{fac}}(z - a_\nu)}{R_{\text{fac}}^2 - \overline{a_\nu}z}.$$

Because $N_0 \leq N_{\text{cert}}$ and the grid has $2^{m_\star-1} + 1$ points with $m_\star = N_{\text{cert}} + 3$, there is an index j such that

$$h_\star \leq ||a_\nu| - \rho_j| \quad (1 \leq \nu \leq N_0).$$

By [Lemma 3](#), M has a zero z_0 with $|z_0| \leq R_0$. Since $R_0 < \rho_j$, this zero lies strictly inside $C_j = \{|z| = \rho_j\}$. If M had a zero z on C_j , then $z \in \{|z| < R_{\text{fac}}\}$, and the factorization $M = Qg$ together with the zero-freeness of g would force $z = a_\nu$ for some ν . This would give

$$||a_\nu| - \rho_j| = 0 < h_\star,$$

contradicting the separation of the chosen grid point. Hence the contour has positive zero count:

$$N_{C_j}(M) \geq 1.$$

It remains to lower-bound $|M|$ on this circle. Put

$$A = 4\psi_\eta^2 R_{\text{fac}}^2.$$

On $|w| = R_{\text{fac}}$, the Blaschke product has $|Q(w)| = 1$, so the boundary envelope for M gives $|g(w)| \leq e^A$. Also $M(0) = 1$ and $|Q(0)| \leq 1$, hence $|g(0)| \geq 1$. Applying Harnack's inequality to the positive harmonic function $A - \log |g|$ gives, for every $|z| \leq R_1$,

$$\exp\left\{-\left(\frac{R_{\text{fac}} + R_1}{R_{\text{fac}} - R_1} - 1\right) A\right\} |Q(z)| \leq |M(z)|.$$

Since $R_{\text{fac}} = R_1 + 1$,

$$\left(\frac{R_{\text{fac}} + R_1}{R_{\text{fac}} - R_1} - 1\right) A = 2R_1 \cdot 4\psi_\eta^2 (R_1 + 1)^2 = 8\psi_\eta^2 R_1 (R_1 + 1)^2.$$

Thus, for every $|z| \leq R_1$,

$$\exp\{-8\psi_\eta^2 R_1 (R_1 + 1)^2\} |Q(z)| \leq |M(z)|.$$

Since

$$8\psi_\eta^2 R_1 (R_1 + 1)^2 \leq 8U_R U_\psi^2 (U_R + 1)^2 = e_\star,$$

and $\log 2 \geq 1/2$, we have

$$2^{-2e_\star} \leq \exp\{-8\psi_\eta^2 R_1 (R_1 + 1)^2\}.$$

Now take $z \in C_j$. For each zero a_ν ,

$$|z - a_\nu| \geq ||z| - |a_\nu|| = |\rho_j - |a_\nu|| \geq h_\star.$$

Also $|z| \leq R_1$, and therefore

$$R_{\text{fac}} + |z| \leq 2R_1 + 1 \leq 2U_R + 1.$$

For each Blaschke factor,

$$\left|\frac{R_{\text{fac}}(z - a_\nu)}{R_{\text{fac}}^2 - \bar{a}_\nu z}\right| \geq \frac{|z - a_\nu|}{R_{\text{fac}} + |z|} \geq \frac{h_\star}{2U_R + 1} = \frac{d_\star}{d_\star^{\text{den}}}.$$

Multiplying over the at most N_{cert} factors and using $(d_\star^{\text{den}})^{-1} \geq 2^{-d_\star^{\text{den}}}$ gives

$$|Q(z)| \geq 2^{-N_{\text{cert}}(m_\star + d_\star^{\text{den}})}.$$

Combining the last two lower bounds,

$$|M(z)| \geq 2^{-2e_\star} 2^{-N_{\text{cert}}(m_\star + d_\star^{\text{den}})} = 2^{-u_\star} = a_\star.$$

Thus for this bank index j ,

$$a_\star \leq |M(z)| \quad (z \in C_j),$$

and the asserted uniform treatment-noise certificate follows. $\square$

Lemma 8 is the population geometric certificate used by the empirical selector in [Definition 18](#). The proof below records the dyadic bookkeeping and the zero-localization argument behind that certificate.

B Empirical process, stability, and lower-bound lemmas

This appendix collects the auxiliary statistical statements used by the fixed-separation contour theorem. The first two results control the analytic transforms that enter the contour statistic: one gives uniform L^2 concentration of the empirical residual and outcome transforms on a fixed disk, and the other gives a deterministic $L^1(P_X)$ stability bound for the nuisance factor. Together they supply the stochastic and population inputs behind the contour-risk bounds in [Theorem 3](#). The concentration statement uses the same sub-Gaussian scale conventions and product-sample structure as [Definition 1](#) and [Assumption 1](#), in line with standard empirical-process and high-dimensional probability arguments [[van der Vaart, 1998](#), [van der Vaart and Wellner, 1996](#), [Vershynin, 2018](#), [Wainwright, 2019](#)].

Lemma 9. [`lem:empirical-transform-uniform-12`] (Uniform transform concentration) *Fix $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, R_1 \in \mathbb{R}$. There is a constant $K > 0$, depending on these displayed constants, such that the following holds. Let p be a parameter tuple and let $P \in \mathcal{P}_{\text{NG},n}$ be represented by a model in [Definition 3](#). Assume:*

- **(Primitive constants.)** *The primitive constants of p are $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$, and its search radius is R_1 .*
- **(Sample size.)** $n \geq 2$.
- **(Sampling.)** $O_1, \dots, O_n$ follow the i.i.d. product law specified in [Assumption 1](#).
- **(Fold.)** $a \in \{0, 1\}$ and the corresponding inference fold I_a is nonempty.

With $Z_n = T - \bar{g}_n(X)$, define

$$\hat{F}_{a,n}(z) = \frac{1}{|I_a|} \sum_{i \in I_a} \exp(z Z_{n,i}), \quad \hat{G}_{a,n}(z) = \frac{1}{|I_a|} \sum_{i \in I_a} Y_i \exp(z Z_{n,i}),$$

and

$$F_n(z) = E_P[\exp(z Z_n)], \quad G_n(z) = E_P[Y \exp(z Z_n)].$$

Then

$$E \left[\sup_{|z| \leq R_1} \left| \hat{F}_{a,n}(z) - F_n(z) \right|^2 \right] + E \left[\sup_{|z| \leq R_1} \left| \hat{G}_{a,n}(z) - G_n(z) \right|^2 \right] \leq \frac{K}{|I_a|}.$$

Proof of [Lemma 9](#). Set

$$A_F = 2 \exp\{8R_1 C_g + 4R_1^2 \psi_\eta^2\},$$

$$A_G = 64\{C_q^4 + 4C_\theta^4 \psi_\eta^4 + 4\psi_\xi^4\} + 2 \exp\{16R_1 C_g + 16R_1^2 \psi_\eta^2\},$$

and define

$$K = 4(A_F + A_G).$$

The constants A_F and A_G are strictly positive, so $K > 0$; it depends only on the displayed primitive constants and R_1 .

Fix a fold for which I_a is nonempty, and let $I = I_a$, $m_I = |I|$, and $Z_n = T - \bar{g}_n(X)$. Since the contour-bank radius in [Definition 17](#) satisfies $R_1 = R_0 + 1$ with $R_0 \geq 0$, we have $R_1 > 0$. Write $D_n(X) = g_0(X) - \bar{g}_n(X)$, so $Z_n = \eta + D_n(X)$. The range bound for g_0 in [Definition 3](#) and

the clipping of $\bar{g}_n$ in [Definition 2](#) give $|D_n(X)| \leq 2C_g$ almost surely. Hence, for almost every observation,

$$4R_1|Z_n| \leq 4R_1|\eta| + 8R_1C_g \leq \frac{\eta^2}{\psi_\eta^2} + 4R_1^2\psi_\eta^2 + 8R_1C_g,$$

where the second inequality is the Young inequality obtained by expanding $(|\eta|/\psi_\eta - 2R_1\psi_\eta)^2 \geq 0$. Therefore

$$\{\exp(2R_1|Z_n|)\}^2 \leq \exp\{8R_1C_g + 4R_1^2\psi_\eta^2\} \exp(\eta^2/\psi_\eta^2).$$

Integrating and using the Luxemburg treatment-tail bound from [Definition 3](#) yields

$$E_P [\exp\{2R_1|Z_n|\}^2] \leq A_F.$$

The outcome-weighted envelope is

$$E_P [\{|Y| \exp(2R_1|Z_n|)\}^2] \leq A_G.$$

Indeed, [Definition 2](#) gives $Y = q_0(X) + \theta_0\eta + \xi$. For almost every observation, the algebraic bound

$$|q_0(X) + \theta_0\eta + \xi|^4 \leq 64\{|q_0(X)|^4 + |\theta_0|^4|\eta|^4 + |\xi|^4\}$$

combines with the bounds in [Definition 3](#), namely $|q_0(X)| \leq C_q$ almost surely and $|\theta_0| \leq C_\theta$, to give

$$|Y|^4 \leq 64\{C_q^4 + C_\theta^4|\eta|^4 + |\xi|^4\}$$

almost surely. The Luxemburg moment bounds give $E_P|\eta|^4 \leq 4\psi_\eta^4$ and $E_P|\xi|^4 \leq 4\psi_\xi^4$, and hence

$$E_P|Y|^4 \leq 64\{C_q^4 + 4C_\theta^4\psi_\eta^4 + 4\psi_\xi^4\}.$$

Together with

$$\{|Y| \exp(2R_1|Z_n|)\}^2 \leq |Y|^4 + \{\exp(4R_1|Z_n|)\}^2,$$

and the preceding residual envelope applied at radius $2R_1$, this gives the stated bound.

We use the following coefficient-series bound. Let U, V be measurable real functions, let $R > 0$, let $C \geq 0$, and suppose

$$E_P [\{|U| \exp(2R|V|)\}^2] \leq C^2.$$

For a nonempty index set I , define

$$c_k(O_1, \dots, O_n) = m_I^{-1} \sum_{i \in I} \left\{ \frac{U(O_i)V(O_i)^k}{k!} - E_P \frac{UV^k}{k!} \right\}.$$

Then

$$E \left[\sup_{|z| \leq R} \left| \sum_{k=0}^{\infty} c_k z^k \right|^2 \right] \leq \left(\frac{2C}{\sqrt{m_I}} \right)^2.$$

The envelope gives

$$\left\| \frac{UV^k}{k!} \right\|_{L^2(P)} \leq \frac{C}{(2R)^k},$$

because $x^k/k! \leq e^{2Rx}/(2R)^k$ for $x \geq 0$; the hypotheses $R > 0$ and $C \geq 0$ make the displayed majorant nonnegative. Under the product sample law, centering over the block I gives

$$\|c_k\|_{L^2(P^{\otimes n})} \leq \frac{C}{(2R)^k \sqrt{m_I}}.$$

The series of majorants is summable, and the centered analytic-series supremum bound yields

$$\left(\sum_{k=0}^{\infty} \frac{C}{(2R)^k \sqrt{m_I}} R^k \right)^2 = \left(\frac{2C}{\sqrt{m_I}} \right)^2.$$

The same coefficient construction represents the empirical transform errors exactly. For $|z| \leq R$,

$$m_I^{-1} \sum_{i \in I} U(O_i) e^{zV(O_i)} - E_P[U e^{zV}] = \sum_{k=0}^{\infty} c_k z^k,$$

with the exchange of summation, finite averaging, and integration justified by the doubled exponential envelope.

Apply the preceding bound first with $U \equiv 1$, $V = Z_n$, $R = R_1$, and $C = \sqrt{A_F}$. Since $A_F > 0$, this choice has $C \geq 0$, and

$$E \left[\sup_{|z| \leq R_1} |\widehat{F}_{a,n}(z) - F_n(z)|^2 \right] \leq \frac{4A_F}{m_I}.$$

Apply it again with $U = Y$, $V = Z_n$, $R = R_1$, and $C = \sqrt{A_G}$. Since $A_G > 0$, this choice has $C \geq 0$, and

$$E \left[\sup_{|z| \leq R_1} |\widehat{G}_{a,n}(z) - G_n(z)|^2 \right] \leq \frac{4A_G}{m_I}.$$

Adding the two displays yields

$$E \left[\sup_{|z| \leq R_1} |\widehat{F}_{a,n}(z) - F_n(z)|^2 \right] + E \left[\sup_{|z| \leq R_1} |\widehat{G}_{a,n}(z) - G_n(z)|^2 \right] \leq \frac{4(A_F + A_G)}{|I_a|} = \frac{K}{|I_a|}.$$

This proves the assertion. $\square$

Lemma 9 is the stochastic input for the selected-contour risk proof. The proof below supplies the disk-uniform L^2 bound used after the pilot fold selects a circle.

The next statement is the population stability input. It uses the direct $L^1(P_X)$ treatment-code radius in [Definition 3](#) to keep the nuisance transform close to one on the contour search region.

Lemma 10. [\[lem:l1-nuisance-zero-free\]](#) (Zero-Free Nuisance Factor) *For every $P \in \mathcal{P}_{\text{NG},n}$ satisfying [Definition 3](#), the associated treatment-code discrepancy D_n obeys*

$$|D_n| \leq 2C_g \quad \text{almost surely,} \quad E_P |D_n| \leq \varepsilon_{1,n}.$$

Consequently, for the corresponding factor H_n and radius R_1 ,

$$\sup_{|z| \leq R_1} |H_n(z) - 1| \leq R_1 e^{2C_g R_1} \varepsilon_{1,n}.$$

If $\varepsilon_{1,n} \leq (4R_1 e^{2C_g R_1})^{-1}$, then $|H_n(z)| \geq 3/4$ for every $|z| \leq R_1$, and F_n and M have the same zeros in $\{|z| \leq R_1\}$, counted with multiplicity.

Proof of [Lemma 10](#). Let

$$D_n(X) = g_0(X) - \bar{g}_n(X), \quad H_n(z) = E_P[e^{zD_n(X)}].$$

Since $|g_0(X)| \leq C_g$ almost surely and $|\bar{g}_n(X)| \leq C_g$ by clipping,

$$|D_n(X)| \leq |g_0(X)| + |\bar{g}_n(X)| \leq 2C_g \quad \text{almost surely.}$$

The treatment-code condition in $\mathcal{P}_{\text{NG},n}$ gives

$$E_P|D_n(X)| = \int |g_0(x) - \bar{g}_n(x)| dP_X(x) \leq \varepsilon_{1,n}.$$

For $|z| \leq R_1$, the exponential remainder bound gives, with $w = zD_n(X)$,

$$|e^{zD_n(X)} - 1| \leq |z| |D_n(X)| e^{|z| |D_n(X)|}.$$

Together with $|D_n(X)| \leq 2C_g$, this yields

$$|H_n(z) - 1| \leq E_P|e^{zD_n(X)} - 1| \leq R_1 e^{2C_g R_1} E_P|D_n(X)| \leq R_1 e^{2C_g R_1} \varepsilon_{1,n}.$$

Taking the supremum over $|z| \leq R_1$ gives the displayed uniform bound.

Assume

$$\varepsilon_{1,n} \leq (4R_1 e^{2C_g R_1})^{-1}.$$

Then the preceding display gives $|H_n(z) - 1| \leq 1/4$ throughout the disk, and therefore

$$|H_n(z)| \geq 1 - |H_n(z) - 1| \geq 3/4 \quad (|z| \leq R_1).$$

It remains to record why the multiplicity comparison is legitimate at each point of the disk. The almost-sure bound $|D_n(X)| \leq 2C_g$ implies that $E_P \exp\{tD_n(X)\} < \infty$ for every real t , so the complex MGF H_n is analytic in a neighborhood of every $|z| \leq R_1$. The treatment-tail condition in $\mathcal{P}_{\text{NG},n}$, under the Luxemburg convention of [Definition 1](#), gives the same local exponential-moment condition for η : for every real t , completing the square bounds $\exp\{t\eta\}$ by a constant multiple of $\exp\{\eta^2/\psi_\eta^2\}$, which is integrable. Hence the treatment-noise MGF $M(z) = E_P[e^{z\eta}]$ is analytic in a neighborhood of every point of the disk as well.

By [Lemma 1](#),

$$F_n(z) = M(z)H_n(z).$$

Fix $|z| \leq R_1$. The preceding lower bound gives $H_n(z) \neq 0$, so the analytic order of H_n at z is zero. Applying additivity of analytic order to the displayed factorization gives

$$\text{ord}_z(F_n) = \text{ord}_z(M) + \text{ord}_z(H_n) = \text{ord}_z(M).$$

Thus F_n and M have exactly the same zeros in $\{|z| \leq R_1\}$, counted with multiplicity. $\square$

[Lemma 10](#) is the deterministic transfer from treatment-transform zeros to observable residual-transform zeros. The proof below uses the clipping bound and the $L^1(P_X)$ radius to control the nuisance factor H_n on the search disk.

A separate tail calculation is used when Gaussian outcome innovations are embedded in lower-bound and benchmark paths. The following elementary scale bound matches the Luxemburg convention fixed in [Definition 1](#).

Lemma 11. [lem:gaussian-exponential-square-quarter-scale] (Gaussian square exponential bound) *Use the Gaussian-law notation of Definition 1. For every positive scale $\psi > 0$, let $\zeta \sim N(0, (\psi/4)^2)$. Then $\exp(\zeta^2/\psi^2)$ is integrable under the law of ζ , and*

$$\mathbb{E} \exp(\zeta^2/\psi^2) \leq 2.$$

Proof of Lemma 11. Let

$$v = \left(\frac{\psi}{4}\right)^2.$$

Since $\psi > 0$, $v > 0$. Under $N(0, v)$, the density calculation gives, for every $x \in \mathbb{R}$,

$$\frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{x^2}{2v}\right) \exp\left(\frac{x^2}{\psi^2}\right) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{7x^2}{\psi^2}\right),$$

because $1/(2v) = 8/\psi^2$.

The coefficient $7/\psi^2$ is positive, so $x \mapsto \exp\{-7x^2/\psi^2\}$ is integrable on $\mathbb{R}$, and hence $\exp(\zeta^2/\psi^2)$ is integrable under $N(0, v)$. The Gaussian integral formula gives

$$\mathbb{E} \exp(\zeta^2/\psi^2) = \frac{1}{\sqrt{2\pi v}} \sqrt{\frac{\pi}{7/\psi^2}}.$$

Finally,

$$\sqrt{\frac{\pi}{7/\psi^2}} \leq 2\sqrt{2\pi v},$$

since after squaring both sides this is

$$\frac{\pi\psi^2}{7} \leq 8\pi v = \frac{\pi\psi^2}{2}.$$

Therefore

$$\mathbb{E} \exp(\zeta^2/\psi^2) \leq 2.$$

□

Lemma 11 is the tail-scale normalization used in the lower-bound construction below.

The final auxiliary result supplies the decision-theoretic lower-bound path. It constructs one-dimensional submodels in which the observed treatment law, regression functions, and supplied clipped codes remain fixed while the coefficient varies locally at $n^{-1/2}$ scale. Such paths are the standard input for parametric lower bounds in semiparametric experiments [Bickel et al., 1993, van der Vaart, 1998].

Lemma 12. [lem:non-gaussian-hard-submodel] (One-dimensional hard submodel) *Fix $r \in \mathbb{N}$ and real constants $\delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$ with $\psi_\xi > 0$. There exist constants $c_0, C_0, \tau, c_{ACE} \in \mathbb{R}$ such that*

$$0 < c_0, \quad 0 < C_0, \quad 0 < \tau \leq \psi_\xi, \quad 0 < c_{ACE}.$$

For every parameter record p whose constants $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$ are the ones just fixed, and every $n \geq 1$, if the non-Gaussian class in Definition 3 is nonempty, then there are a base law $P^\circ \in \mathcal{P}_{\text{NG}, n}$ and a family $\{P_{n, \vartheta} : |\vartheta| \leq c_0\}$ such that:

- *(Class membership and target.) For every $|\vartheta| \leq c_0$, $P_{n, \vartheta} \in \mathcal{P}_{\text{NG}, n}$ and $\theta_0(P_{n, \vartheta}) = \vartheta$.*

- **(Fixed observed treatment law.)** For every $|\vartheta| \leq c_0$, the law of (X, T) under $P_{n, \vartheta}$ equals its law under P° .
- **(Fixed regressions and supplied codes.)** For every $|\vartheta| \leq c_0$, the functions g_0, q_0 and the clipped supplied codes $\bar{g}_n, \bar{q}_n$ under $P_{n, \vartheta}$ equal those under P° .
- **(Gaussian outcome innovation.)** For every $|\vartheta| \leq c_0$, the outcome innovation $\xi = Y - q_0(X) - \theta_0 \eta$ under $P_{n, \vartheta}$ has law $N(0, \tau^2)$ and is independent of (X, T) .
- **(Outcome equation.)** For every $|\vartheta| \leq c_0$,

$$Y = q_0(X) + \vartheta \eta + \xi$$

almost surely under $P_{n, \vartheta}$.

- **(Product relative entropy.)** For all real $\vartheta_0, \vartheta_1, h$ with $|\vartheta_0| \leq c_0, |\vartheta_1| \leq c_0$, and

$$|\vartheta_1 - \vartheta_0| = \frac{h}{\sqrt{n}},$$

one has

$$D_{\text{KL}}(P_{n, \vartheta_0}^{\otimes n} \parallel P_{n, \vartheta_1}^{\otimes n}) \leq C_0 h^2.$$

Moreover, for every starting law $P^\circ \in \mathcal{P}_{\text{ACE}, n}^{\text{JMS}}$ in the class of [Definition 5](#), there is a family $\{P_{n, \vartheta}^{\text{ACE}} : |\vartheta| \leq c_0\}$ such that:

- **(JMS ACE membership and target.)** For every $|\vartheta| \leq c_0$, $P_{n, \vartheta}^{\text{ACE}} \in \mathcal{P}_{\text{ACE}, n}^{\text{JMS}}$ and $\theta_0(P_{n, \vartheta}^{\text{ACE}}) = \vartheta$.
- **(Fixed observed treatment law.)** For every $|\vartheta| \leq c_0$, the law of (X, T) under $P_{n, \vartheta}^{\text{ACE}}$ equals its law under P° .
- **(Fixed nuisance objects.)** For every $|\vartheta| \leq c_0$, the functions g_0, q_0 , the clipped supplied codes $\bar{g}_n, \bar{q}_n$, and the treatment innovation η under $P_{n, \vartheta}^{\text{ACE}}$ equal those under P° , and the treatment and outcome code-radius conditions in [Definition 5](#) hold at $P_{n, \vartheta}^{\text{ACE}}$.
- **(Gaussian outcome innovation.)** For every $|\vartheta| \leq c_0$, the outcome innovation $\xi = Y - q_0(X) - \theta_0 \eta$ under $P_{n, \vartheta}^{\text{ACE}}$ has law $N(0, \tau^2)$ and is independent of (X, T) .
- **(Outcome equation.)** For every $|\vartheta| \leq c_0$,

$$Y = q_0(X) + \vartheta \eta + \xi$$

almost surely under $P_{n, \vartheta}^{\text{ACE}}$.

- **(Product relative entropy.)** For all real $\vartheta_0, \vartheta_1, h$ with $|\vartheta_0| \leq c_0, |\vartheta_1| \leq c_0$, and

$$|\vartheta_1 - \vartheta_0| = \frac{h}{\sqrt{n}},$$

one has

$$D_{\text{KL}}((P_{n, \vartheta_0}^{\text{ACE}})^{\otimes n} \parallel (P_{n, \vartheta_1}^{\text{ACE}})^{\otimes n}) \leq C_0 h^2.$$

Finally, for every pair of clipped-code functions $g_\star, q_\star : \mathcal{X} \rightarrow \mathbb{R}$, if there is a law $P^\circ \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ with $\bar{g}_n = g_\star$ and $\bar{q}_n = q_\star$, then the fixed-code JMS ACE minimax risk satisfies

$$\frac{c_{\text{ACE}}}{n} \leq \inf_{\tilde{\theta}(O_1, \dots, O_n; \bar{g}_n, \bar{q}_n)} \sup_{\substack{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} : \\ \bar{g}_n = g_\star, \bar{q}_n = q_\star}} E_P \left[\{\tilde{\theta} - \theta_0(P)\}^2 \right].$$

Proof of Lemma 12. Set

$$c_0 = \begin{cases} \min\{C_\theta, 1\}, & C_\theta > 0, \\ 1, & C_\theta \leq 0, \end{cases} \quad \tau = \frac{\psi_\xi}{4}, \quad C_0 = \max \left\{ 1, \frac{16\psi_\eta^2}{\psi_\xi^2} \right\}.$$

Then $c_0 > 0$, $C_0 > 0$, and $0 < \tau \leq \psi_\xi$. For parameter records whose C_θ -coordinate is the fixed constant C_θ , admissibility gives $C_\theta > 0$, and hence $c_0 \leq C_\theta$.

Let

$$K = C_0 c_0^2.$$

The two-point testing bound supplies a constant $c_K > 0$, depending only on K , such that whenever two n -sample laws have relative entropy at most K , every measurable estimator T satisfies

$$c_K(\theta_1 - \theta_0)^2 \leq \max \{E_0[(T - \theta_0)^2], E_1[(T - \theta_1)^2]\}.$$

This is the usual Le Cam reduction from estimation to testing, after symmetrically clipping the estimator to a range containing both target values. Define

$$c_{\text{ACE}} = c_K c_0^2 > 0.$$

Fix now an admissible parameter record p with the displayed primitive constants and a class index $n \geq 1$. Let $P^\circ \in \mathcal{P}_{\text{NG},n}$, represented by a base model m° . For each $\vartheta \in \mathbb{R}$, define $P_{n,\vartheta}$ by preserving the joint law of (X, T) under P° , drawing an independent

$$\xi_\star \sim N(0, \tau^2),$$

and setting

$$Y = q_0(X) + \vartheta\{T - g_0(X)\} + \xi_\star.$$

Equivalently, the path is the product of the retained (X, T) -law and the centered Gaussian innovation, transported through this affine outcome map.

Along this path,

$$\theta_0(P_{n,\vartheta}) = \vartheta, \quad g_0(P_{n,\vartheta}) = g_0(P^\circ), \quad q_0(P_{n,\vartheta}) = q_0(P^\circ),$$

and the clipped supplied codes satisfy

$$\bar{g}_n(P_{n,\vartheta}) = \bar{g}_n(P^\circ), \quad \bar{q}_n(P_{n,\vartheta}) = \bar{q}_n(P^\circ).$$

The covariate-treatment marginal is also retained:

$$(P_{n,\vartheta})_{X,T} = P_{X,T}^\circ.$$

The treatment innovation is unchanged along the path:

$$\eta_{P_{n,\vartheta}} = \eta_{P^\circ}.$$

The outcome innovation is the fresh Gaussian coordinate:

$$\xi_{P_{n,\vartheta}} = Y - q_0(X) - \vartheta\eta = \xi_\star,$$

so

$$(P_{n,\vartheta}) \circ \xi_{P_{n,\vartheta}}^{-1} = N(0, \tau^2), \quad \xi_{P_{n,\vartheta}} \perp (X, T).$$

For $|\vartheta| \leq c_0$, the coefficient range holds because $c_0 \leq C_\theta$. All treatment-side assumptions, the regression ranges, the cumulant separation, and the treatment $L^1(P_X)$ code-radius condition are inherited from P° . The conditional mean condition follows from the displayed outcome equation and the independence and centering of $\xi_\star$. Finally, [Lemma 11](#) applied at scale ψ_ξ gives

$$E \exp\{\xi_\star^2/\psi_\xi^2\} \leq 2,$$

and the independence of $\xi_\star$ from X gives the corresponding conditional Luxemburg bound. Hence

$$|\vartheta| \leq c_0 \implies P_{n,\vartheta} \in \mathcal{P}_{\text{NG},n}.$$

The same construction gives the asserted outcome equation:

$$Y = q_0(X) + \vartheta\eta + \xi \quad P_{n,\vartheta}\text{-a.s.}$$

It remains, for this branch, to record the product relative-entropy bound. For the affine Gaussian path, one observation satisfies

$$D_{\text{KL}}(P_{n,\vartheta_0} \| P_{n,\vartheta_1}) \leq \frac{(\vartheta_1 - \vartheta_0)^2 2\psi_\eta^2}{2\tau^2}.$$

The moment bound used here is the Luxemburg consequence

$$E_{P^\circ}[\eta^2] \leq 2\psi_\eta^2.$$

Tensorization over n independent observations gives

$$D_{\text{KL}}\left(P_{n,\vartheta_0}^{\otimes n} \parallel P_{n,\vartheta_1}^{\otimes n}\right) \leq n \frac{(\vartheta_1 - \vartheta_0)^2 2\psi_\eta^2}{2\tau^2}.$$

If $|\vartheta_1 - \vartheta_0| = h/\sqrt{n}$ and $\tau = \psi_\xi/4$, then

$$n \frac{(\vartheta_1 - \vartheta_0)^2 2\psi_\eta^2}{2\tau^2} = \frac{16\psi_\eta^2}{\psi_\xi^2} h^2 \leq C_0 h^2.$$

Thus the required entropy display holds.

Now let $P^\circ \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$. Define $P_{n,\vartheta}^{\text{ACE}}$ by the same affine Gaussian outcome construction from this starting law:

$$Y = q_0(X) + \vartheta\{T - g_0(X)\} + \xi_\star, \quad \xi_\star \sim N(0, \tau^2), \quad \xi_\star \perp (X, T).$$

The displayed preservation identities remain valid:

$$(P_{n,\vartheta}^{\text{ACE}})_{X,T} = P_{X,T}^\circ,$$

$$g_0(P_{n,\vartheta}^{\text{ACE}}) = g_0(P^\circ), \quad q_0(P_{n,\vartheta}^{\text{ACE}}) = q_0(P^\circ), \quad \eta_{P_{n,\vartheta}^{\text{ACE}}} = \eta_{P^\circ},$$

and

$$\bar{g}_n(P_{n,\vartheta}^{\text{ACE}}) = \bar{g}_n(P^\circ), \quad \bar{q}_n(P_{n,\vartheta}^{\text{ACE}}) = \bar{q}_n(P^\circ).$$

Consequently the two JMS ACE code-radius inequalities are transported unchanged:

$$\|\bar{g}_n - g_0\|_{L^r(P_X)} \leq \varepsilon_{1,n}, \quad \|\bar{q}_n - q_0\|_{L^r(P_X)} \leq \varepsilon_{2,n}.$$

Together with the transported treatment-side and range conditions, the coefficient bound $|\vartheta| \leq c_0 \leq C_\theta$, and the Gaussian conditional Luxemburg calculation above, this gives

$$|\vartheta| \leq c_0 \implies P_{n,\vartheta}^{\text{ACE}} \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}.$$

The Gaussian residual law, residual independence, outcome equation, and product entropy bound are the same as in the non-Gaussian branch:

$$D_{\text{KL}}((P_{n,\vartheta_0}^{\text{ACE}})^{\otimes n} \parallel (P_{n,\vartheta_1}^{\text{ACE}})^{\otimes n}) \leq C_0 h^2$$

whenever $|\vartheta_1 - \vartheta_0| = h/\sqrt{n}$.

It remains to prove the fixed-code ACE lower bound. Suppose a starting law $P^\circ \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ has clipped codes $g_\star, q_\star$. Set

$$\vartheta_- = -\frac{c_0}{2\sqrt{n}}, \quad \vartheta_+ = \frac{c_0}{2\sqrt{n}},$$

and let $P_- := P_{n,\vartheta_-}^{\text{ACE}}$, $P_+ := P_{n,\vartheta_+}^{\text{ACE}}$. Since $n \geq 1$,

$$|\vartheta_-| \leq c_0, \quad |\vartheta_+| \leq c_0, \quad |\vartheta_+ - \vartheta_-| = \frac{c_0}{\sqrt{n}}.$$

Both laws belong to the fixed-code JMS ACE class with clipped codes $g_\star, q_\star$, and the entropy bound gives

$$D_{\text{KL}}(P_-^{\otimes n} \parallel P_+^{\otimes n}) \leq C_0 c_0^2 = K.$$

Applying the two-point lower bound with target values ϑ_- and ϑ_+ yields, for every measurable estimator,

$$\max_{\nu \in \{-,+\}} E_{P_\nu} \left[(\tilde{\theta} - \theta_0(P_\nu))^2 \right] \geq c_K (\vartheta_+ - \vartheta_-)^2 = \frac{c_K c_0^2}{n} = \frac{c_{\text{ACE}}}{n}.$$

Taking the supremum over the fixed-code JMS ACE class and then the infimum over estimators gives

$$\frac{c_{\text{ACE}}}{n} \leq \inf_{\tilde{\theta}} \sup_{\substack{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}: \\ \bar{g}_n = g_\star, \bar{q}_n = q_\star}} E_P \left[\{\tilde{\theta} - \theta_0(P)\}^2 \right].$$

□

Lemma 12 is the lower-bound ingredient for the n^{-1} minimax statements. The proof below builds the fixed-code local path and applies the relative-entropy calculation used in the testing-to-estimation step.

The ACE branch of the proof verifies the same fixed-code path inside $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$, with the treatment innovation and clipped nuisance codes preserved along the path.

C Certified construction and executable correspondence

The statistical statements in [Theorems 3](#) and [4](#) use a sample statistic that is defined through finite arithmetic records. This appendix specifies the conditional executable correspondence theorem for implementations that satisfy the displayed bounded-build contract. Under that contract, the ordinary-sample Borel statistic and the represented-data execution refer to the same primitive records, the same contour bank, and the same rational interval operations. The background is the standard computable-analysis and interval-arithmetic representation of real-number computation [[Weihrauch, 2000](#), [Ko, 1991](#), [Moore, 1966](#)].

The first definition specifies the arithmetic substrate. It records the certified real representation, the interval operations for complex rectangles, the bounded evaluations of elementary functions, and the finite quadrature rule used later on contour integrands.

Definition 16. `[def:certified-contour-arithmetic-substrate]` (Certified contour arithmetic substrate $\text{CCIA}_{\text{build}}$) *The paper-independent bounded-build obligation $\text{CCIA}_{\text{build}}$ is the following finite implementation specification. A certified real record consists of rational closed intervals $I_m = [I_m^-, I_m^+]$, an executable modulus $\mu : \mathbb{Q}_{>0} \rightarrow \mathbb{N}$, and the semantic contract*

$$x \in I_{m+1} \subseteq I_m, \quad I_{\mu(e)}^+ - I_{\mu(e)}^- \leq e.$$

A positive record additionally supplies a rational $p > 0$ such that $p \leq x$. Executable code is restricted to inspecting the interval, modulus, and positivity fields.

For rational intervals $A = [a_-, a_+]$ and $B = [b_-, b_+]$, define

$$A + B = [a_- + b_-, a_+ + b_+], \quad A - B = [a_- - b_+, a_+ - b_-],$$

and

$$AB = [\min\{a_-b_-, a_-b_+, a_+b_-, a_+b_+\}, \max\{a_-b_-, a_-b_+, a_+b_-, a_+b_+\}].$$

A complex rectangle is $A + iB$. Complex addition, subtraction, conjugation, and multiplication are implemented by these endpoint operations and

$$(A + iB)(C + iD) = (AC - BD) + i(AD + BC).$$

For $A = [a_-, a_+]$, set

$$\text{sq}_-(A) = \begin{cases} 0, & a_- \leq 0 \leq a_+, \\ \min\{a_-^2, a_+^2\}, & \text{otherwise,} \end{cases} \quad \text{sq}_+(A) = \max\{a_-^2, a_+^2\}.$$

Then $|A + iB|^2$ is enclosed by the rational interval obtained by adding the corresponding square bounds. Rational bisection between nonnegative rational endpoints gives lower and upper square-root enclosures: after q bisections their width is at most the initial width times 2^{-q} . This supplies certified modulus lower and upper bounds. Division is the guarded operation

$$\frac{A + iB}{C + iD} = (A + iB)(C - iD)(C^2 + D^2)^{-1},$$

executed when the modulus routine has returned a rational lower bound $m > 0$; outward division by the positive rational interval enclosing $C^2 + D^2$ is then sound.

The certified name of π is constructed from Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

For $0 < x < 1$, the first $N + 1$ terms of

$$\arctan x = \sum_{j=0}^N (-1)^j \frac{x^{2j+1}}{2j+1} + R_N(x), \quad |R_N(x)| \leq \frac{x^{2N+3}}{2N+3},$$

give rational nested enclosures after intersecting the first $m+1$ requested enclosures. If $e = p/q > 0$ is in lowest terms, the explicit choice $N = 5q + 5$ makes the propagated Machin remainder smaller than e , giving an effective modulus with finite fuel.

For a rational interval contained in $[-B, B]$, with rational $B \geq 0$, evaluate the Taylor polynomials for $\exp$, $\sin$, and $\cos$ by the preceding interval operations. The Lagrange remainders are bounded respectively by

$$3^{\lceil B \rceil} \frac{B^{N+1}}{(N+1)!}, \quad \frac{B^{N+1}}{(N+1)!}, \quad \frac{B^{N+1}}{(N+1)!}.$$

For a requested rational error $e = p/q > 0$, take

$$N = \lceil 2B \rceil + q + 2\lceil B \rceil + 2.$$

After the index $\lceil 2B \rceil$, successive absolute Taylor terms decrease by at least a factor $1/2$; since $3^{\lceil B \rceil} < 2^{2\lceil B \rceil}$ and $2^{-q} \leq q^{-1} \leq e$, this displayed N gives a sound error at most e . Finite intersections make the output rectangles nested. Hence

$$\exp(x + iy) = \exp(x) \{ \cos y + i \sin y \}$$

has a nested complex-rectangle extension with an effective modulus on every bounded input rectangle. Combining the constructed π name with these trigonometric operations gives, from a positive radius record $\mathfrak{r}$ and integers $N \geq 1$ and $0 \leq q < N$, certified circle-node records for

$$\mathfrak{r} \exp(2\pi i q / N) \quad \text{and} \quad 2\pi i \mathfrak{r} \exp(2\pi i q / N).$$

Finite rectangle sums and products are primitive recursive. If a complex-valued function on $[0, 1]$ has a supplied rational Lipschitz bound L , a uniform mesh with

$$N_{\text{mesh}} = \lceil L/e \rceil + 1$$

has oscillation at most e on each cell. Evaluating every node to a rational error budget divided by the displayed finite number of operations therefore yields sound finite infimum, supremum, and trapezoidal-integral enclosures. Their fuel consists of $N_{\text{mesh}} + 1$ node calls, the displayed Taylor cutoffs, square-root bisection cutoffs, and a finite number of endpoint operations. Absolute value of a real interval is implemented by the square-free endpoint rule implicit in sq_- and sq_+ , and has the same compositional modulus property.

A concrete implementation discharges $\text{CCIA}_{\text{build}}$ exactly when it implements these algorithms, exports their soundness and effective-modulus contracts, and compiles the resulting module. This definition records the explicit bounded-build obligation. Later executable conclusions are conditional on a compiled build satisfying this specification.

Definition 16 is the finite numerical language used by the construction. Later definitions invoke this substrate to represent the contour quantities from **Definition 13** by finite interval computations.

The next definition builds the deterministic contour bank from the primitive certified records attached to a parameter record. The bank radii cover the annulus just outside the zero-localization disk from **Lemma 3**, and the dyadic certificate supplies the modulus scale used by the selector.

Definition 17. `[def:contour-bank-handle]` (Certified contour-bank handle $\mathbf{p}_\star$) *Let p be a parameter record with sample size $n \in \mathbb{N}$, $1 \leq n$, orders $r, k \in \mathbb{N}$ satisfying $k = r + 1$ and $2 \leq r$, exponent $s \in [0, \infty]$ satisfying $r \leq s$, probability level $\gamma \in (1/2, 1)$, constants $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma > 0$, and sequences $\varepsilon_1, \varepsilon_2 : \mathbb{N} \rightarrow \mathbb{R}$ satisfying $\varepsilon_\ell(t) \geq 0$ for $t \geq 1$ and $\varepsilon_\ell(j) \leq \varepsilon_\ell(i)$ whenever $1 \leq i \leq j$, for $\ell = 1, 2$. Let $\text{Name}(x)$ denote a supplied certified-real record with value x , and let $\text{Name}_+(x)$ denote such a record together with a rational lower-bound witness $\ell > 0$ satisfying $\ell \leq x$. As part of the experiment convention, fix the supplied record*

$$\mathbf{p}_\star = (k_\star, \mathfrak{d}_\star, \mathfrak{p}_{\eta,\star}, \mathfrak{r}_{0,\star}, \mathfrak{r}_{1,\star}),$$

where $k_\star$ is exact with $k_\star = k$ and

$$\mathfrak{d}_\star : \text{Name}_+(\delta), \quad \mathfrak{p}_{\eta,\star} : \text{Name}_+(\psi_\eta), \quad \mathfrak{r}_{0,\star} : \text{Name}_+(R_0), \quad \mathfrak{r}_{1,\star} : \text{Name}_+(R_1).$$

The value contracts are

$$R_0 = (2^{k+4}k!)^{1/(k-2)}(\psi_\eta^2/\delta)^{1/(k-2)}, \quad R_1 = R_0 + 1.$$

Writing $A_k = (2^{k+4}k!)^{1/(k-2)}$ for the numerical factor, this reads $R_0 = A_k(\psi_\eta^2/\delta)^{1/(k-2)}$. The record $\mathbf{p}_\star$ is the primitive certified input shared by the bank and the estimator for this parameter record. The bank data assembled below from p and $\mathbf{p}_\star$ is written B .

Invoke the supplied moduli of $\mathfrak{p}_{\eta,\star}$ and $\mathfrak{r}_{1,\star}$ at error one. If the returned intervals have rational upper endpoints u_ψ and u_R , define

$$U_\psi = \lceil \max\{1, u_\psi\} \rceil, \quad U_R = \lceil \max\{1, u_R\} \rceil, \quad N_{\text{cert}} = 4U_\psi^2(U_R + 2)^3.$$

Set

$$m_\star = N_{\text{cert}} + 3, \quad d_\star = 2^{-m_\star}, \quad h_\star = d_\star/3, \quad J_{\text{base}} = 2^{m_\star-1}.$$

The bank therefore contains

$$J = J_{\text{base}} + 1$$

circles, indexed by $j \in \{0, \dots, J_{\text{base}}\}$. For each $j = 0, \dots, J_{\text{base}}$, use certified addition on the fixed record $\mathfrak{r}_{0,\star}$ and the rational point record for $1/4 + j2^{-m_\star}$ to form

$$\rho_j = \mathfrak{r}_{0,\star} + \text{Name}(1/4 + j2^{-m_\star}).$$

The associated real-valued radius field is

$$\rho_j = (\rho_j). \text{value} = R_0 + 1/4 + j2^{-m_\star}.$$

Define the integers

$$e_\star = 8U_R U_\psi^2 (U_R + 1)^2, \quad d_\star^{\text{den}} = 3(2U_R + 1),$$

$$u_\star = 2e_\star + N_{\text{cert}}(m_\star + d_\star^{\text{den}}).$$

Define the rational dyadic certificate and its real value by

$$a_{\star, \mathbb{Q}} = 2^{-u_\star} > 0, \quad a_\star = (a_{\star, \mathbb{Q}} : \mathbb{R}).$$

Every output datum is obtained by the displayed finite modulus calls, rational endpoint operations, certified additions with rational point records, and primitive recursion over the displayed integer bounds. The construction is a finite certified procedure in the supplied record $\mathbf{p}_\star$.

Definition 17 is the shared primitive input for the bank and estimator. The positive dyadic certificate $a_\star$ is the modulus scale used by the selection and evaluation guards.

The estimator definition now combines the arithmetic substrate and the bank with the observed sample. It has two layers: an ordinary real-sample map, which defines the Borel statistic used in risk statements, and a represented-data map, which records how a compiled adapter returns the same finite-rational trace on certified inputs.

Definition 18. `[def:adaptive-contour-estimator]` (Adaptive contour statistic $\widehat{\theta}_{\text{spec}, n}$) *The construction has a mathematical layer and a represented-data layer, both relative to exactly the same experiment-wide parameter and primitive records.*

Let $\mathcal{X}$ be a measurable covariate space. Fix a parameter record p consisting of a positive sample size n , natural numbers r and k with $k = r + 1$ and $2 \leq r$, an exponent $s \in [0, \infty]$ with $r \leq s$, a probability level $\gamma \in (1/2, 1)$, strictly positive constants $C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, \delta, \sigma$, and two sequences $\varepsilon_{1n}, \varepsilon_{2n} : \mathbb{N} \rightarrow \mathbb{R}$ that are nonnegative on positive indices and nonincreasing there: for $\ell \geq 1$, $0 \leq \varepsilon_{1n}(\ell)$ and $0 \leq \varepsilon_{2n}(\ell)$, and whenever $1 \leq a \leq b$, $\varepsilon_{1n}(b) \leq \varepsilon_{1n}(a)$ and $\varepsilon_{2n}(b) \leq \varepsilon_{2n}(a)$. Fix a deterministic treatment-regression code sequence; the estimator records measurability of its current n th code as its public-domain proposition.

Fix the record $\mathbf{p}_\star$ from the contour-bank handle: its natural-name field equals k , its positive certified-real fields represent δ , ψ_η , R_0 , and R_1 , and its value contracts are

$$R_0 = A_k \left(\frac{\psi_\eta^2}{\delta} \right)^{(k-2)^{-1}}, \quad R_1 = R_0 + 1.$$

Once for all, fix one actual range record

$$\mathbf{c}_{\theta, \star} : \text{Name}_+(C_\theta).$$

These records are fixed for the estimator before the law or sample varies, and they remain the primitive names used throughout the construction.

At the mathematical layer, for an ordinary real observation coordinate set

$$\mathcal{I}_m(x) = [2^{-m} \lfloor 2^m x \rfloor, 2^{-m} \{ \lfloor 2^m x \rfloor + 1 \}], \quad m \geq 1.$$

Canonicalize T_i , Y_i , and $\bar{g}_n(X_i)$ through these floor-dyadic records, where $\bar{g}_n$ is the projection of the current treatment-regression code onto $[-C_g, C_g]$. Run the finite rational procedure below with those observation records and with the already fixed, unchanged records $\mathbf{p}_\star$ and $\mathbf{c}_{\theta, \star}$. The ordinary program is the resulting finite-rational result, the full trace is its trace field, and the ordinary value is the projection of its raw rational output onto the fixed certified range $[-C_\theta, C_\theta]$. This defines $\widehat{\theta}_{\text{spec}, n}$ as a total map on ordinary real samples relative to the fixed full primitive record. The canonical observation digits are part of this mathematical interpretation.

At the represented-data layer, callers supply the observation record

$$\tilde{\mathbf{s}} = ((\mathbf{t}_i, \mathbf{y}_i, \mathbf{g}_i)_{i=1}^n),$$

where $\mathbf{t}_i$, $\mathbf{y}_i$, and $\mathbf{g}_i$ are supplied certified digital records for T_i , Y_i , and $\bar{g}_n(X_i)$. The represented program forms the full internal input

$$\mathbf{s}_\star = ((\mathbf{t}_i, \mathbf{y}_i, \mathbf{g}_i)_{i=1}^n, \mathbf{p}_\star, \mathbf{c}_{\theta, \star})$$

by adjoining the displayed fixed primitive components. For each supplied compiled bounded spectral adapter, consisting of the build together with pilot-boundary, winding-quadrature, and evaluation-quadrature entry points satisfying their exact correspondence contracts with the built finite programs, the represented program applies those entry points to this full internal input. Its output-correspondence proposition asserts that, for every ordinary data set, the represented result on the canonical represented input equals the ordinary finite-rational result, the two traces are equal, and the represented output name certifies the ordinary clipped value.

Invoke the contour-bank handle on $\mathbf{p}_\star$, and form $\mathbf{z}_i = \mathbf{t}_i - \mathbf{g}_i$ by certified subtraction. At every certified circle node, use the substrate's complex rectangle operations and nested complex-exponential extension to construct, by finite sums, enclosures of

$$\hat{F}_{a,n}^{(p)}(z) = |I_a|^{-1} \sum_{i \in I_a} Z_{n,i}^p e^{z Z_{n,i}}, \quad p \in \{0, 1, 2\},$$

and

$$\hat{G}_{a,n}^{(p)}(z) = |I_a|^{-1} \sum_{i \in I_a} Y_i Z_{n,i}^p e^{z Z_{n,i}}, \quad p \in \{0, 1\}.$$

Every node precision is obtained by applying the compositional modulus to a rational error budget.

For explicit mesh fuel, refine each observation record at error one and take the resulting rational upper magnitude bounds $U_{Z,i}$, $U_{Y,i}$, and $U_{\rho,j}$. Use $e^x \leq 3^{\lceil x \rceil}$ for $x \geq 0$ and $\pi < 4$ to turn the finite empirical derivative sums into rational bounds for all functions below. With $z_j(t) = \rho_j e^{2\pi i t}$, put

$$W_{0j}(t) = \rho_j e^{2\pi i t} \frac{\hat{F}'_{0,n}(z_j(t))}{\hat{F}_{0,n}(z_j(t))},$$

and

$$V_j(t) = \rho_j e^{2\pi i t} \frac{\hat{G}_{1,n}(z_j(t))}{\hat{F}_{1,n}(z_j(t))}.$$

The integrand V_j is NOT normalized at this stage: the certified zero count N_j divides the finished mesh enclosure, not the integrand. If a rational lower modulus certificate is $m > 0$, use

$$|W'_{0j}(t)| \leq 64 \left[\rho_j \frac{\sup |\hat{F}'_{0,n}|}{m} + \rho_j^2 \left\{ \frac{\sup |\hat{F}''_{0,n}|}{m} + \frac{\sup |\hat{F}'_{0,n}|^2}{m^2} \right\} \right],$$

and

$$|V'_j(t)| \leq 64 \left[\rho_j \frac{\sup |\hat{G}_{1,n}|}{m} + \rho_j^2 \left\{ \frac{\sup |\hat{G}'_{1,n}|}{m} + \frac{\sup |\hat{G}_{1,n}| \sup |\hat{F}'_{1,n}|}{m^2} \right\} \right].$$

Both displays carry the rational constant 64 rather than the analytically tight factor coming from dz_j/dt : the whole mesh schedule is evaluated in $\mathbb{Q}$, so the certified Lipschitz constant must itself

be rational, and enlarging it only refines the mesh. For a rational tolerance $e > 0$ and any resulting rational bound L , take the mesh size $\lceil L/e \rceil + 1$, and allocate node error by the finite number of operations. Thus every precision and mesh has displayed finite fuel.

On the pilot fold, compute for every j a sound interval of width at most $a_*/64$ enclosing $\inf_{C_j} |\widehat{F}_{0,n}|$. If its lower endpoint m_j is positive, guarded division and finite quadrature enclose

$$\int_0^1 W_{0j}(t) dt = \frac{1}{2\pi i} \oint_{C_j} \frac{\widehat{F}'_{0,n}(z)}{\widehat{F}_{0,n}(z)} dz$$

in a complex rectangle whose coordinate widths are strictly below $1/4$. That integral is the winding number of $\widehat{F}_{0,n}$ along C_j ; write

$$N_j = \frac{1}{2\pi i} \oint_{C_j} \frac{\widehat{F}'_{0,n}(z)}{\widehat{F}_{0,n}(z)} dz,$$

which by the argument principle counts, with multiplicity, the zeros of $\widehat{F}_{0,n}$ enclosed by C_j . Record N_j exactly when the real interval contains a unique nonnegative integer and the imaginary interval contains zero; otherwise reject the circle. A circle is admissible exactly when $m_j \geq a_*/2$ and $N_j \geq 1$. Select an admissible circle having the greatest rational lower endpoint, with smallest-index tie-break, and return the certified zero name if the admissible set is empty.

On the evaluation fold, compute the analogous modulus enclosure and return the certified zero name if its lower endpoint is below $a_*/4$. Otherwise guarded division and finite quadrature enclose the unnormalized integral

$$\int_0^1 V_j(t) dt = \frac{1}{2\pi i} \oint_{C_j} \frac{\widehat{G}_{1,n}(z)}{\widehat{F}_{1,n}(z)} dz,$$

and dividing that enclosure by N_j afterwards gives a rectangle of coordinate radius at most $1/n$ enclosing

$$\frac{1}{N_j 2\pi i} \oint_{C_j} \frac{\widehat{G}_{1,n}(z)}{\widehat{F}_{1,n}(z)} dz.$$

Let $y \in \mathbb{Q}$ be the rational midpoint of the real interval of that normalized rectangle, represented by its rational point name. Projection uses certified addition, subtraction, multiplication by $1/2$, and the interval absolute-value operation to construct a certified name whose value is

$$\text{clip}_{C_\theta}(y) = \frac{|y + C_\theta| - |y - C_\theta|}{2} = \Pi_{[-C_\theta, C_\theta]}(y).$$

This certified-real name is the program output.

An execution trace is the finite list of input-modulus calls, returned precisions, Taylor and square-root cutoffs, mesh sizes, rational rectangle operations, guarded divisions, rational endpoint comparisons, fallback decisions, and the tie-break. Identical ordinary samples with the same parameter, treatment-regression code sequence, and fixed primitive records have identical ordinary traces and output records; identical represented observation records with the same fixed primitive records and the same compiled bounded spectral adapter have identical represented traces and output records. The value represented by the ordinary output is, by definition, the ordinary-sample statistic evaluated with the same fixed primitive records. The procedure uses $\bar{g}_n$. The ordinary program, ordinary value, and full trace are defined directly by the canonical represented input and the finite rational adapter; the represented output correspondence is the stated contract for each supplied compiled bounded spectral adapter.

[Definition 18](#) is the exact statistic used in the main risk theorem. Its ordinary-sample branch is the Borel statistic in the risk displays, and its represented-data branch is the conditional correspondence target for compiled adapters satisfying the finite-arithmetic contract.

The last statement is the deterministic perturbation bound used after a circle has been selected. It compares the certified statistic with a population contour target on the selected circle under uniform denominator and numerator errors.

Lemma 13. [lem:selected-contour-perturbation-bound] (Selected contour perturbation bound) *Fix p as in [Definition 1](#), the primitive certified record $\mathbf{p}_\star$ of [Definition 17](#), the range certified record $\mathbf{c}_{\theta,\star}$ of [Definition 18](#), a treatment-regression code sequence, and observations $O_1, \dots, O_n$ with $O = (X, T, Y)$ as in [Definition 2](#). Let B be the contour bank generated from p and $\mathbf{p}_\star$, with radii ρ_j and lower-modulus certificate $a_\star$, and let $\hat{\theta}_{\text{spec},n}$ be the canonical certified statistic of [Definition 18](#). Let $F, G : \mathbb{C} \rightarrow \mathbb{C}$, let $d, e, C_G, \mu, \nu \in \mathbb{R}$, and let $\vartheta \in \mathbb{R}$ satisfy $|\vartheta| \leq C_\theta$. Assume the following conditions.*

- **(Selector event.)** *The canonical selector event holds: $d \geq 0$, $e \geq 0$, $C_G \geq 0$, $\mu > 0$, $\nu > 0$,*

$$\mu = \frac{31a_\star}{64}, \quad \nu = \frac{23a_\star}{64}, \quad d \leq \mu - \nu,$$

and, for the canonical represented input built from the sample, there are a bank index j and an integer $N \geq 1$ such that the selector chooses C_j , the pilot pass at C_j decodes to N , the evaluation-fold modulus lower endpoint at C_j is at least $a_\star/4$, and every $z \in C_j$ satisfies

$$\mu \leq |F(z)|, \quad |G(z)| \leq C_G, \quad |\hat{F}_{1,n}(z) - F(z)| \leq d, \quad |\hat{G}_{1,n}(z) - G(z)| \leq e.$$

- **(Selected-circle integrability.)** *For every bank index j selected by the canonical program, the functions*

$$z \mapsto \frac{\hat{G}_{1,n}(z)}{\hat{F}_{1,n}(z)} \quad \text{and} \quad z \mapsto \frac{G(z)}{F(z)}$$

are contour integrable on C_j .

- **(Selected-circle identification.)** *For every bank index j selected by the canonical program and every decoded pilot integer N , the target is identified by*

$$\vartheta = \Re \left[\{N 2\pi i\}^{-1} \oint_{C_j} \frac{G(z)}{F(z)} dz \right].$$

Then

$$\left| \hat{\theta}_{\text{spec},n} - \vartheta \right| \leq R_1 \max\{\nu^{-1}, C_G(\mu\nu)^{-1}\}(d + e) + \frac{1}{\max\{n, 1\}}.$$

Proof of [Lemma 13](#). On the selector event, choose the selected bank index j and decoded integer $N \geq 1$. Write C_j for the selected circle and let

$$\hat{T}_j = \Re \left[\{N 2\pi i\}^{-1} \oint_{C_j} \frac{\hat{G}_{1,n}(z)}{\hat{F}_{1,n}(z)} dz \right], \quad T_j = \Re \left[\{N 2\pi i\}^{-1} \oint_{C_j} \frac{G(z)}{F(z)} dz \right].$$

The selected-contour identification hypothesis gives $T_j = \vartheta$.

For each $z \in C_j$, the selector assumptions give

$$\mu \leq |F(z)|, \quad |G(z)| \leq C_G, \quad |\hat{F}_{1,n}(z) - F(z)| \leq d, \quad |\hat{G}_{1,n}(z) - G(z)| \leq e.$$

Since $d \leq \mu - \nu$, the reverse triangle inequality gives

$$|\hat{F}_{1,n}(z)| \geq |F(z)| - |\hat{F}_{1,n}(z) - F(z)| \geq \mu - d \geq \nu.$$

Thus the pointwise ratio perturbation bound is

$$\left| \frac{\hat{G}_{1,n}(z)}{\hat{F}_{1,n}(z)} - \frac{G(z)}{F(z)} \right| \leq \frac{e}{\nu} + \frac{C_G d}{\mu \nu}.$$

Indeed,

$$\frac{\hat{G}_{1,n}}{\hat{F}_{1,n}} - \frac{G}{F} = \frac{\hat{G}_{1,n} - G}{\hat{F}_{1,n}} + \frac{G(F - \hat{F}_{1,n})}{\hat{F}_{1,n}F},$$

and the two terms are bounded by e/ν and $C_G d/(\mu \nu)$, respectively.

The two selected-circle integrability assumptions allow integration of this pointwise bound around C_j , so

$$\left| \oint_{C_j} \frac{\hat{G}_{1,n}(z)}{\hat{F}_{1,n}(z)} dz - \oint_{C_j} \frac{G(z)}{F(z)} dz \right| \leq 2\pi \rho_j \left(\frac{e}{\nu} + \frac{C_G d}{\mu \nu} \right).$$

After multiplication by $\{N2\pi i\}^{-1}$ and taking real parts,

$$|\hat{T}_j - \vartheta| \leq \frac{\rho_j}{N} \left(\frac{e}{\nu} + \frac{C_G d}{\mu \nu} \right).$$

The certified evaluation rectangle used by the statistic contains $\hat{T}_j$, and its real-coordinate width is at most $2/\max\{n, 1\}$. If y is the rational midpoint used before clipping, then

$$|y - \hat{T}_j| \leq \frac{1}{\max\{n, 1\}}.$$

Because $|\vartheta| \leq C_\theta$, clipping to $[-C_\theta, C_\theta]$ is contractive relative to ϑ . Hence

$$|\hat{\theta}_{\text{spec},n} - \vartheta| \leq |y - \vartheta| \leq |y - \hat{T}_j| + |\hat{T}_j - \vartheta|.$$

Finally, $\rho_j < R_1$ by the bank geometry in [Definition 17](#) and $N \geq 1$. Since d, e, C_G, μ, ν have the stated signs,

$$\frac{\rho_j}{N} \left(\frac{e}{\nu} + \frac{C_G d}{\mu \nu} \right) \leq R_1 \max\{\nu^{-1}, C_G(\mu \nu)^{-1}\}(d + e).$$

Combining the last three displays proves the asserted inequality. $\square$

[Lemma 13](#) is the deterministic accuracy link between the finite construction and the statistical risk proof. The proof below combines the selected-circle event with the population contour identity in [Theorem 2](#).

D Verification scope and crosswalk

This appendix records how the reader-facing statistical claims in the paper align with the formal statements displayed in the main text and technical appendices. The crosswalk is organized by mathematical role: model primitives and law classes, analytic identification, fixed-separation estimation, comparison results, local benchmarks, and certified execution.

The model primitives are fixed by [Definitions 1](#) and [2](#). Sampling and structural restrictions are the assumptions in [Assumptions 1](#) to [16](#). The law classes and risk criteria used in the statistical comparisons are the objects in [Definitions 3](#) to [11](#).

The analytic identification chain is the sequence [Definitions 12](#) and [13](#), [Theorems 1](#) and [2](#), and [Lemma 1](#). These statements fix the treatment-innovation transform, the observable residual transforms, and the contour ratio that identifies the partially linear coefficient under the stated zero-free and positive-count conditions.

The fixed-separation estimation result is anchored by [Lemma 3](#) and [Theorems 3](#) and [4](#). These statements provide the zero-localization radius, the finite-bank contour statistic, the n^{-1} minimax mean-squared-error scale on the fixed-code non-Gaussian class, the corresponding generalized-quantile bound, and the Gaussian-class diagnostic.

The comparison with ACE is recorded in [Propositions 1](#) and [2](#) and [Lemma 5](#). The published order- r ACE guarantee enters through the cited Jin–Mackey–Syrkanis result [[Jin et al., 2025](#)]; the displayed proposition states the exact imported bound and the class relations used to compare it with the contour guarantee.

The explicit mixture and local benchmark statements are [Definition 14](#), [Proposition 3](#), [Lemma 7](#), and [Theorem 5](#). The future-work formulation in [Definition 15](#) and [Remark 1](#) records the sample-level contour information and the triangular local-to-Gaussian adaptation target.

The certified execution layer is specified by [Definitions 16](#) to [18](#). These statements connect the ordinary Borel statistic used in the statistical risk bounds with the finite represented-data construction based on the same primitive records.

Verification scope. The theorem statements and proofs corresponding to the displayed formal environments are machine-checked in Lean 4 under their stated assumptions and citation interfaces. The ACE-alignment theorem in [Proposition 1](#) verifies the class reduction, the application of the imported bound, and the comparison algebra conditional on the Jin–Mackey–Syrkanis conclusion cited in its theorem-local footnote.

E Proofs of the main results

Proof of [Theorem 1](#). Put

$$D_n(X) = g_0(X) - \bar{g}_n(X), \quad b_n(X) = q_0(X) - \theta_0 D_n(X),$$

as in [Definition 7](#), and write

$$J(w) = J_{z_0, \ell}(w) = w^{\ell-1} e^{z_0 w}, \quad w \in \mathbb{C},$$

as in [Definition 12](#). From the definitions of η and Z_n in [Definitions 2](#) and [13](#),

$$Z_n = T - \bar{g}_n(X) = \eta + D_n(X).$$

The definitions of ξ in [Definition 2](#) and b_n above also give, by algebra,

$$Y = \theta_0 Z_n + b_n(X) + \xi.$$

Membership in $\mathcal{P}_{\text{NG},n}$, together with the clipping convention in [Definition 2](#), supplies the integrability and boundedness used below. Pulled back to the observation space, [Definition 3](#) gives

$$|g_0(X)| \leq C_g, \quad |q_0(X)| \leq C_q, \quad |\theta_0| \leq C_\theta \quad P\text{-a.s.},$$

and the clipping gives $|\bar{g}_n(X)| \leq C_g$. Hence

$$|D_n(X)| \leq 2C_g, \quad |b_n(X)| \leq C_q + C_\theta(2C_g) \quad P\text{-a.s.}$$

The treatment and outcome sub-Gaussian clauses in [Definition 3](#) imply real exponential moments of every order for η and ξ ; the displayed bound on D_n then gives the corresponding real exponential moments for $Z_n = \eta + D_n(X)$. These facts justify the differentiated transforms, conditional expectations, and products appearing below.

1. Since M is analytic at z_0 and has analytic order $\ell \geq 1$ there,

$$M^{(\alpha)}(z_0) = 0 \quad \text{for every } 0 \leq \alpha < \ell.$$

Moreover, for each such α ,

$$M^{(\alpha)}(z_0) = E_P[\eta^\alpha e^{z_0 \eta}],$$

by differentiating the moment-generating function under the expectation. Hence, for any $d \in \mathbb{R}$,

$$\begin{aligned} E_P[J(\eta + d)] &= E_P[(\eta + d)^{\ell-1} e^{z_0(\eta+d)}] \\ &= e^{z_0 d} \sum_{\alpha=0}^{\ell-1} \binom{\ell-1}{\alpha} d^{\ell-1-\alpha} E_P[\eta^\alpha e^{z_0 \eta}] \\ &= e^{z_0 d} \sum_{\alpha=0}^{\ell-1} \binom{\ell-1}{\alpha} d^{\ell-1-\alpha} M^{(\alpha)}(z_0) = 0. \end{aligned}$$

This is the shifted annihilation identity.

2. We next show the conditional identity. Let $\mathcal{A}_X$ be any measurable covariate event. Since $Z_n = \eta + D_n(X)$, the binomial expansion gives

$$\begin{aligned} &\int_{\{X \in \mathcal{A}_X\}} J(Z_n) dP \\ &= \sum_{\alpha=0}^{\ell-1} \binom{\ell-1}{\alpha} \int \eta^\alpha e^{z_0 \eta} D_n(X)^{\ell-1-\alpha} e^{z_0 D_n(X)} \mathbf{1}_{\mathcal{A}_X}(X) dP. \end{aligned}$$

The treatment-noise independence in [Definition 3](#) factors each summand as

$$\binom{\ell-1}{\alpha} E_P[\eta^\alpha e^{z_0 \eta}] E_P[D_n(X)^{\ell-1-\alpha} e^{z_0 D_n(X)} \mathbf{1}_{\mathcal{A}_X}(X)],$$

and the first factor is $M^{(\alpha)}(z_0) = 0$. Thus the integral over every covariate event is zero, which characterizes

$$E_P[J(Z_n) \mid X] = 0 \quad P\text{-a.s.}$$

3. Define, for $z \in \mathbb{C}$,

$$F_n(z) = E_P[e^{zZ_n}].$$

Since $Z_n = \eta + D_n(X)$, the same independence gives the factorization

$$F_n(z) = E_P[e^{z\eta}] E_P[e^{zD_n(X)}] = M(z)H_n(z),$$

with H_n as in [Definition 13](#). Also,

$$E_P[Z_n J(Z_n)] = E_P[Z_n^\ell e^{z_0 Z_n}] = F_n^{(\ell)}(z_0).$$

Differentiating $F_n = MH_n$ ℓ times at z_0 yields

$$F_n^{(\ell)}(z_0) = \sum_{\alpha=0}^{\ell} \binom{\ell}{\alpha} M^{(\alpha)}(z_0) H_n^{(\ell-\alpha)}(z_0).$$

All terms with $\alpha < \ell$ vanish by the analytic-order calculation above, and the remaining term is

$$F_n^{(\ell)}(z_0) = M^{(\ell)}(z_0) H_n(z_0).$$

Therefore

$$E_P[Z_n J(Z_n)] = M^{(\ell)}(z_0) H_n(z_0).$$

4. Finally, using the algebraic decomposition $Y = \theta_0 Z_n + b_n(X) + \xi$,

$$E_P[Y J(Z_n)] = \theta_0 E_P[Z_n J(Z_n)] + E_P[b_n(X) J(Z_n)] + E_P[\xi J(Z_n)].$$

The second term is zero because $b_n(X)$ is covariate-measurable and the conditional identity above gives

$$E_P[b_n(X) J(Z_n)] = E_P[b_n(X) E_P[J(Z_n) \mid X]] = 0.$$

The third term is zero because $J(Z_n)$ is measurable with respect to (X, T) , while [Definition 3](#) gives $E_P[\xi \mid X, T] = 0$:

$$E_P[\xi J(Z_n)] = E_P[J(Z_n) E_P[\xi \mid X, T]] = 0.$$

Thus

$$E_P[Y J(Z_n)] = \theta_0 E_P[Z_n J(Z_n)].$$

Whenever $E_P[Z_n J(Z_n)] \neq 0$, division gives

$$\theta_0 = \frac{E_P[Y J(Z_n)]}{E_P[Z_n J(Z_n)]}.$$

□

Proof of [Lemma 1](#). Fix $z \in \mathbb{C}$. The definitions give

$$Z_n = T - \bar{g}_n(X) = \eta + \{g_0(X) - \bar{g}_n(X)\} = \eta + D_n(X).$$

Moreover, [Definitions 2](#) and [3](#) give the pointwise clipping bound $|\bar{g}_n(X)| \leq C_g$ and the treatment-regression range bound $|g_0(X)| \leq C_g$ almost surely, hence

$$|D_n(X)| = |g_0(X) - \bar{g}_n(X)| \leq |g_0(X)| + |\bar{g}_n(X)| \leq 2C_g \quad P\text{-a.s.}$$

Together with the Luxemburg treatment-tail condition for η , this bound gives the real exponential moments needed for Z_n and for the transforms below. Since $D_n(X)$ is a function of X and η is independent of X under [Definition 3](#),

$$\begin{aligned} F_n(z) &= E_P \exp\{z(\eta + D_n(X))\} \\ &= E_P [\exp(z\eta) \exp\{zD_n(X)\}] \\ &= E_P \exp(z\eta) E_P \exp\{zD_n(X)\} = M(z)H_n(z). \end{aligned}$$

For the second identity, first note the pointwise residual decomposition

$$Y = \theta_0 Z_n + b_n(X) + \xi,$$

because $b_n(X) = q_0(X) - \theta_0 D_n(X)$, $Z_n = \eta + D_n(X)$, and $\xi = Y - q_0(X) - \theta_0 \eta$. The preceding exponential-moment envelope for Z_n also gives differentiation under the expectation:

$$F'_n(z) = E_P [Z_n e^{zZ_n}].$$

Thus

$$\begin{aligned} G_n(z) &= E_P [Y e^{zZ_n}] \\ &= \theta_0 E_P [Z_n e^{zZ_n}] + E_P [b_n(X) e^{zZ_n}] + E_P [\xi e^{zZ_n}]. \end{aligned}$$

The contamination weight is bounded almost surely by

$$\begin{aligned} |b_n(X)| &= |q_0(X) - \theta_0 D_n(X)| \\ &\leq |q_0(X)| + |\theta_0| |D_n(X)| \quad P\text{-a.s.} \\ &\leq C_q + C_\theta(2C_g), \end{aligned}$$

where the last line uses the outcome-regression and coefficient range bounds in [Definition 3](#) together with the displayed bound on D_n . This boundedness and the same exponential envelope for Z_n give integrability of the contamination term.

The weight e^{zZ_n} is measurable with respect to (X, T) . Writing its real and imaginary parts separately and using $E_P[\xi \mid X, T] = 0$, one obtains

$$E_P [\xi e^{zZ_n}] = 0.$$

Finally, independence of η and X factors the contamination term:

$$\begin{aligned} E_P [b_n(X) e^{zZ_n}] &= E_P [b_n(X) e^{zD_n(X)} e^{z\eta}] \\ &= E_P e^{z\eta} E_P [b_n(X) e^{zD_n(X)}] = M(z)B_n(z). \end{aligned}$$

Substituting the last three displays and the derivative identity gives

$$G_n(z) = \theta_0 F'_n(z) + M(z)B_n(z).$$

Since z was arbitrary, both identities hold for every $z \in \mathbb{C}$. □

Proof of [Theorem 2](#). Write ρ_j for the radius of the selected positively oriented bank circle $C_j = \{z \in \mathbb{C} : |z| = \rho_j\}$. The bank construction gives $\rho_j > 0$. Put

$$D_n(X) = g_0(X) - \bar{g}_n(X), \quad b_n(X) = q_0(X) - \theta_0(P)D_n(X),$$

and

$$B_n(z) = E_P \left[b_n(X) e^{z D_n(X)} \right], \quad z \in \mathbb{C}.$$

By [Lemma 10](#),

$$|D_n(X)| \leq 2C_g \quad P\text{-a.s.}$$

The clipping defining $\bar{g}_n$, the range conditions in $\mathcal{P}_{\text{NG},n}$, and $|\theta_0(P)| \leq C_\theta$ give

$$|b_n(X)| \leq |q_0(X)| + |\theta_0(P)| |D_n(X)| \leq C_q + 2C_\theta C_g \quad P\text{-a.s.}$$

Consequently H_n and B_n are complex differentiable at every point. Indeed, on the unit neighborhood of any $z_0 \in \mathbb{C}$,

$$|D_n(X) e^{z D_n(X)}| \leq 2C_g \exp\{(|z_0| + 1)2C_g\},$$

and

$$|b_n(X) D_n(X) e^{z D_n(X)}| \leq (C_q + 2C_\theta C_g)(2C_g) \exp\{(|z_0| + 1)2C_g\},$$

so dominated differentiation under the integral applies to the weighted transforms. Separately, the sub-Gaussian exponential-moment condition for the treatment innovation η , together with $Z_n = \eta + D_n(X)$ and the almost-sure bound on D_n , gives $E_P e^{t Z_n} < \infty$ for every real t . The standard moment-generating-function analyticity criterion therefore gives analyticity of F_n on a neighborhood of the closed disk $\{|z| \leq \rho_j\}$.

On the closed disk $\{|z| \leq \rho_j\}$, the nuisance zero-freeness assumption gives $H_n(z) \neq 0$. Hence the quotient B_n/H_n is well-defined on the closed disk, differentiable in the open disk, and continuous on the closed disk. Cauchy's theorem on the circle gives

$$\oint_{C_j} \frac{B_n(z)}{H_n(z)} dz = 0.$$

For $z \in C_j$, [Lemma 1](#) gives

$$F_n(z) = M(z)H_n(z), \quad G_n(z) = \theta_0(P)F'_n(z) + M(z)B_n(z),$$

where $M(z) = E_P[e^{z\eta}]$. Since $F_n(z) \neq 0$ and $H_n(z) \neq 0$ on C_j , also $M(z) \neq 0$ on C_j . Therefore, pointwise for $z \in C_j$,

$$\frac{G_n(z)}{F_n(z)} = \theta_0(P) \frac{F'_n(z)}{F_n(z)} + \frac{B_n(z)}{H_n(z)}.$$

Integrating around C_j and using the Cauchy identity for the quotient B_n/H_n yields

$$\oint_{C_j} \frac{G_n(z)}{F_n(z)} dz = \theta_0(P) \oint_{C_j} \frac{F'_n(z)}{F_n(z)} dz.$$

The function F_n is analytic near the closed disk and zero-free on C_j , so the argument principle gives

$$\frac{1}{2\pi i} \oint_{C_j} \frac{F'_n(z)}{F_n(z)} dz = N_{C_j}(F_n).$$

Thus

$$\oint_{C_j} \frac{G_n(z)}{F_n(z)} dz = \theta_0(P) N_{C_j}(F_n) 2\pi i.$$

The positive-count assumption gives $N_{C_j}(F_n) \geq 1$, so $N_{C_j}(F_n) 2\pi i \neq 0$. Dividing by this nonzero factor gives

$$\theta_0(P) = \{N_{C_j}(F_n) 2\pi i\}^{-1} \oint_{C_j} \frac{G_n(z)}{F_n(z)} dz.$$

By [Definition 13](#), the right-hand side is $\Theta_{C_j}(F_n, G_n)$. Therefore

$$(\theta_0(P) : \mathbb{C}) = \Theta_{C_j}(F_n, G_n) = \{N_{C_j}(F_n) 2\pi i\}^{-1} \oint_{C_j} \frac{G_n(z)}{F_n(z)} dz.$$

□

Proof of [Lemma 2](#). Fix $t \in \mathbb{R}$ and put

$$\lambda = t\psi, \quad v = \frac{W}{\psi}.$$

The square-exponential envelope first gives integrability of e^{tW} : for every outcome point,

$$tW \leq \frac{W^2}{\psi^2} + \frac{t^2\psi^2}{4},$$

because $(W/\psi - t\psi/2)^2 \geq 0$. Hence e^{tW} is dominated by $e^{t^2\psi^2/4} e^{W^2/\psi^2}$.

Consider first the case $|\lambda| \leq 1$. For every real u ,

$$e^u \leq 1 + u + u^2 e^{|u|}.$$

Indeed, when $|u| \leq 1$, the Taylor-remainder bound $|e^u - 1 - u| \leq u^2$ gives $e^u - 1 - u \leq u^2 \leq u^2 e^{|u|}$. When $|u| > 1$ and $u \geq 0$, one has $u > 1$, so $e^u \leq u^2 e^{|u|} \leq 1 + u + u^2 e^{|u|}$. When $|u| > 1$ and $u < 0$, one has $u \leq -1$, whence $e^u \leq 1 \leq 1 + u + u^2 \leq 1 + u + u^2 e^{|u|}$. With $u = \lambda v$,

$$|\lambda v| \leq \frac{\lambda^2 + v^2}{2} \leq \frac{1 + v^2}{2}.$$

Also,

$$v^2 e^{(1+v^2)/2} = e^{1/2} v^2 e^{v^2/2} \leq e^{1/2} e^{v^2} \leq 2e^{v^2},$$

where the two inequalities use $v^2 \leq e^{v^2/2}$ and $e^{1/2} \leq 2$. Combining these estimates with monotonicity of the exponential gives

$$(\lambda v)^2 e^{|\lambda v|} \leq 2\lambda^2 e^{v^2}.$$

Therefore

$$e^{tW} = e^{\lambda v} \leq 1 + tW + 2\lambda^2 \exp(W^2/\psi^2).$$

Integrating and using $E_P W = 0$ and $E_P \exp(W^2/\psi^2) \leq 2$,

$$E_P e^{tW} \leq 1 + 4\lambda^2 \leq \exp(4\lambda^2) = \exp(4\psi^2 t^2).$$

It remains to treat $|\lambda| > 1$. Completing the square gives

$$e^{tW} \leq \exp(\lambda^2/4) \exp(W^2/\psi^2).$$

After integration,

$$E_P e^{tW} \leq 2 \exp(\lambda^2/4).$$

Since $2 \leq e$ and $|\lambda| > 1$ implies $1 < \lambda^2$,

$$2 \exp(\lambda^2/4) \leq \exp(1 + \lambda^2/4) \leq \exp(4\lambda^2) = \exp(4\psi^2 t^2).$$

The two cases prove the asserted bound for every $t \in \mathbb{R}$. □

Proof of Lemma 3. Let

$$M(z) = E_P e^{z\eta}.$$

From $k = r + 1$ and $r \geq 2$, one has $k \geq 3$. The treatment-tail assumption gives

$$E_P \exp(\eta^2/\psi_\eta^2) \leq 2,$$

and the conditional-mean identity for g_0 gives

$$E_P \eta = E_P \{T - g_0(X)\} = 0.$$

Applying Lemma 2 to η gives, for every real t ,

$$E_P e^{t\eta} \leq \exp(4\psi_\eta^2 t^2).$$

Consequently, for every $z \in \mathbb{C}$,

$$|M(z)| \leq E_P e^{\operatorname{Re}(z)\eta} \leq \exp\{4\psi_\eta^2 |z|^2\}.$$

Moreover M is entire and $M(0) = 1$, since all real exponential moments are finite.

Suppose, toward a contradiction, that $M(z) \neq 0$ for every $|z| \leq R_0$. Then $\log |M|$ is harmonic on the open disk $\{|z| < R_0\}$. Choose an analytic function F on this disk whose real part is $\log |M|$, and set

$$f(z) = F(z) - F(0).$$

Then $f(0) = 0$ and, for $|z| < R_0$,

$$\operatorname{Re} f(z) = \log |M(z)| \leq 4\psi_\eta^2 R_0^2.$$

The Borel–Carathéodory estimate on the disk of radius R_0 , applied to the normalized function f , gives on the sphere $|z| = R_0/2$

$$|f(z)| \leq 8\psi_\eta^2 R_0^2.$$

Cauchy’s estimate on the disk of radius $R_0/2$ therefore yields

$$|f^{(k)}(0)| \leq \frac{k! 8\psi_\eta^2 R_0^2}{(R_0/2)^k} = 2^{k+3} k! \psi_\eta^2 R_0^{2-k}.$$

Near the origin, $M(0) = 1$, so M takes values in the slit plane on a small disk and the principal logarithm $\log M$ is analytic there. The analytic function

$$z \mapsto f(z) - \log M(z)$$

has identically zero real part on that small connected disk and vanishes at 0; hence it is identically zero there. Thus

$$f^{(k)}(0) = \left. \frac{d^k}{dz^k} \log M(z) \right|_{z=0}.$$

Taking real parts gives

$$|\kappa_k(\eta)| \leq 2^{k+3} k! \psi_\eta^2 R_0^{2-k}.$$

By the definition

$$R_0 = (2^{k+4} k!)^{1/(k-2)} \left(\frac{\psi_\eta^2}{\delta} \right)^{1/(k-2)},$$

we have

$$R_0^{k-2} = 2^{k+4} k! \frac{\psi_\eta^2}{\delta}.$$

Substitution in the preceding bound gives

$$|\kappa_k(\eta)| \leq 2^{k+3} k! \psi_\eta^2 \left(2^{k+4} k! \frac{\psi_\eta^2}{\delta} \right)^{-1} = \frac{\delta}{2} < \delta.$$

This contradicts the cumulant separation condition $|\kappa_k(\eta)| \geq \delta$. Therefore there exists $z_0 \in \mathbb{C}$ with $|z_0| \leq R_0$ and $M(z_0) = 0$. $\square$

Proof of Lemma 4. Put $R = R_1$, $Z_n = T - \bar{g}_n(X)$, and

$$A = 64\{C_q^4 + 4C_\theta^4 \psi_\eta^4 + 4\psi_\xi^4\} + 2 \exp\{16RC_g + 16R^2 \psi_\eta^2\}.$$

The radius is nonnegative, since $R = R_0 + 1$ and $R_0 \geq 0$ by its displayed construction in Definition 17.

First,

$$E_P |Y|^4 \leq 64\{C_q^4 + 4C_\theta^4 \psi_\eta^4 + 4\psi_\xi^4\}.$$

Indeed, $Y = q_0(X) + \theta_0 \eta + \xi$; the range bounds give $|q_0(X)| \leq C_q$ and $|\theta_0| \leq C_\theta$, while the Luxemburg bounds give $E_P |\eta|^4 \leq 4\psi_\eta^4$ and, after integrating the conditional bound, $E_P |\xi|^4 \leq 4\psi_\xi^4$. Applying the elementary fourth-power bound to the three summands gives the display.

Next,

$$E_P \exp\{8R|Z_n|\} \leq 2 \exp\{16RC_g + 16R^2 \psi_\eta^2\}.$$

This is the residual exponential envelope applied with radius $2R$. In detail, $Z_n = \eta + D_n$, where $D_n = g_0(X) - \bar{g}_n(X)$ satisfies $|D_n| \leq 2C_g$ almost surely. Hence $8R|Z_n| \leq 8R|\eta| + 16RC_g$, and Young's inequality at this radius gives

$$8R|\eta| \leq \eta^2 / \psi_\eta^2 + 16R^2 \psi_\eta^2.$$

The Luxemburg bound $E_P \exp(\eta^2 / \psi_\eta^2) \leq 2$ then yields the displayed exponential bound.

Therefore

$$E_P [\{|Y| \exp(2R|Z_n|)\}^2] \leq A.$$

The pointwise inequality used here is

$$\{|Y| \exp(2R|Z_n|)\}^2 \leq |Y|^4 + \exp(8R|Z_n|),$$

and the preceding two displays bound the two expectations.

Let

$$E(o) = |Y(o)| \exp\{2R|Z_n(o)|\}.$$

Then $E \in L^2(P)$ and $\|E\|_2 \leq \sqrt{A}$. For $|z| \leq R$,

$$|Y \exp(zZ_n)| = |Y| \exp\{\operatorname{Re}(zZ_n)\} \leq |Y| \exp\{2R|Z_n|\} = E,$$

because $\operatorname{Re}(zZ_n) \leq |z||Z_n| \leq 2R|Z_n|$. Hence

$$|G_n(z)| = |E_P\{Y \exp(zZ_n)\}| \leq E_P E \leq \|E\|_2 \leq \sqrt{A}.$$

Since $\sqrt{A} = C_G(p)$, the claimed bound follows. $\square$

Proof of Theorem 3. 1. Define the primitive search radius

$$R_{\text{fix}} = A_{r+1} \left(\frac{\psi_\eta^2}{\delta} \right)^{1/(r-1)} + 1,$$

where A_{r+1} is the numerical factor in Definition 17. Let $K_{\text{emp}} > 0$ be the constant supplied by Lemma 9 for $(C_\theta, C_g, C_q, \psi_\eta, \psi_\xi, R_{\text{fix}})$. For $y \in \mathbb{R}$, put

$$u_{\text{ceil}}(y) = \lceil \max\{1, y + 1\} + 1 \rceil,$$

and define

$$\begin{aligned} U_\psi^\circ &= u_{\text{ceil}}(\psi_\eta), & U_R^\circ &= u_{\text{ceil}}(R_{\text{fix}}), \\ N_{\text{cert}}^\circ &= 4(U_\psi^\circ)^2(U_R^\circ + 2)^3, & m_\star^\circ &= N_{\text{cert}}^\circ + 3, \\ e_\star^\circ &= 8U_R^\circ(U_\psi^\circ)^2(U_R^\circ + 1)^2, & d_{\star, \text{den}}^\circ &= 3(2U_R^\circ + 1), \\ u_\star^\circ &= 2e_\star^\circ + N_{\text{cert}}^\circ(m_\star^\circ + d_{\star, \text{den}}^\circ), & a_\star^\circ &= 2^{-u_\star^\circ}. \end{aligned}$$

Thus $a_\star^\circ > 0$. Set

$$C_G^\circ = [64\{C_q^4 + 4C_\theta^4\psi_\eta^4 + 4\psi_\xi^4\} + 2\exp\{16R_{\text{fix}}C_g + 16R_{\text{fix}}^2\psi_\eta^2\}]^{1/2},$$

$$L_{\text{fix}} = R_{\text{fix}} \max \left\{ \left(\frac{23a_\star^\circ}{64} \right)^{-1}, C_G^\circ \left(\frac{31a_\star^\circ}{64} \frac{23a_\star^\circ}{64} \right)^{-1} \right\},$$

and

$$C_{\text{fix}} = 12L_{\text{fix}}^2 K_{\text{emp}} + \frac{1536C_\theta^2 K_{\text{emp}}}{(a_\star^\circ)^2} + 2.$$

Since $R_{\text{fix}} > 0$, $K_{\text{emp}} > 0$, and $a_\star^\circ > 0$, one has $C_{\text{fix}} > 0$. The two fixed-code converse arguments below give positive constants $\mathfrak{c}_{\text{NG}}$ and $\mathfrak{c}_{\text{JMS}}$, depending only on the same primitive tuple: $\mathfrak{c}_{\text{NG}}$ comes from the fixed-code non-Gaussian converse, and $\mathfrak{c}_{\text{JMS}}$ comes from the separate fixed-code JMS ACE converse. We take

$$c = \min\{\mathfrak{c}_{\text{NG}}, \mathfrak{c}_{\text{JMS}}\}, \quad C = C_{\text{fix}}.$$

Then $c > 0$ and $C > 0$.

2. Fix the covariate measurable space, p_0 , the fixed records, p , and the base model as in the statement. The transported-record hypothesis gives

$$p.C_\theta = C_\theta, \quad p.C_g = C_g, \quad p.C_q = C_q, \quad p.\psi_\eta = \psi_\eta, \quad p.\psi_\xi = \psi_\xi, \quad \text{searchRadius}(p) = R_{\text{fix}}.$$

Let $\mathbf{p}_{\star,p}$ and $\mathbf{c}_{\theta,\star,p}$ denote the transported bank and range records. If $a_{\star,p}$ is the lower-modulus certificate of the actual contour bank, then the width-one certified refinement containing ψ_η and R_{fix} gives

$$u_{\star,p} \leq u_{\star}^\circ, \quad a_{\star}^\circ = 2^{-u_{\star}^\circ} \leq 2^{-u_{\star,p}} = a_{\star,p}.$$

Consequently the actual selector constant

$$C_{\text{sel}}(p, \mathbf{p}_{\star,p}) = 12L_p^2 K_{\text{emp}} + \frac{1536p.C_\theta^2 K_{\text{emp}}}{a_{\star,p}^2} + 2,$$

with

$$L_p = \text{searchRadius}(p) \max \left\{ \left(\frac{23a_{\star,p}}{64} \right)^{-1}, C_G(p) \left(\frac{31a_{\star,p}}{64} \frac{23a_{\star,p}}{64} \right)^{-1} \right\},$$

satisfies

$$C_{\text{sel}}(p, \mathbf{p}_{\star,p}) \leq C.$$

The monotonicity is exactly the monotonicity of the bank exponent, the map $u \mapsto 2^{-u}$, and positive reciprocals.

3. The statistic is measurable. Indeed, $\hat{\theta}_{\text{spec},n}$ is obtained by running the finite rational contour program of [Definition 18](#) and then clipping its raw real output:

$$\hat{\theta}_{\text{spec},n}(O_1, \dots, O_n) = \min \{ \max \{ Y_{\text{raw}}(O_1, \dots, O_n), -C_\theta \}, C_\theta \}.$$

On each finite branch trace, the queried observation endpoints are floor-dyadic functions of the sample coordinates, and the remaining operations are rational arithmetic and finite comparisons. The terminal traces are countable and exhaustive, so Y_{raw} is Borel measurable; clipping by constants preserves measurability.

4. The represented execution clause follows from the canonical compilation contract. For every sample,

$$\text{Rep}_{\text{compiled}}(\text{canonicalInput}_{p,\mathbf{p}_{\star,p},\mathbf{c}_{\theta,\star,p}}) = \text{Ord}_{p,\mathbf{p}_{\star,p},\mathbf{c}_{\theta,\star,p}},$$

the represented trace equals the ordinary trace, and the represented output record certifies

$$\min \{ \max \{ Y_{\text{raw}}, -C_\theta \}, C_\theta \}.$$

The last identity is the elementary three-case equality

$$\frac{|y + C_\theta| - |y - C_\theta|}{2} = \min \{ \max \{ y, -C_\theta \}, C_\theta \},$$

used with the positive range record for C_θ .

5. The class relations are those of [Lemma 5](#): for every model m ,

$$m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \implies m \in \mathcal{P}_{\text{NG},n},$$

and, if $s = r$,

$$m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \iff m \in \mathcal{P}_{\text{NG},n}^{\text{ACE}}.$$

The proof uses monotonicity of $L^p(P_X)$ -norms on a probability space to pass from the published $L^r(P_X)$ treatment-code condition to the $L^1(P_X)$ treatment-code condition, and when $s = r$ the comparison subclass imposes the same code-radius requirement.

6. We record the pointwise risk bound used in both fixed-code classes. Let $m \in \mathcal{P}_{\text{NG},n}$, assume $n \geq 2$, and assume

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

where $R_1 = \text{searchRadius}(p)$. For $a \in \{0, 1\}$ and every sample define

$$d_a = \sup_{|z| \leq R_1} |\hat{F}_{a,n}(z) - F_n(z)|, \quad e_a = \sup_{|z| \leq R_1} |\hat{G}_{a,n}(z) - G_n(z)|.$$

Set

$$\mathcal{A} = \{d_0 \leq a_{\star,p}/8, d_1 \leq a_{\star,p}/8\}.$$

On $\mathcal{A}$, the canonical selector packet supplies exactly the hypotheses of [Lemma 13](#). The packet is obtained as follows. First, [Lemma 8](#) gives a bank contour C_{j_0} for the treatment-innovation transform with positive zero count and boundary modulus at least $a_{\star,p}$. By [Lemma 10](#), the smallness condition gives $|H_n(z)| \geq 3/4$ on the search disk and preserves the zero count from the treatment-innovation transform to $F_n = MH_n$; hence C_{j_0} has positive F_n -zero count, H_n is zero-free on and inside the contour, and $|F_n(z)| \geq 3a_{\star,p}/4$ on its boundary.

The pilot-fold inequality $d_0 \leq a_{\star,p}/8$, together with the certified modulus width $a_{\star,p}/64$ and the certified winding enclosure in [Definition 18](#), makes this population contour admissible: its pilot lower endpoint is at least $39a_{\star,p}/64$, and its decoded pilot integer is the positive population zero count. The selected contour maximizes the pilot lower endpoint among admissible contours, so for the selected contour C_j and decoded integer N ,

$$\mu_p := \frac{31a_{\star,p}}{64} \leq |F_n(z)| \quad (z \in C_j),$$

while the evaluation-fold inequality $d_1 \leq a_{\star,p}/8$ gives the evaluation modulus margin required by the selector and

$$\nu_p := \frac{23a_{\star,p}}{64} > 0, \quad d_1 \leq \mu_p - \nu_p.$$

The same certified node specifications give

$$|\hat{F}_{1,n}(z) - F_n(z)| \leq d_1, \quad |\hat{G}_{1,n}(z) - G_n(z)| \leq e_1 \quad (z \in C_j),$$

and [Lemma 4](#) gives $|G_n(z)| \leq C_G(p)$ on $|z| \leq R_1$. Continuity and the positive denominator margins give the two contour-integrability hypotheses for the empirical and population

ratios. Finally, [Theorem 2](#) applies to the selected contour, because F_n is zero-free on C_j , H_n is zero-free on and inside C_j , and the decoded integer N is the positive population zero count; therefore the selected population contour ratio has real part $\theta_0(m)$.

Applying [Lemma 13](#) with

$$\mu_p = \frac{31a_{*,p}}{64}, \quad \nu_p = \frac{23a_{*,p}}{64}$$

gives, on $\mathcal{A}$,

$$|\hat{\theta}_{\text{spec},n} - \theta_0(m)| \leq L_p(d_1 + e_1) + \frac{1}{n}.$$

Hence, on $\mathcal{A}$,

$$(\hat{\theta}_{\text{spec},n} - \theta_0(m))^2 \leq 4L_p^2(d_1^2 + e_1^2) + \frac{2}{n^2}.$$

The folds have cardinality at least $n/3$, and [Lemma 9](#) gives

$$E(d_a^2 + e_a^2) \leq \frac{3K_{\text{emp}}}{n} \quad (a = 0, 1).$$

Markov's inequality therefore yields

$$P(\mathcal{A}^c) \leq \frac{384K_{\text{emp}}}{a_{*,p}^2 n}.$$

Since the estimator and the target both lie in $[-C_\theta, C_\theta]$,

$$(\hat{\theta}_{\text{spec},n} - \theta_0(m))^2 \leq 4C_\theta^2$$

on $\mathcal{A}^c$. Integrating the two displays gives

$$E_m \left[(\hat{\theta}_{\text{spec},n} - \theta_0(m))^2 \right] \leq \frac{C_{\text{sel}}(p, \mathbf{p}_{*,p})}{n} \leq \frac{C}{n}.$$

7. The quantile bound is the Markov consequence of the preceding mean-square bound. For

$$W(O_1, \dots, O_n) = |\hat{\theta}_{\text{spec},n}(O_1, \dots, O_n) - \theta_0(m)|, \quad t = \sqrt{\frac{C}{\gamma n}},$$

the risk bound gives

$$P(W > t) \leq \frac{C/n}{t^2} = \gamma.$$

Equivalently,

$$P(W \leq t) \geq 1 - \gamma,$$

and the defining property of the generalized lower quantile in [Definition 6](#) gives

$$Q_{P,1-\gamma}(W) \leq \sqrt{\frac{C}{\gamma n}}.$$

8. It remains to identify the two lower constants. Put

$$b_\theta = \min\{C_\theta, 1\}, \quad V_{\text{KL}} = \max\left\{1, \frac{16\psi_\eta^2}{\psi_\xi^2}\right\}, \quad \tau_{\text{lo}} = \frac{\psi_\xi}{4}.$$

For a nonempty fixed-code non-Gaussian class, choose m° in that class. For $n \geq 2$, define

$$\theta_- = -\frac{b_\theta}{2\sqrt{n}}, \quad \theta_+ = \frac{b_\theta}{2\sqrt{n}}.$$

The affine Gaussian perturbations m_- and m_+ keep the law of (X, T) , g_0, q_0 , and the supplied clipped codes from m° , and use the outcome equation

$$Y = q_0(X) + \theta_\pm \eta + \xi_\tau, \quad \xi_\tau \sim N(0, \tau_{\text{lo}}^2),$$

with ξ_τ independent of (X, T) . Since $|\theta_\pm| \leq b_\theta \leq C_\theta$, and since [Lemma 11](#) gives the required outcome-innovation Luxemburg envelope at scale ψ_ξ , both m_- and m_+ remain in the same fixed-code non-Gaussian class. Their target separation is

$$|\theta_+ - \theta_-| = \frac{b_\theta}{\sqrt{n}}.$$

The Gaussian likelihood ratio calculation gives

$$D_{\text{KL}}(m_-^{\otimes n} \| m_+^{\otimes n}) \leq n \frac{(\theta_+ - \theta_-)^2 2\psi_\eta^2}{2\tau_{\text{lo}}^2} = \frac{16\psi_\eta^2}{\psi_\xi^2} b_\theta^2 \leq V_{\text{KL}} b_\theta^2.$$

Le Cam's two-point MSE inequality therefore supplies a constant $\lambda_{\text{LC}} > 0$, depending only on $V_{\text{KL}} b_\theta^2$, such that every measurable estimator $\tilde{\theta}$ satisfies

$$\max_{v \in \{-, +\}} E_{m_v} \left[(\tilde{\theta} - \theta_0(m_v))^2 \right] \geq \lambda_{\text{LC}} \frac{b_\theta^2}{n}.$$

Thus the fixed-code non-Gaussian minimax risk is at least

$$\frac{\mathfrak{c}_{\text{NG}}}{n}, \quad \mathfrak{c}_{\text{NG}} = \lambda_{\text{LC}} b_\theta^2 > 0.$$

This is the fixed-code non-Gaussian converse.

For a nonempty fixed-code JMS ACE class, the separate fixed-code ACE converse applies the same two-point argument starting from a law in that JMS ACE class. The affine perturbations preserve JMS ACE membership, the target values θ_- and θ_+ , the law of (X, T) , the nuisance functions g_0, q_0 , and both clipped supplied codes. The same product relative-entropy calculation and Le Cam inequality give a constant $\mathfrak{c}_{\text{JMS}} > 0$, depending only on the primitive tuple, such that

$$\frac{\mathfrak{c}_{\text{JMS}}}{n} \leq \inf_{\tilde{\theta}} \sup_{P \in \mathcal{P}_{\text{ACE}, n}^{\text{JMS}}(p; \text{base})} E_P \left[(\tilde{\theta} - \theta_0(P))^2 \right].$$

9. For the fixed-code non-Gaussian class, assume the hypotheses in the theorem. The lower bound from the preceding step and $c \leq \mathfrak{c}_{\text{NG}}$ give

$$\frac{c}{n} \leq \mathfrak{R}_n^{\text{NG}}.$$

The statistic $\widehat{\theta}_{\text{spec},n}$ is measurable by Step 3, so it is an admissible rule in the minimax infimum, and therefore

$$\mathfrak{R}_n^{\text{NG}} \leq \sup_{P \in \mathcal{P}_{\text{NG},n}(p;\text{base})} E_P \left[(\widehat{\theta}_{\text{spec},n} - \theta_0(P))^2 \right].$$

For every P in this fixed-code class, the current clipped treatment code agrees with the base clipped treatment code. Since the canonical inputs in [Definition 18](#) use the current clipped treatment code, the statistic assembled from the base treatment-code sequence agrees pointwise with the statistic assembled from the class member's treatment-code sequence. Applying Step 6 to the latter statistic and then using this pointwise equality gives

$$\sup_{P \in \mathcal{P}_{\text{NG},n}(p;\text{base})} E_P \left[(\widehat{\theta}_{\text{spec},n} - \theta_0(P))^2 \right] \leq \frac{C}{n}.$$

The same pointwise equality reduces the absolute-error statistic in the generalized-quantile display to the statistic covered by Step 7, giving the displayed generalized-quantile bound over the same supremum.

10. For the fixed-code JMS ACE class, assume its nonemptiness, $n \geq 2$, and the same smallness condition on $\varepsilon_{1,n}$. The ACE lower bound from Step 8 and $c \leq \mathfrak{c}_{\text{JMS}}$ give

$$\frac{c}{n} \leq \inf_{\tilde{\theta}} \sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p;\text{base})} E_P \left[(\tilde{\theta} - \theta_0(P))^2 \right].$$

The estimator is measurable, hence admissible, so the infimum is bounded above by its worst-case risk. By Step 5, every JMS ACE law lies in $\mathcal{P}_{\text{NG},n}$. The fixed-code restriction gives the same current clipped treatment code as the base model, so the pointwise congruence from [Definition 18](#) identifies the statistic built from the base treatment-code sequence with the statistic built from the class member's treatment-code sequence. Step 6 therefore yields

$$\sup_{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}(p;\text{base})} E_P \left[(\widehat{\theta}_{\text{spec},n} - \theta_0(P))^2 \right] \leq \frac{C}{n}.$$

The same pointwise equality and Step 7 give the asserted generalized-quantile bound on this class as well. □

Proof of [Proposition 1](#). Fix the displayed constants and choose the constants c and C supplied by [Theorem 3](#). Set

$$C_{\text{spec}} = C.$$

Then $C_{\text{spec}} > 0$, and its dependence is only through $r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi$.

1. For the class relations, apply [Lemma 5](#) at the displayed parameter record and the displayed sample size n . It gives, law by law,

$$P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \implies P \in \mathcal{P}_{\text{NG},n}$$

and

$$P \in \mathcal{P}_{\text{NG},n}^{\text{ACE}} \implies P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}.$$

Since these pointwise implications are exactly the membership predicates for the displayed law classes, they yield

$$\mathcal{P}_{\text{ACE},n}^{\text{JMS}} \subseteq \mathcal{P}_{\text{NG},n}, \quad \mathcal{P}_{\text{NG},n}^{\text{ACE}} \subseteq \mathcal{P}_{\text{ACE},n}^{\text{JMS}}.$$

The same result also gives, when $s = r$,

$$P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \iff P \in \mathcal{P}_{\text{NG},n}^{\text{ACE}},$$

hence the asserted equality of the two classes under the common clipped-code convention.

2. Fix $P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ using the clipped supplied codes $\bar{g}_n, \bar{q}_n$, and suppose $\mathbf{E}_n^{\text{JMS}}$, $\varepsilon_{1,n} > 0$, and $\varepsilon_{2,n} \geq 0$. If $\varepsilon_{2,n} > 0$, the positive-radius part of the assumed published order- r ACE generalized-quantile guarantee, with the objects and gates stated in [Proposition 1](#), applied with the supplied handle and the same clipped codes, gives

$$Q_{P,1-\gamma}(|\hat{\theta}_{\text{ACE},r,n} - \theta_0(P)|) \leq B_{\text{ACE},n,\gamma}.$$

It remains to treat the boundary case $\varepsilon_{2,n} = 0$. For any $\chi > 0$, let $p^{(\chi)}$ be the parameter record obtained from p by replacing the whole second radius sequence by

$$\varepsilon_{2,m}^{(\chi)} = \max\{\varepsilon_{2,m}, \chi\}, \quad m \geq 1,$$

and leaving every other coordinate unchanged. This sequence is again nonnegative and nonincreasing. Read the same observed-data law, regressions, and supplied code sequences as a law $P^{(\chi)}$ for $p^{(\chi)}$. If $0 < \chi < \varepsilon_{1,n}$, then the class membership is preserved: every condition in $\mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ is unchanged except the outcome-code radius, and at index n

$$\varepsilon_{2,n}^{(\chi)} = \chi \quad \text{with} \quad 0 \leq \varepsilon_{2,n} \leq \chi.$$

Moreover, with

$$\omega_n = (\gamma n)^{-1/2}(\psi_\xi + C_\theta \psi_\eta),$$

the denominator entering the second JMS eligibility quantity is unchanged:

$$\max\{\varepsilon_{1,n}, \max\{\varepsilon_{2,n}^{(\chi)}, \omega_n\}\} = \max\{\varepsilon_{1,n}, \max\{\varepsilon_{2,n}, \omega_n\}\},$$

because $\varepsilon_{2,n} = 0$ and $0 < \chi < \varepsilon_{1,n}$. All other eligibility quantities are unchanged by construction, so $\mathbf{E}_n^{\text{JMS}}$ holds for $p^{(\chi)}$ at the same index.

Let

$$\Lambda_n = C_\gamma r! 16^r \delta^{-1} \varepsilon_{1,n}^r.$$

The assumed ACE guarantee includes $C_\gamma > 0$, and the displayed hypotheses give $\Lambda_n > 0$. For any real $\beta > B_{\text{ACE},n,\gamma}$, set

$$\chi_\beta = \min \left\{ \frac{\varepsilon_{1,n}}{2}, \frac{\beta - B_{\text{ACE},n,\gamma}}{2\Lambda_n} \right\}.$$

Then $0 < \chi_\beta < \varepsilon_{1,n}$, and the positive-radius result applied to $p^{(\chi_\beta)}$ gives

$$Q_{P,1-\gamma}(|\widehat{\theta}_{\text{ACE},r,n} - \theta_0(P)|) \leq B_{\text{ACE},n,\gamma} + \Lambda_n \chi_\beta.$$

By the definition of χ_β ,

$$\Lambda_n \chi_\beta \leq \frac{\beta - B_{\text{ACE},n,\gamma}}{2},$$

so the same quantile is at most β . Since this holds for every $\beta > B_{\text{ACE},n,\gamma}$, the desired boundary inequality follows:

$$Q_{P,1-\gamma}(|\widehat{\theta}_{\text{ACE},r,n} - \theta_0(P)|) \leq B_{\text{ACE},n,\gamma}.$$

3. For the spectral guarantee, assume the displayed nonemptiness condition and choose a law $P^\circ \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$ using the clipped supplied codes $\bar{g}_n, \bar{q}_n$. Define the fixed-code ACE class

$$\mathcal{A}_n^\star = \{P \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} : \bar{g}_n(P) = \bar{g}_n, \bar{q}_n(P) = \bar{q}_n\}.$$

This is the ACE-class set associated with the base law P° in [Theorem 3](#). The statistic computed from the supplied treatment-code sequence agrees pointwise with the statistic computed from the base law's supplied treatment-code sequence, because both constructions use the current treatment code only through the residuals

$$T_i - \bar{g}_n(X_i), \quad i = 1, \dots, n.$$

The ACE-class part of [Theorem 3](#), applied with the fixed primitive and range records, $n \geq 2$, and

$$\varepsilon_{1,n} \leq \{4R_1 \exp(2C_g R_1)\}^{-1},$$

therefore yields

$$\sup_{P \in \mathcal{A}_n^\star} Q_{P,1-\gamma}(|\widehat{\theta}_{\text{spec},n} - \theta_0(P)|) \leq \sqrt{\frac{C_{\text{spec}}}{\gamma n}}.$$

Since $\mathcal{A}_n^\star$ is exactly the class of laws in the statement with the clipped supplied codes $\bar{g}_n, \bar{q}_n$, this is the asserted spectral upper guarantee.

4. Finally consider a sequence (p_n) satisfying the hypotheses in the last item of the proposition. Define

$$\mathcal{L}_n = \varepsilon_{1,n}^T \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1}$$

and

$$\lambda_{\text{ACE}} = C_\gamma r! 16^r \delta^{-1}.$$

The assumed ACE guarantee includes $C_\gamma > 0$, so $\lambda_{\text{ACE}} > 0$. Along the sequence, the fixed-constant identities in the hypotheses rewrite the ACE bound as

$$B_{\text{ACE},n,\gamma} = \lambda_{\text{ACE}} \left[\mathcal{L}_n + 64(C_g + \psi_\eta)^r \{r^2(C_g + \psi_\eta) + \psi_\xi + C_\theta \psi_\eta\} (\gamma n)^{-1/2} \right].$$

The second summand in brackets is nonnegative, hence eventually

$$\lambda_{\text{ACE}} \mathcal{L}_n \leq B_{\text{ACE},n,\gamma}.$$

The assumed divergence is

$$\frac{\mathcal{L}_n}{n^{-1/2}} \longrightarrow \infty.$$

For $n \geq 1$,

$$\sqrt{\frac{C_{\text{spec}}}{\gamma n}} = \sqrt{\frac{C_{\text{spec}}}{\gamma}} n^{-1/2}.$$

Once $\mathcal{L}_n > 0$ and $\lambda_{\text{ACE}} \mathcal{L}_n \leq B_{\text{ACE},n,\gamma}$, we have

$$0 \leq \frac{\sqrt{C_{\text{spec}}/(\gamma n)}}{B_{\text{ACE},n,\gamma}} \leq \frac{\sqrt{C_{\text{spec}}/\gamma}}{\lambda_{\text{ACE}}} \left(\frac{\mathcal{L}_n}{n^{-1/2}} \right)^{-1}.$$

The right-hand side tends to 0, so the squeeze theorem gives

$$\frac{\sqrt{C_{\text{spec}}/(\gamma n)}}{B_{\text{ACE},n,\gamma}} \longrightarrow 0.$$

□

Proof of Theorem 4. 1. Apply Theorem 3 with the fixed constants

$$r, \delta, C_\theta, C_g, C_q, \psi_\eta, \psi_\xi.$$

It gives constants $c, C > 0$, depending only on these displayed constants, with the fixed-code non-Gaussian lower and upper bounds stated there. Also apply Proposition 2 on the chosen measurable covariate space. It gives, for every parameter record p , writing n for the sample size of p , and every $m \in \mathcal{P}_{G,n}^{\text{JMS}}$,

$$\theta_0(m) = 0,$$

and it gives the two zero Gaussian minimax conclusions for every nonempty clipped-code intersection at that same sample size.

2. Fix p_0 , the fixed primitive records, the sequence (p_j) , the supplied measurable deterministic codes $(\hat{g}, \hat{q})$, and the displayed eventual non-Gaussian event. Write n_j for the sample size of p_j , and put

$$\mathfrak{p}_{*,j} := \text{the transport of } \mathfrak{p}_* \text{ from } p_0 \text{ to } p_j, \quad \mathfrak{c}_{\theta,*,j} := \text{the transport of } \mathfrak{c}_{\theta,*} \text{ from } p_0 \text{ to } p_j.$$

For every sufficiently large j , the displayed event in the theorem supplies a base model m_j^{anc} such that

$$m_j^{\text{anc}} \in \mathcal{P}_{\text{NG},n_j}(p_j), \quad \bar{g}_{m_j^{\text{anc}},n_j} = \bar{g}_{n_j}, \quad \bar{q}_{m_j^{\text{anc}},n_j} = \bar{q}_{n_j},$$

together with

$$n_j \geq 2, \quad \varepsilon_{1,n_j} \leq \left(4R_1(p_j) \exp\{2C_g R_1(p_j)\} \right)^{-1}.$$

Let g_j^{anc} and q_j^{anc} be the supplied code sequences carried by m_j^{anc} . The two clipped-code equalities above imply equality of the two fixed-code law sets,

$$\begin{aligned} & \{m : m \in \mathcal{P}_{\text{NG},n_j}(p_j), \bar{g}_{m,n_j} = \bar{g}_{n_j}, \bar{q}_{m,n_j} = \bar{q}_{n_j}\} \\ &= \{m : m \in \mathcal{P}_{\text{NG},n_j}(p_j), \bar{g}_{m,n_j} = \bar{g}_{m_j^{\text{anc}},n_j}, \bar{q}_{m,n_j} = \bar{q}_{m_j^{\text{anc}},n_j}\}. \end{aligned}$$

The statistic also agrees pointwise when the supplied treatment code is replaced by g_j^{anc} :

$$\hat{\theta}_{\text{spec},n_j}(p_j, \mathbf{p}_{\star,j}, \mathbf{c}_{\theta,\star,j}; \hat{g}) = \hat{\theta}_{\text{spec},n_j}(p_j, \mathbf{p}_{\star,j}, \mathbf{c}_{\theta,\star,j}; g_j^{\text{anc}}),$$

because [Definition 18](#) forms the represented observation record with the current clipped values $\bar{g}_{n_j}(X_i)$. Applying [Theorem 3](#) to p_j with base model m_j^{anc} , and using the i.i.d. product sampling law of that base model, yields the minimax bounds

$$\frac{c}{n_j} \leq \mathfrak{R}_{n_j}^{\text{NG}}(p_j; \hat{g}, \hat{q}) \leq \frac{C}{n_j},$$

and, after the preceding law-set and statistic equalities, the estimator upper bound

$$\sup_{P \in \mathcal{P}_{\text{NG},n_j}(p_j; \bar{g}_{n_j}, \bar{q}_{n_j})} \mathbb{E}_P \left[(\hat{\theta}_{\text{spec},n_j} - \theta_0(P))^2 \right] \leq \frac{C}{n_j}.$$

The measurability of $\hat{\theta}_{\text{spec},n_j}$ follows from the same pointwise equality and the measurability conclusion of [Theorem 3](#). The transported primitive bank record reuses the same certified names for $k, \delta, \psi_\eta, R_0$, and R_1 . By [Definition 17](#), the translated-dyadic contour-bank data are a deterministic function of those names and their error-one refinements. Thus, writing B for the bank generated from p_0 and $\mathbf{p}_\star$, the bank generated from p_j and $\mathbf{p}_{\star,j}$ is also B .

For the compiled statement, fix a compiled bounded spectral adapter satisfying the full canonical build-and-compilation specification for $(p_j, \mathbf{p}_{\star,j}, \mathbf{c}_{\theta,\star,j}, \hat{g})$. The represented-data soundness implication for compiled bounded spectral adapters states that this full canonical build-and-compilation specification entails the represented-execution contract for the same parameter record, primitive bank record, range record, and treatment-code sequence. Its proof rewrites the pilot-boundary, winding-quadrature, and evaluation-quadrature entry points to the reference finite programs, identifies the compiled represented program with the ordinary finite-rational program on canonical represented inputs, and then applies the clipping certificate for $\mathbf{c}_{\theta,\star,j}$. Therefore the compiled represented-data execution realizes

$$(p_j, \mathbf{p}_{\star,j}, \mathbf{c}_{\theta,\star,j}, \hat{g})$$

with the same ordinary finite-rational result, trace, and certified clipped output required by the represented-execution clause of the theorem.

3. For the Gaussian part, fix a parameter record p , write n for its sample size, and fix $P \in \mathcal{P}_{\text{G},n}^{\text{JMS}}$. The first conclusion of [Proposition 2](#) gives

$$\theta_0(P) = 0.$$

Now fix measurable deterministic codes $(\widehat{g}, \widehat{q})$ whose clipped-code Gaussian intersection at this sample size is nonempty. The second conclusion of [Proposition 2](#), applied to this same code pair, gives

$$\mathfrak{R}_n^G(p; \widehat{g}, \widehat{q}) = 0, \quad \mathfrak{M}_{n,1-\gamma}^G(p; \widehat{g}, \widehat{q}) = 0.$$

These are precisely the two Gaussian assertions. □

Proof of [Proposition 3](#). Set

$$t = \frac{\pi}{2}, \quad A_{\sin} = \exp\left(-\frac{\pi^2}{8}\right),$$

and

$$K_{\sin} = 4(C_q + 2C_\theta C_g)^2 + 4\psi_\xi^2 + C_\theta^2(8 + 16C_g^2).$$

Choose

$$C = 1 + \frac{16}{A_{\sin}^2} K_{\sin}.$$

Since $A_{\sin} > 0$ and $K_{\sin} \geq 0$, this constant is positive and depends only on $C_\theta, C_g, C_q, \psi_\xi$. Fix p, P , and the model objects satisfying the hypotheses. Define, on the observation space,

$$Z_n = T - \bar{g}_n(X), \quad D_n(X) = g_0(X) - \bar{g}_n(X),$$

so that $Z_n = \eta + D_n(X)$.

1. The stipulated law gives the treatment-innovation transform explicitly. Indeed, for every $z \in \mathbb{C}$,

$$\begin{aligned} M(z) &= \frac{1}{2} \exp\left(-z + \frac{z^2}{2}\right) + \frac{1}{2} \exp\left(z + \frac{z^2}{2}\right) \\ &= \exp\left(\frac{z^2}{2}\right) \cosh z. \end{aligned}$$

Therefore

$$M(it) = \exp\left(-\frac{\pi^2}{8}\right) \cos\left(\frac{\pi}{2}\right) = 0.$$

Near zero,

$$\frac{d}{dz} \log M(z) = z + \tanh z.$$

The third derivative at zero of z is zero, while

$$(\tanh z)' = \operatorname{sech}^2 z, \quad (\tanh z)'' = -2 \tanh z \operatorname{sech}^2 z, \quad (\tanh z)'''(0) = -2.$$

Hence the fourth logarithmic derivative of M at zero is -2 , which is the asserted fourth cumulant:

$$\kappa_4(\eta) = -2.$$

All differentiations are justified by the Gaussian mixture's finite exponential moments.

2. The same transform computation gives the weighted characteristic moment used for the denominator. Since

$$M'(z) = \exp(z^2/2) z \cosh z + \exp(z^2/2) \sinh z,$$

evaluating at $z = it$ yields

$$E_P[\eta e^{it\eta}] = M'(it) = iA_{\sin}.$$

Using $Z_n = \eta + D_n(X)$, the independence of η and X , and the identity $M(it) = 0$,

$$\begin{aligned} E_P[Z_n e^{itZ_n}] &= E_P[\eta e^{it\eta}] E_P[e^{itD_n(X)}] + E_P[e^{it\eta}] E_P[D_n(X) e^{itD_n(X)}] \\ &= iA_{\sin} E_P[e^{itD_n(X)}]. \end{aligned}$$

Taking imaginary parts gives

$$E_P[Z_n \sin(tZ_n)] = A_{\sin} E_P[\cos(tD_n(X))],$$

which is the displayed identity after substituting $t = \pi/2$.

3. The $L^1(P_X)$ treatment-code condition gives

$$E_P|D_n(X)| = \|\bar{g}_n - g_0\|_{L^1(P_X)} \leq \varepsilon_{1,n}.$$

Since $\varepsilon_{1,n} \leq 1/\pi$,

$$t\varepsilon_{1,n} \leq \frac{1}{2}.$$

Using $|1 - \cos u| \leq |u|$,

$$E_P[\cos(tD_n(X))] \geq 1 - tE_P|D_n(X)| \geq 1 - t\varepsilon_{1,n} \geq \frac{1}{2}.$$

Combining this with the denominator identity gives

$$\frac{A_{\sin}}{2} \leq E_P[Z_n \sin(tZ_n)].$$

4. Define the residual sine score

$$\text{Rem}_n(O) = \{Y - \theta_0(P)Z_n\} \sin(tZ_n)$$

and the denominator sine score

$$\text{Den}_n(O) = Z_n \sin(tZ_n).$$

With

$$b_n(X) = q_0(X) - \theta_0(P)D_n(X),$$

the outcome equation gives

$$Y - \theta_0(P)Z_n = b_n(X) + \xi.$$

The zero $M(it) = 0$ implies

$$E_P[\sin(t\eta)] = 0, \quad E_P[\cos(t\eta)] = 0.$$

By independence of η and X ,

$$E_P[b_n(X) \sin\{t(\eta + D_n(X))\}] = 0.$$

The conditional mean restriction for the outcome innovation gives

$$E_P[\xi \sin(tZ_n)] = E_P[E_P(\xi \mid X, T) \sin(tZ_n)] = 0.$$

Thus

$$E_P[\text{Rem}_n(O)] = 0.$$

5. The mixture second moment is bounded by

$$E_P[\eta^2] \leq 4.$$

The range condition for g_0 , together with clipping of $\bar{g}_n$, gives

$$|D_n(X)| \leq 2C_g \quad P\text{-a.s.}$$

Since $|\sin u| \leq 1$,

$$E_P[\text{Den}_n(O)^2] \leq E_P[Z_n^2] \leq 2E_P[\eta^2] + 2E_P[D_n(X)^2] \leq 8 + 16C_g^2.$$

Also,

$$|b_n(X)| \leq C_q + 2C_\theta C_g.$$

The Luxemburg sub-Gaussian condition in [Assumption 9](#) gives

$$E_P[\xi^2] \leq 2\psi_\xi^2.$$

Therefore

$$E_P[\text{Rem}_n(O)^2] \leq 4(C_q + 2C_\theta C_g)^2 + 4\psi_\xi^2.$$

6. By [Definition 14](#), the statistic $\hat{\theta}_{\sin, n}$ is the clipped ratio with denominator score Den_n , remainder score Rem_n , frequency $t = \pi/2$, and threshold $A_{\sin}/4$, because

$$\mathbb{P}_n\{Y \sin(tZ_n)\} = \theta_0(P) \mathbb{P}_n \text{Den}_n + \mathbb{P}_n \text{Rem}_n.$$

Let

$$\mu_{\text{Den}} = E_P[\text{Den}_n(O)].$$

The preceding lower bound gives $\mu_{\text{Den}} \geq A_{\sin}/2$. Applying [Lemma 6](#) pathwise and then averaging under the i.i.d. product law yields

$$E_P\left[(\hat{\theta}_{\sin, n} - \theta_0(P))^2\right] \leq \frac{16}{A_{\sin}^2 n} \left\{ E_P[\text{Rem}_n(O)^2] + C_\theta^2 E_P[\text{Den}_n(O)^2] \right\}.$$

Using the two score bounds,

$$E_P\left[(\hat{\theta}_{\sin, n} - \theta_0(P))^2\right] \leq \frac{16K_{\sin}}{A_{\sin}^2 n} \leq \frac{C}{n}.$$

Together with the identities and denominator lower bound proved above, this proves all claims.

□

Proof of Lemma 6. Write

$$\bar{W} = \mathbb{P}_n W, \quad \bar{R} = \mathbb{P}_n R, \quad \Delta_W = \mathbb{P}_n(W - \mu_W).$$

Since $n \geq 1$,

$$\Delta_W = \bar{W} - \mu_W.$$

Consider first the branch $\bar{W} \geq A/4$. Then $\bar{W} > 0$, and the unclipped ratio obeys

$$\frac{\theta \bar{W} + \bar{R}}{\bar{W}} - \theta = \frac{\bar{R}}{\bar{W}}.$$

The projection onto $[-C_\theta, C_\theta]$ fixes θ , because $|\theta| \leq C_\theta$, and is 1-Lipschitz. Hence

$$|\hat{\theta} - \theta| \leq \left| \frac{\bar{R}}{\bar{W}} \right| \leq \frac{4|\bar{R}|}{A}.$$

Squaring and adding the nonnegative term involving Δ_W gives

$$(\hat{\theta} - \theta)^2 \leq \frac{16}{A^2} \bar{R}^2 \leq \frac{16}{A^2} \{ \bar{R}^2 + C_\theta^2 \Delta_W^2 \}.$$

It remains to handle the branch $\bar{W} < A/4$, where $\hat{\theta} = 0$. Since $\mu_W \geq A/2$,

$$\mu_W - \bar{W} \geq A/4, \quad \text{so} \quad A^2 \leq 16(\bar{W} - \mu_W)^2 = 16\Delta_W^2.$$

Also $|\theta| \leq C_\theta$, hence $\theta^2 \leq C_\theta^2$. Therefore

$$(\hat{\theta} - \theta)^2 = \theta^2 \leq C_\theta^2 \leq \frac{16}{A^2} C_\theta^2 \Delta_W^2 \leq \frac{16}{A^2} \{ \bar{R}^2 + C_\theta^2 \Delta_W^2 \}.$$

The two branches give the claimed pathwise bound. □

Proof of Theorem 5. 1. Fix $C_\theta, C_g, C_q, \psi_\xi$. By Lemma 7, there is a constant $C > 0$, depending only on these four displayed constants, such that whenever an admissible parameter record has primitive entries $C_\theta, C_g, C_q, \psi_\xi$ equal to the displayed constants and cumulant order $k = 4$, every $a \in (0, 1]$ satisfying the Gaussian–Rademacher law assumption on η , the sampling and model restrictions, and the $L^1(P_X)$ treatment-code radius satisfies the full Gaussian–Rademacher path conclusion with constant C . Applying this rule to the theorem’s records with $r = 3$ and $k = 4$ gives the first assertion.

2. Now fix a published ACE handle, $\gamma \in \mathbb{R}$, and $C_\gamma \in \mathbb{R}$, and assume that the handle satisfies the theorem’s published ACE interface at (γ, C_γ) . Unpacking that hypothesis gives first

$$C_\gamma > 0.$$

It also gives the following specialization rule: for every parameter record p whose probability-level coordinate is γ , every real $\Delta > 0$, and every pair of strictly positive current code radii

$$\varepsilon_{1,n} > 0, \quad \varepsilon_{2,n} > 0,$$

every supplied code pair $(\widehat{g}, \widehat{q})$ and every model satisfying the clipped-code identities, the local ACE class conditions at Δ , and the local JMS eligibility inequalities at Δ obey

$$Q_{P,1-\gamma} \left(\left| \widehat{\theta}_{\text{ACE},r,n} - \theta_0(P) \right| \right) \leq B_{\text{ACE},n,\gamma}(\Delta; C_\gamma).$$

For the theorem, take

$$\Delta = \delta_n.$$

The standing assumption $\delta_n > 0$ supplies the required positivity of Δ . Together with the displayed strict positivity conditions $\varepsilon_{1,n} > 0$ and $\varepsilon_{2,n} > 0$, the theorem's clipped-code identities, local ACE class conditions, and local JMS eligibility inequalities are exactly the premises of the specialization rule at this Δ . Substitution therefore yields

$$Q_{P,1-\gamma} \left(\left| \widehat{\theta}_{\text{ACE},r,n} - \theta_0(P) \right| \right) \leq B_{\text{ACE},n,\gamma}(\delta_n; C_\gamma),$$

where

$$B_{\text{ACE},n,\gamma}(\Delta; C_\gamma) = C_\gamma r! 16^r \Delta^{-1} \left(\varepsilon_{1,n}^r \varepsilon_{2,n} + C_\theta \varepsilon_{1,n}^{r+1} + 64(C_g + \psi_\eta)^r (r^2(C_g + \psi_\eta) + \psi_\xi + C_\theta \psi_\eta)(\gamma n)^{-1/2} \right).$$

□

Proof of Proposition 2. 1. Fix a model m in the displayed Gaussian class, and write P for its observed-data law. Define

$$\eta = T - g_0(X), \quad \xi = Y - q_0(X) - \theta_0(P)\eta.$$

The conditional mean restrictions in Definitions 2 and 4 give

$$E_P[\xi \mid X, T] = 0,$$

and $q_0(X)$ and $\theta_0(P)\eta$ are measurable with respect to (X, T) . Hence conditional-expectation linearity yields

$$E_P[Y \mid X, T] = q_0(X) + \theta_0(P)\eta \quad P\text{-a.s.}$$

The bounded-outcome condition gives $-C_q \leq Y \leq C_q$ P -a.s. Conditioning the two inequalities on (X, T) gives

$$-C_q \leq E_P[Y \mid X, T] \leq C_q \quad P\text{-a.s.},$$

and therefore

$$|q_0(X) + \theta_0(P)\eta| \leq C_q \quad P\text{-a.s.}$$

Conditioning the same bounded-outcome inequalities on X , and using

$$E_P[Y \mid X] = q_0(X),$$

gives

$$|q_0(X)| \leq C_q \quad P\text{-a.s.}$$

Combining the two displayed bounds,

$$|\theta_0(P)\eta| = |(q_0(X) + \theta_0(P)\eta) - q_0(X)| \leq |q_0(X) + \theta_0(P)\eta| + |q_0(X)| \leq 2C_q \quad P\text{-a.s.}$$

2. Suppose, for contradiction, that $\theta_0(P) \neq 0$, and define

$$\Lambda_\theta = \frac{2C_q}{|\theta_0(P)|} \in \mathbb{R}.$$

The previous step implies

$$|\eta| \leq \Lambda_\theta \quad P\text{-a.s.},$$

so

$$P\{\eta > \Lambda_\theta\} = 0.$$

By [Definition 4](#), $\eta \sim N(0, \sigma^2)$ with $\sigma > 0$. Thus

$$N(0, \sigma^2)((\Lambda_\theta, \infty)) = 0.$$

A nondegenerate Gaussian law is equivalent to Lebesgue measure on $\mathbb{R}$, while

$$\lambda((\Lambda_\theta, \infty)) = \infty.$$

This contradiction proves

$$\theta_0(P) = 0.$$

3. Now fix deterministic code sequences $\hat{g}, \hat{q}$ and assume the fixed-code Gaussian class in the statement is inhabited. Consider the estimator

$$\tilde{\theta}_0(O_1, \dots, O_n; \hat{g}_n, \hat{q}_n) = 0.$$

For every model in that fixed-code class, the first part gives $\theta_0(P) = 0$, and hence

$$E_{P^{\otimes n}} \left[\{\tilde{\theta}_0(O_1, \dots, O_n; \hat{g}_n, \hat{q}_n) - \theta_0(P)\}^2 \right] = 0.$$

Taking the supremum over the fixed-code class and then the infimum over all estimators in [Definition 9](#) gives

$$\mathfrak{R}_n^G(p; \hat{g}, \hat{q}) \leq 0.$$

Since this risk is nonnegative by definition,

$$\mathfrak{R}_n^G(p; \hat{g}, \hat{q}) = 0.$$

4. It remains to evaluate the generalized-quantile criterion for the same zero estimator. Since $\gamma \in (1/2, 1)$,

$$0 < 1 - \gamma < 1.$$

For every model in the fixed-code Gaussian class,

$$|\tilde{\theta}_0(O_1, \dots, O_n; \hat{g}_n, \hat{q}_n) - \theta_0(P)| = 0 \quad P^{\otimes n}\text{-a.s.}$$

Thus the law of this loss is the point mass at zero. Its distribution function is 0 for negative thresholds and 1 at every nonnegative threshold, so the generalized lower $(1 - \gamma)$ -quantile of [Definition 6](#) is

$$Q_{P, 1-\gamma}(0) = 0.$$

Consequently the supremum of the fixed-code quantile loss for the zero estimator is 0, and the minimax infimum is at most 0. Nonnegativity in [Definition 9](#) gives

$$\mathfrak{M}_{n, 1-\gamma}^G(p; \hat{g}, \hat{q}) = 0.$$

□

Proof of Lemma 5. Fix n and a model m . For every code index j , the clipped-code discrepancies

$$x \mapsto \bar{g}_j(x) - g_0(x), \quad x \mapsto \bar{q}_j(x) - q_0(x)$$

are P_X -measurable, because the supplied codes and the regressions are measurable and clipping preserves measurability. Also P_X is a probability law.

Assume first that $m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$. All structural, range, tail, and cumulant requirements of $\mathcal{P}_{\text{NG},n}$ are fields of the JMS ACE class in Definition 5. It remains to convert the $L^r(P_X)$ treatment-code condition into the $L^1(P_X)$ condition in Definition 3. Put

$$f(x) = \bar{g}_n(x) - g_0(x).$$

Since P_X is a probability law and $1 \leq r$, norm monotonicity gives

$$\|f\|_{L^1(P_X)} \leq \|f\|_{L^r(P_X)}.$$

The JMS ACE condition gives $\|f\|_{L^r(P_X)} \leq \varepsilon_{1,n}$, hence

$$\int |\bar{g}_n(x) - g_0(x)| dP_X(x) \leq \varepsilon_{1,n}.$$

Thus $m \in \mathcal{P}_{\text{NG},n}$.

Next assume $m \in \mathcal{P}_{\text{NG},n}^{\text{ACE}}$. The underlying non-Gaussian membership supplies the common structural, range, tail, cumulant, and L^1 treatment-code requirements. The comparison-subclass restrictions give

$$\|\bar{g}_n - g_0\|_{L^s(P_X)} \leq \varepsilon_{1,n}, \quad \|\bar{q}_n - q_0\|_{L^s(P_X)} \leq \varepsilon_{2,n}.$$

Because $r \leq s$ and P_X is a probability law,

$$\|\bar{g}_n - g_0\|_{L^r(P_X)} \leq \|\bar{g}_n - g_0\|_{L^s(P_X)}, \quad \|\bar{q}_n - q_0\|_{L^r(P_X)} \leq \|\bar{q}_n - q_0\|_{L^s(P_X)}.$$

These are exactly the two JMS ACE code-radius requirements, so $m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$.

Finally suppose $s = r$. The preceding paragraph already proves

$$m \in \mathcal{P}_{\text{NG},n}^{\text{ACE}} \implies m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}.$$

Conversely, if $m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}}$, the first paragraph gives $m \in \mathcal{P}_{\text{NG},n}$, and the two displayed JMS ACE $L^r(P_X)$ code bounds become exactly the two $L^s(P_X)$ subclass bounds when $s = r$. Hence

$$m \in \mathcal{P}_{\text{ACE},n}^{\text{JMS}} \iff m \in \mathcal{P}_{\text{NG},n}^{\text{ACE}}.$$

□

Proof of Lemma 7. Choose

$$K = 4(C_q + 2C_\theta C_g)^2 + 4\psi_\xi^2 + C_\theta^2(8 + 16C_g^2), \quad C = 1 + 64K + 4C_\theta^2.$$

Then $C > 0$. Fix p, a, m satisfying the assumptions, and write

$$t = t_a = \frac{\pi}{2a}, \quad A = A_a = a \exp\left\{-\frac{(1-a^2)\pi^2}{8a^2}\right\}.$$

Since $0 < a \leq 1$, $t > 0$ and $A > 0$.

Independence of $G_\circ$ and S gives the product transform

$$M_a(z) = E \exp\{z\sqrt{1-a^2}G_\circ\} E \exp\{zaS\} = \exp\left(\frac{1-a^2}{2}z^2\right) \cosh(az).$$

The fourth logarithmic derivative at zero of this transform is the Rademacher contribution, hence

$$\kappa_4(\eta_a) = -2a^4.$$

On the imaginary axis,

$$M_a(iu) = \exp\left(-\frac{1-a^2}{2}u^2\right) \cos(au).$$

The exponential factor is nonzero. Therefore the first positive zero occurs when $au = \pi/2$, namely

$$t_a = \frac{\pi}{2a}, \quad M_a(it_a) = 0, \quad M_a(iu) \neq 0 \quad (0 < u < t_a).$$

For any $d \in \mathbb{R}$,

$$E_P[\sin\{t_a(\eta + d)\}] = \Im\left(e^{it_ad} E_P[e^{it_a\eta}]\right) = \Im\left(e^{it_ad} M_a(it_a)\right) = 0.$$

Let

$$Z_n = T - \bar{g}_n(X), \quad D_n = g_0(X) - \bar{g}_n(X),$$

so that

$$Z_n = \eta + D_n(X).$$

The derivative of the transform at the first zero is

$$M'_a(it_a) = iA_a,$$

equivalently

$$E_P[\eta e^{it_a\eta}] = iA_a.$$

Using $Z_n = \eta + D_n(X)$, independence of η and X , and $M_a(it_a) = 0$,

$$\begin{aligned} E_P[Z_n e^{it_a Z_n}] &= E_P[\eta e^{it_a \eta}] E_P[e^{it_a D_n(X)}] + E_P[e^{it_a \eta}] E_P[D_n(X) e^{it_a D_n(X)}] \\ &= iA_a E_P[e^{it_a D_n(X)}]. \end{aligned}$$

Taking imaginary parts gives

$$E_P[Z_n \sin(t_a Z_n)] = A_a E_P[\cos(t_a D_n)].$$

Assume now that $t_a \varepsilon_{1,n} \leq 1/2$. The $L^1(P_X)$ condition gives

$$E_P|D_n(X)| \leq \varepsilon_{1,n}.$$

Since $1 - \cos u \leq |u|$,

$$E_P[\cos(t_a D_n)] \geq 1 - t_a E_P|D_n| \geq 1 - t_a \varepsilon_{1,n} \geq \frac{1}{2}.$$

Together with the preceding identity and $A_a > 0$, this yields

$$E_P[Z_n \sin(t_a Z_n)] \geq A_a/2.$$

Define the two scores

$$W = Z_n \sin(t_a Z_n), \quad R = (Y - \theta_0 Z_n) \sin(t_a Z_n).$$

The zero-shift identity and the conditional mean condition imply

$$E_P[R] = 0.$$

Indeed, writing

$$Y - \theta_0 Z_n = q_0(X) - \theta_0 D_n(X) + \xi,$$

the covariate term is killed by the sine-shift identity after conditioning on X , while

$$E_P[\xi \sin(t_a Z_n)] = E_P[\sin(t_a Z_n) E_P(\xi \mid X, T)] = 0.$$

Thus

$$E_P[Y \sin(t_a Z_n)] = \theta_0 E_P[Z_n \sin(t_a Z_n)].$$

The Gaussian–Rademacher path has the second-moment envelope

$$E_P[\eta^2] \leq 4.$$

With $|g_0| \leq C_g$ and $|\bar{g}_n| \leq C_g$, we have

$$|D_n| \leq 2C_g,$$

and therefore

$$E_P[W^2] \leq 8 + 16C_g^2.$$

Similarly, the range conditions give

$$|q_0(X) - \theta_0 D_n(X)| \leq C_q + 2C_\theta C_g,$$

and the conditional Luxemburg bound for ξ gives

$$E_P[\xi^2] \leq 2\psi_\xi^2.$$

Consequently

$$E_P[R^2] \leq 4(C_q + 2C_\theta C_g)^2 + 4\psi_\xi^2.$$

The two displayed square-moment bounds are exactly the components entering K .

Let

$$\bar{W} = \mathbb{P}_n W, \quad \bar{R} = \mathbb{P}_n R, \quad \mu_W = E_P W.$$

The estimator in the statement is the clipped ratio based on

$$\mathbb{P}_n[Y \sin(t_a Z_n)] = \theta_0 \bar{W} + \bar{R}$$

with threshold $A_a/4$. Since $\mu_W \geq A_a/2$, [Lemma 6](#) gives the pathwise inequality

$$(\hat{\theta}_{a,n} - \theta_0)^2 \leq \frac{16}{A_a^2} \left\{ \bar{R}^2 + C_\theta^2 (\bar{W} - \mu_W)^2 \right\}.$$

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